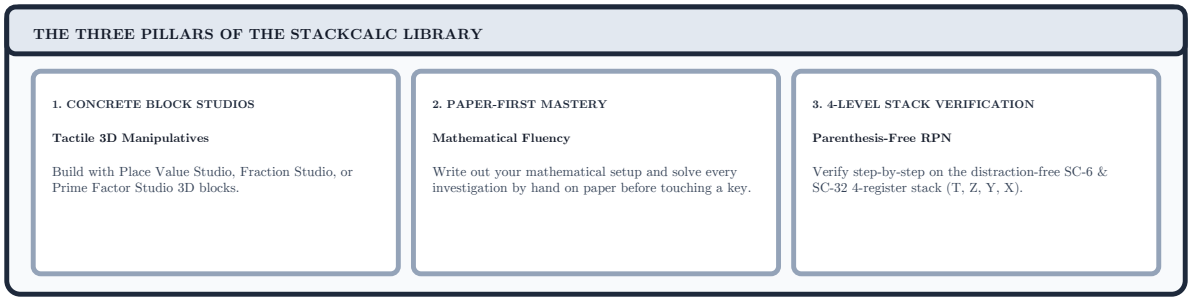
THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 2B

# UPPER ELEMENTARY MATH LAB

## Multi-Digit Operations, 84 mm Fraction Towers, Prime Factors & Order of Operations (Grades 3–5)

A self-contained, open-and-go Grades 3–5 student exploration and practice workbook for homeschooling families and classrooms. Every 11-page unit connects tactile manipulative models—Fraction Studio proportional towers, Prime Factor Studio logarithmic blocks, and Place Value trays—with Core & Extension Concept Labs, Misconception Spotters, 4-Problem Lab Investigations, six leveled Worked-Example & Practice Worksheets (Practice A through F), a 100-Point Unit Mastery Check, an Applied STEM Lab Case, and 4 Cumulative Mastery Exams (840 active problems), verified step-by-step on the StackCalc SC-6 calculator.




**StackCalc**

stackcalc.io

First Edition

## BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

**Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)**

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Solving Equations from the Inside Out (Grades 7–12, AP & Engineers / StackCalc SC-32)

**K–12 Concept-First Curriculum & Grade-Band Student Workbooks (US Letter 8.5" × 11" Editions)**

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Facilitation & Coaching Notes
- Book 2A — Early Elementary Math Lab: Grades K–2 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 2B — Upper Elementary Math Lab: Grades 3–5 SC-6 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method SC-6 Student Workbook (10 Units • 1,000 Problems • 128 Pages)
- Book 4 — High School & STEM Lab: Grades 9–12 & AP SC-32 Student Workbook (10 Eleven-Page Units + 4 Exams • 840 Problems • 128 Pages)
- Book 5 — Stack Speed & Fluency: Full-Year Speed Lessons, Playbooks & 180 Progressive Practice Tests (3,690 Problems • 128 Pages)
- Complete Online Answer Keys & Keystroke Verification Portal: https://stackcalc.io/curriculum/answers

**Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)**

- Place Value Studio: 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio: 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

**Automated Unit-Test Verification & 36-Week Open-Education Pacing Architecture**

- 100% Unit-Tested Problems: Every mathematical solution and RPN keystroke sequence across the curriculum is verified by automated Python and Swift RPNCore unit tests.
- Flexible Pacing: Compatible with Singapore Math, Saxon, RightStart, Beast Academy/AoPS, Eureka, Illustrative Math, Big Ideas, and Bridges.

## COPYRIGHT, STATE STANDARDS ALIGNMENT & OPEN-EDUCATION LICENSE

**UPPER ELEMENTARY MATH LAB: Multi-Digit Operations, 84 mm Fraction Towers, Prime Factors & Order of Operations (Grades 3–5)**

Copyright © 2026 by StackCalc (https://stackcalc.io). Some rights reserved. Published by StackCalc Learning Lab, San Francisco, California. First Edition (US Letter 8.50 × 11.00 in / 215.9 × 279.4 mm).

**State & District Curriculum Alignment:**

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

**Open-Education License (CC BY-NC-SA 4.0) & Perpetual Classroom Reproduction Permission:**

Classroom teachers, math coaches, interventionists, homeschool families, and schools are expressly permitted to photocopy or print student lab worksheets, speed trials, reference cards, and parent welcome letters for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

**Regulatory & Physical Safety Notice (Reese's Law / P.L. 117-171 / 16 CFR Part 1263 & ASTM F963):**

Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

**Trademark Notice & Print Colophon:**

StackCalc, SC-32, and SC-6 are trademarks of StackCalc. All curriculum and textbook names are trademarks of their respective owners, none of whom endorse or sponsor this independent publication. Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, and STIX Math in high-contrast monochrome ink on archival white paper.

# Table of Contents

Check off each 11-page unit as you complete the Labs, Practice A–F, Mastery Quiz, and STEM Case.


| Section / Unit & Mathematical Topic | Grades & Standards | 11-Page Architecture (80 Probs/Unit) | Pages / Done |
|---|---|---|---|
| **Preface: Welcome to the Hands-On Math Lab!** | Students & Families | Paper & Blocks First, Zero Crutch Rule | **p. iv [ ]** |
| **RPN Stack Dynamics & Hardware Tour Bootcamp** | SC-6 Orientation | CRA Framework, Stack Dynamics & Pre-Test | **pp. 1–4 [ ]** |
| **Unit 01: Multiplication & Area Models**<br>Focus: Rectangular Area Grid Diagram & Place Value Studio | **Grade 3**<br>CCSS.MATH.3.OA.A.1 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Why 24 Is the Champion Packing Number & 90° Arrays | **pp. 5–15 [ ]** |
| **Unit 02: Division Concepts & Remainders**<br>Focus: Place Value Studio & Prime Factor Studio | **Grade 3**<br>CCSS.MATH.3.OA.A.2 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Fair-Share Division & The Baker's Remainder Check | **pp. 16–26 [ ]** |
| **Unit 03: Unit Fractions & Proportional Heights**<br>Block Studio: Fraction Studio (84 mm Proportional Towers) | **Grade 3 – 4**<br>CCSS.MATH.3.NF.A.1 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>The Pharaoh's Bakery & Why Fraction Towers Stand 84 mm | **pp. 27–37 [ ]** |
| **Unit 04: Multi-Digit Multiplication & Partial Products**<br>Focus: Partial-Product Area Grid Diagram & Place Value Studio | **Grade 4**<br>CCSS.MATH.4.NBT.B.5 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Slicing Wide Tile Grids Along the Ten-Lines | **pp. 38–48 [ ]** |
| **Unit 05: Long Division & Remainders**<br>Block Studio: Place Value Studio (6×2 Whole Number Tray) | **Grade 4**<br>CCSS.MATH.4.NBT.B.6 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Unpacking Multi-Story Crates & The Q × D + R Audit | **pp. 49–59 [ ]** |
| **Unit 06: Fraction Equivalence & Denominators**<br>Block Studio: Fraction Studio (Comparator Well & Proportional Towers) | **Grade 4**<br>CCSS.MATH.4.NF.A.1 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Hot Dogs vs. Buns, Drum Polyrhythms & The 84 mm Well | **pp. 60–70 [ ]** |
| **Unit 07: Prime Factors, GCF & LCM**<br>Block Studio: Prime Factor Studio (Logarithmic Prime Blocks 2, 3, 5, 7) | **Grade 4 – 5**<br>CCSS.MATH.4.OA.B.4 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>17-Year Prime Cicadas & Hunting-Tooth Bicycle Gears | **pp. 71–81 [ ]** |
| **Unit 08: Order of Operations & The Stack**<br>Focus: Expression Tree Diagram & 4-Register Stack | **Grade 4 – 5**<br>CCSS.MATH.5.OA.A.1 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Beating the Equals-Sign Trap & The 4-Floor Elevator | **pp. 82–92 [ ]** |
| **Unit 09: Adding & Subtracting Fractions**<br>Block Studio: Fraction Studio (84 mm Proportional Towers) | **Grade 5**<br>CCSS.MATH.5.NF.A.1 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Overflowing the 84 mm Well & Artisan Bakery Scaling | **pp. 93–103 [ ]** |
| **Unit 10: Fraction Multiplication & Division**<br>Block Studio: Fraction Studio (84 mm Proportional Towers) | **Grade 5**<br>CCSS.MATH.5.NF.B.4 | **Core/Ext • Inv • Practice A–F • Quiz • STEM**<br>Pythagoras's Musical Strings & Split-and-Rebuild Towers | **pp. 104–114 [ ]** |

**APPENDICES A–D: CERTIFICATE & TRACKER (PP. 115–116), CUMULATIVE EXAMS (PP. 117–120) & 3D BLOCK STUDIOS (PP. 121–124)**

Complete verified answer keys & keystroke traces are hosted online at https://stackcalc.io/curriculum/answers#upper-elementary!

# Preface: Welcome to the Hands-On Math Lab!

Why we build with blocks and solve on paper first—and how each 11-page unit builds complete mastery.

---

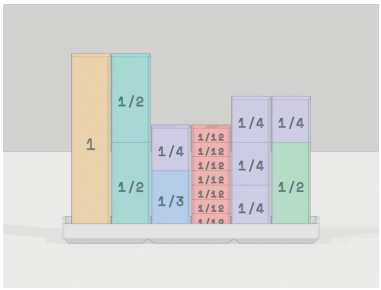
### 1. PAPER & BLOCKS FIRST: WHY THIS LAB MAKES YOU STRONGER AT MATH (NEVER A CRUTCH!)

Whether you are learning at your homeschool kitchen table or in a classroom, this workbook was built on one simple rule: a calculator should NEVER replace your brain! In every unit, you build Place Value, Fraction, and Prime Factor concepts with physical blocks and solve every problem by hand on paper first. Only after you write your paper prediction do you pick up your distraction-free StackCalc calculator to check your work.

Why does your calculator have an [ENTER] key instead of [=]? Think about how you calculate on paper: first you write your first number on the top line (Waiting Shelf, Line Y). Next you write your second number below it (Workbench, Line X). Only after both quantities are in place do you apply your operation (+, −, ×, ÷) to combine them—just like paper math!

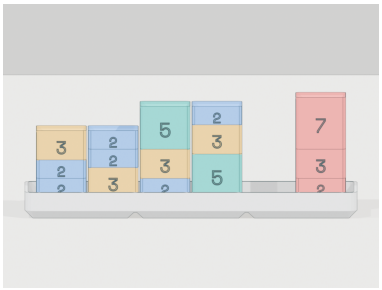
### 2. HANDS-ON TACTILE MODELS & VERIFICATION FOR UPPER ELEMENTARY MATH LAB

**1. Fraction Towers**



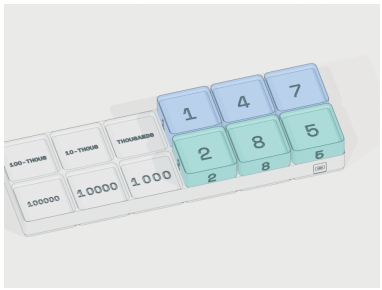

84 mm Proportional Towers (1 Whole to 1/12)

**2. Prime Factor Blocks**




Logarithmic Prime Blocks (2, 3, 5, 7)

**3. Two-Row Stack Tray**




Parenthesis-Free Inner Operations First

### 3. 5-DAY PACING & 3-TIER PROGRESSION ACROSS EVERY 11-PAGE UNIT

Every unit in Upper Elementary Math Lab follows a structured 11-page progression (ideal for a 5-day rhythm or 3-tier differentiation):
- Day 1 (Pages 1–3): Core Concept Lab, Extension & Misconception Spotter, and 4-Problem Hands-On Lab Investigation.
- Day 2 (Pages 4–5 • Tier 1 Foundation): Practice A & B — Physical block pairing, step-by-step paper setups, and core 2-step fluency.
- Day 3 (Pages 6–7 • Tier 2 Grade-Level Core): Practice C & D — Word problems, mental math shortcuts, and error diagnostics.
- Day 4 (Pages 8–9 • Tier 3 Challenge): Practice E & F — Multi-step synthesis and fast-finisher championship problems.
- Day 5 (Pages 10–11): 100-Point Unit Mastery Check Assessment & Applied Real-World STEM Engineering Case Study.

• Online Answer Key & Keystroke Portal: https://stackcalc.io/curriculum/answers#upper-elementary

### Recommended Kit for Book 2B (Grades 3–5) & Try-Before-You-Buy Simulator (stackcalc.io)

- Core Kit for Book 2B: StackCalc SC-6 Calculator + Fraction, Prime Factor & Place Value Studios + Volume I: From Blocks to the Stack.
- Waiting for your physical SC-6 to arrive? Practice every lesson today on our free interactive web simulator or iOS/watchOS app at https://stackcalc.io!

# Hands-On Starter: Touch the Math Before the Keys

Why tactile models make arithmetic, fractions, and algebra click before using a calculator.

## 1. THE CONCRETE-REPRESENTATIONAL-ABSTRACT (CRA) PATHWAY

Mathematics is not a collection of abstract symbols to memorize—it is the study of real patterns and quantities. In this workbook, you always follow the three-stage CRA learning sequence before touching a calculator key:

**STAGE 1: CONCRETE**

**Build with 3D Block Studios**

Snap Digit Blocks, 84 mm Fraction Towers, or Logarithmic Prime Blocks into the studio tray.

**STAGE 2: REPRESENTATIONAL**

**Solve by Hand on Paper**

Write your mathematical setup and paper prediction step-by-step.

**STAGE 3: ABSTRACT**

**Verify on SC-6 Stack**

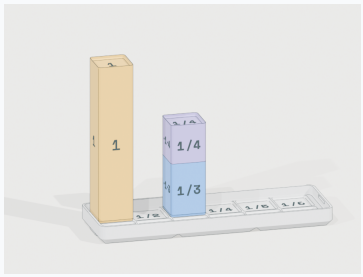
Push quantities onto Registers Y and X to confirm your paper math matches!

## 2. YOUR 3D MANIPULATIVE BLOCK STUDIOS & PAPER-FIRST PROBLEM SOLVING

- Fraction Studio (84.0 mm Proportional Towers): 1 Whole stands 84.0 mm tall. Halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), and twelfths (7 mm) stand side-by-side in the 12-Socket Comparator Lid to prove equivalence by touch.
- Multiplying & Dividing Fractions: Scale towers directly or split into equal parts and rebuild 1 Full Tower (proving why $\div(c/d) = \times(d/c)$).
- Prime Factor Studio (Logarithmic Blocks 2, 3, 5, 7): Equal products stand at the exact same physical height ($2 \times 2 \times 3 = 12$ matches $3 \times 4$).

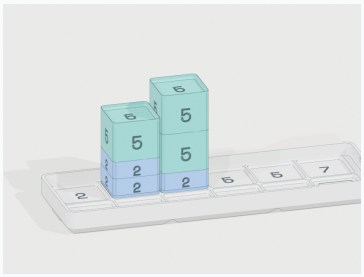
**TACTILE ECOSYSTEM BRIDGE: 84.0 mm Flush Ceiling = Equivalent Fractions • Two-Dot Entry [.][a][.][b] = Fraction on Stack • Log Prime Height = Product**

**Fraction Towers (84 mm)**




Fraction Studio • 84 mm Comparator Well

**Prime Factor Towers**




Prime Factor Studio • Log Prime Towers

**Area Partial Products**

**Area Partial Products: $24 \times 6 = 120 + 24$**

| Tens Box: $20 \times 6$ | $4 \times 6$ |
|---|---|
| = 120 | = 24 |

20 [ENTER] 6 [×] 4 [ENTER] 6 [×] [+] → 144

### Paper Steps First — The Universal Lab Rule

Never touch a calculator key until you have built with blocks or written your paper steps. The StackCalc instrument is your personal laboratory verification tool—not an answer generator!

# SC-6 Calculator Dynamics: Calculating Without Parentheses

How the Waiting Shelf [Y] and Workbench [X] make calculations natural and transparent.

## 1. MEET THE FOUR-LEVEL STACK (T, Z, Y, X)

Ordinary calculators force you to type parentheses or rely on invisible hidden memory. The StackCalc calculator displays up to 4 lines on screen simultaneously (T, Z, Y, X):

- Register X (Floor 1): Active Workbench where you type your current number.
- Register Y (Floor 2): Waiting Shelf holding your first number after [ENTER].
- Registers Z & T (Floors 3 & 4): Upper memory levels that lift & drop automatically.




## 2. ESSENTIAL SC-6 STACK ELEVATOR DYNAMICS

**1. Two-Dot Fraction Entry (1/2 + 1/4 = 3/4):**

[ . ] [ 1 ] [ . ] [ 2 ] [ ENTER ] [ . ] [ 1 ] [ . ] [ 4 ] [ + ]

Two-dot entry [.][1][.][2] builds 1/2 directly on the stack!

**2. Inside-Out Grouping ((14 + 6) × 3 = 60):**

[ 1 ] [ 4 ] [ ENTER ] [ 6 ] [ + ] [ 3 ] [ × ]

Add inner 14 + 6 = 20 first, then multiply by 3 with zero parentheses.

**3. Dividing Fractions (1/2 ÷ 3/4 = 2/3):**

[ . ] [ 1 ] [ . ] [ 2 ] [ ENTER ] [ . ] [ 3 ] [ . ] [ 4 ] [ ÷ ]

Verifies paper split-and-rebuild (1/2 ÷ 3 × 4 = 2/3) in one press.

**4. Prime Tower Product (2 × 2 × 3 = 12):**

[ 2 ] [ ENTER ] [ 2 ] [ × ] [ 3 ] [ × ]

Chains prime factors left-to-right just like stacking Prime Factor Studio blocks.

### Elevator Lift & Drop Rule • Try It Live on Your SC-6 or Web/iOS Simulator

Every binary operation (+, −, ×, ÷) combines Waiting Shelf Y and Workbench X into X while upper registers drop automatically. Key the 4 sequences above on your physical SC-6 calculator (or the free interactive simulator & StackCalc 32 iOS/watchOS app at stackcalc.io) before turning to Page 3!

# Hardware Tour: Meet Your SC-6 Instrument

Understanding keycaps, shift layers, LCD annunciators, and error recovery.

## 1. THE SC-6 KEYBOARD ARCHITECTURE & LAYERS

The StackCalc SC-6 features a streamlined distraction-free interface designed for primary and elementary mathematics:
- Primary Digits & Decimal: High-contrast mechanical keys that map 1-to-1 with Digit Blocks [0]–[9] on the Place Value Tray.
- [ENTER] Key: The stack-lift key that pushes the current number from Workbench X (Tray Row 0) up to Waiting Shelf Y (Tray Row 1).
- Four Basic Operations (+, −, ×, ÷): Combine registers Y and X instantly with no hidden equal sign or parentheses.
- Two-Dot Fraction Entry: Typing [.] [1] [.] [2] enters 1/2 directly (matching the 42.0 mm Fraction Studio block); [2] [.] [1] [.] [3] enters 2 1/3!

**PHYSICAL BLOCK & KEYPAD BRIDGE: Tray Row 1 = Register Y • Tray Row 0 = Register X • Two Dots [.][a][.][b] = 84 mm Fraction Tower**

**Stack Lift & Digits**




**Four Operations**




**Fraction & Edit**




## 2. ESSENTIAL HARDWARE RECOVERY & STACK MANAGEMENT RULES

- Stack Swap [$x \rightleftarrows y$]: Made a mistake in operand order (for subtraction or division)? Press [$x \rightleftarrows y$] to swap Workbench X and Shelf Y!
- Stack Drop [DROP]: Discards the number in Register X; Register Y drops down to take its place.
- Stack Duplication: Pressing [ENTER] when no new number is typed copies Register X into Register Y (great for doubling and squaring!).
- Division by Zero Protection: Dividing by zero displays 'Error: Div by 0' without clearing your upper stack memory.

**2-SECOND STACK AUDIT: Always verify Waiting Shelf *Y* = First Quantity (Minuend / Dividend) and Workbench *X* = Second Quantity!**

**SWAP ORDER [$x \rightleftarrows y$]**



Flips Register Y and Register X before pressing [−] or [÷].

**ROLL DOWN [DROP]**




Removes an unwanted entry in X and pulls Shelf Y down to X.

**DUPLICATE [ENTER]**




Copies X into Y so [+] doubles or [×] squares instantly.

### Simulator Quick-Start

You can practice all hardware key sequences right now in your web browser at stackcalc.io before proceeding to the diagnostic test.

# Diagnostic Baseline: Test Your Readiness

Solve these four baseline problems on paper first, then verify on your calculator.

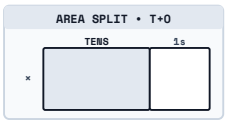
---

**DIAGNOSTIC BASELINE WARM-UP — SOLVE ON PAPER FIRST, THEN VERIFY ON THE STACK**

Complete these 4 diagnostic problems on paper first in the dot-grid workspace, then run the verification keycap sequence on your stack.

### Diagnostic 1: Multi-Digit Regrouping

Solve 148 + 75 on paper by regrouping tens and hundreds:

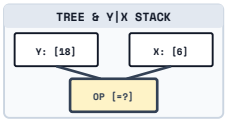



My Paper Prediction: _____ [ ] Verified on SC-6

1 4 8 ENTER 7 5 +

### Diagnostic 2: Inside-Out Order of Operations

Evaluate (18 − 6) × 4 on paper first, then verify with zero parentheses:




My Paper Prediction: _____ [ ] Verified on SC-6

1 8 ENTER 6 - 4 ×

### Diagnostic 3: Equivalent Fraction Sum

Add 1/2 + 3/8 on paper using a common denominator of 8:

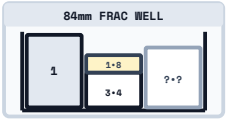



My Paper Prediction: _____ [ ] Verified on SC-6

• 1 • 2 ENTER • 3 • 8 +

### Diagnostic 4: Fraction Division by Rebuilding

Evaluate 3/4 ÷ 1/8 on paper (how many eighths fit in 3/4?):




My Paper Prediction: _____ [ ] Verified on SC-6

• 3 • 4 ENTER • 1 • 8 ÷

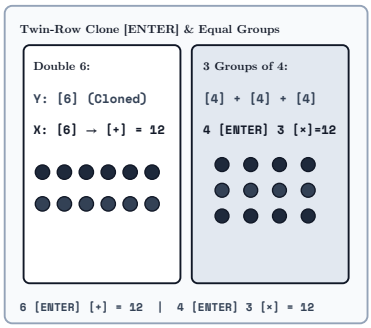
# Multiplication Concepts, Commutativity & Area Models

Mathematical Focus: Rectangular Area Grid Diagram & Place Value Studio • Standards: CCSS.MATH.3.OA.A.1, CCSS.MATH.3.OA.B.5

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Rectangular Area Grids & Commutativity: 6 × 8 = 48):**

Multiplication computes two-dimensional area; order does not change the total product.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Tile Arrays, Skip-Counting & Area Models) • Why 24 Is the Champion Packing Number & 90° Arrays: Beverage crates love 24 because 24 tiles pack into 1×24, 2×12, 3×8, and 4×6 rectangles—and spinning any array 90° preserves its product!**




**2. KEY MATHEMATICAL PRINCIPLES: UNDERSTANDING MULTIPLICATION STRUCTURALLY**

- Multiplication scales equal groups: 6 × 8 means 6 groups of 8, or a rectangular area grid 6 rows by 8 columns.
- Commutative property: 6 × 8 = 48 and 8 × 6 = 48 (rotating a rectangular array preserves its total area).
- On the SC-6: enter first factor into Waiting Shelf Y, second factor into Workbench X, then press [×].
- Scaling by 10*s* on the Place Value Tray: 4 × 30 is 4 × 3 Tens = 12 Tens = 120!

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Mathematical Setup: Sketch a 6-row by 8-column rectangular area grid on paper, with height 6 for Waiting Shelf Y and width 8 for Workbench X. Question: Rotate the area grid 90 degrees to 8 rows by 6 columns. Does the total square count change? (Prediction: 48)

**Verification Keystrokes:** [6] [ENTER] [8] [×]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [6] | 6 | Height (6) enters Register X |
| [ENTER] | 6 | 6 lifts to Waiting Shelf Register Y |
| [8] | 8 | Width (8) enters Workbench Register X |
| [×] | 48 | 6 × 8 = 48 blocks! Rotating gives 8 ENTER 6 × = 48 |

**3. PAPER STEPS & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Mathematical Rule: Multiplication scales equal groups: 6 × 8 means 6 groups of 8, or a rectangular area grid 6 rows by 8 columns.
- Stack Shortcut & Pattern: [6] [ENTER] [8] [×] = 48 — Scaling by 10*s* on the Place Value Tray: 4 × 30 is 4 × 3 Tens = 12 Tens = 120!
- 2-Second Paper-to-Screen Audit: Confirm your paper setup and inner values match Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: A garden bed is 7 feet long by 9 feet wide. What is its total square footage?**

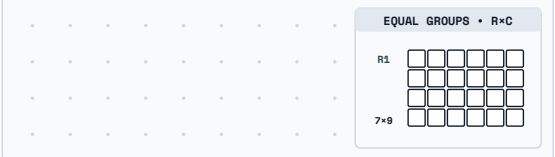
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



# Extension: Multiplying Multiples of 10 on the Place Value Tray: 4 × 30 = 120

Deepening Concepts • Standards: CCSS.MATH.3.NBT.A.3

**1. EXTENSION CONCEPT & SETUP (Multiplying Multiples of 10 on the Place Value Tray: 4 × 30 = 120):**

Factoring out 10: 4 × 30 = (4 × 3) × 10 = 12 × 10 = 120.

**Extension Setup: Place [3]|[0] in the Tens and Ones sockets of the Place Value Tray. Scale the 3 Tens by 4. • Question: 4 × 3 Tens = 12 Tens. Regroup 10 Tens into [1] Hundred + [2] Tens + [0] Ones = 120. (Prediction: 120)**

**2. EXTENSION STACK SEQUENCE (SC-6):** [4] [ENTER] [3] [0] [×]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [4] | 4 | 4 enters Register X |
| [ENTER] | 4 | 4 lifts to Waiting Shelf Register Y |
| [3] [0] | 30 | 30 enters Workbench Register X |
| [×] | 120 | 4 × 30 = 120! 12 tens |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS**

**Misconception 1 — Confusing Area with Perimeter: In 6 × 8, student adds 6 + 8 = 14 or doubles to 28.**

Mathematical & Stack Remediation: Emphasize that area covers the interior with 6 rows of 8 blocks, not just the border edges.

**Misconception 2 — Dropping Zeroes in Multiples of 10: 4 × 30 is answered as 12 instead of 120.**

Mathematical & Stack Remediation: On the Place Value Tray, 30 is 3 Tens; 4 × 3 Tens = 12 Tens = 120.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [6] [ENTER] [8] [×] = 48**

**Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Solve 6 × 40 using place value tens on paper and verify on the stack.**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

**[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)**

Mathematical Scratchpad:



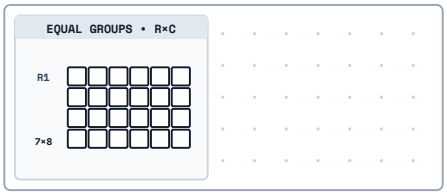
# Unit 01 Lab: Multiplication Concepts, Commutativity & Area Models

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your place-value / bar model in Box 1. (2) Show your step-by-step paper math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: A solar array has 7 rows of 8 panels. How many panels in all?**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

_____

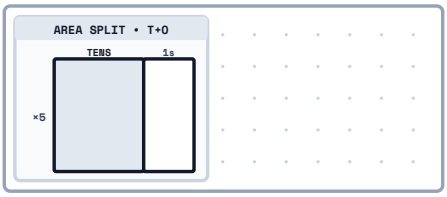
**Final Verified Answer:**

_____

**[ ] Verified on SC-6 Stack**

**Investigation 2: Multiply a single digit by a multiple of 10: 5 × 40 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

_____

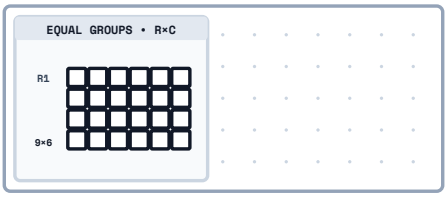
**Final Verified Answer:**

_____

**[ ] Verified on SC-6 Stack**

**Investigation 3: A storage rack holds 9 shelves with 6 bins each. How many bins in all?**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

_____

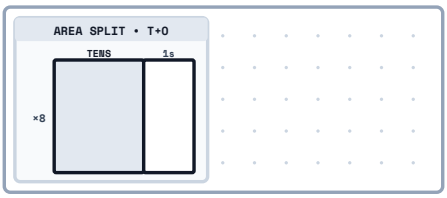
**Final Verified Answer:**

_____

**[ ] Verified on SC-6 Stack**

**Investigation 4: Solve 8 × 30 using place value scaling: 8 × 30 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

_____

**Final Verified Answer:**

_____

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your area grid or expression tree sketch show why your paper answer matched the SC-6 stack?**

_____

# Unit 01 Practice A — Rectangular Area Grids & Equal Groups (0–81)

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Build a 6 by 8 rectangular grid of tiles.**

6 rows of 8 tiles = 48 tiles. Key: [6] [ENTER] [8] [×].

Verified Answer: 48

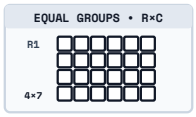
**EXAMPLE 2** **Find the product of 7 equal groups of 9.**

7 × 9 = 63. Key: [7] [ENTER] [9] [×].

Verified Answer: 63

**1** Tier 1 • Builder

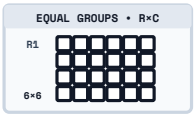
**Area of a 4 by 7 tile rectangle: 4 × 7 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

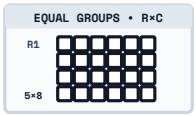
**Area of a 6 by 6 square grid: 6 × 6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

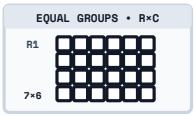
**Area of a 5 by 8 tile rectangle: 5 × 8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

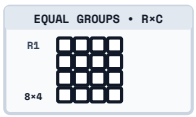
**Area of a 7 by 6 tile rectangle: 7 × 6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

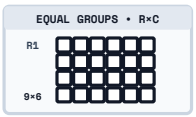
**Area of an 8 by 4 tile rectangle: 8 × 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

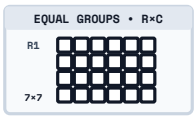
**Area of a 9 by 6 tile rectangle: 9 × 6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

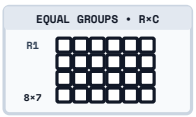
**Area of a 7 by 7 square grid: 7 × 7 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

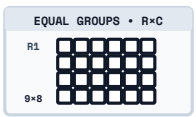
**Area of an 8 by 7 tile rectangle: 8 × 7 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

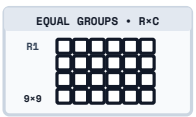
**Area of a 9 by 8 tile rectangle: 9 × 8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Area of a 9 by 9 square grid: 9 × 9 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 01 Practice B — Commutative Turn-Arounds & Missing Factors

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Compare 6 × 8 and 8 × 6 on the stack.**

Rotating the grid 90° keeps 48 tiles: 8 × 6 = 48.

Verified Answer: 48

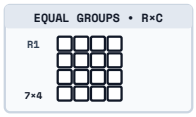
**EXAMPLE 2** **Find the missing factor: 7 × [ ? ] = 56.**

Divide 56 by 7: [56] [ENTER] [7] [÷] = 8.

Verified Answer: 8

**1** Tier 1 • Builder

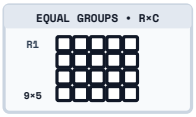
**Verify commutative turn-around for 7 × 4: 4 × 7 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

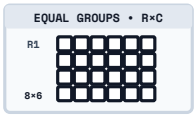
**Verify commutative turn-around for 9 × 5: 5 × 9 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

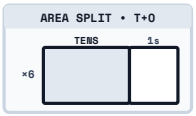
**Verify commutative turn-around for 8 × 6: 6 × 8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

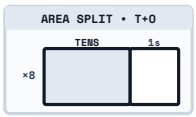
**Find the missing width: 6 × [ ? ] = 42**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

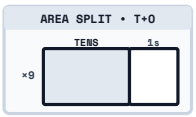
**Find the missing height: [ ? ] × 8 = 64**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

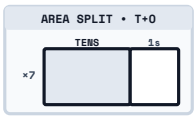
**Find the missing width: 9 × [ ? ] = 63**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

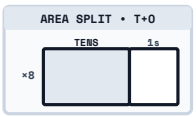
**Find the missing height: [ ? ] × 7 = 35**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Find the missing width: 8 × [ ? ] = 72**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

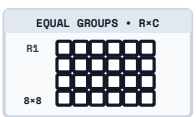
**Find the missing height: [ ? ] × 6 = 54**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Difference between an 8×8 square and a 7×9 rectangle: (8 × 8) − (7 × 9) =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 01 Practice C — Real-World Area, Garden Plots & Array Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** A rooftop solar array has 6 rows of 9 panels.

$6 \times 9 = 54$ solar panels.

Verified Answer: 54 panels

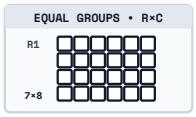
**EXAMPLE 2** A garden bed is 8 ft by 7 ft. Find the area.

$8 \times 7 = 56$ sq ft.

Verified Answer: 56 sq ft

**1** Tier 2 • Detective

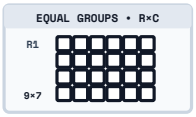
An orchard has 7 rows of 8 apple trees. How many trees in all?




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

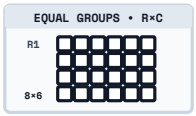
A classroom floor takes 9 rows of 7 carpet squares. Total squares?




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

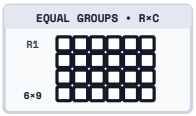
A muffin bakery bakes 8 trays with 6 jumbo muffins each. Total muffins?




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

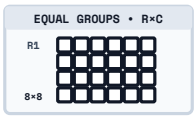
A parking lot has 6 lanes with 9 spaces in each lane. Total spaces?




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

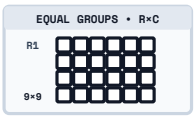
A marching band stands in 8 rows of 8 musicians. How many musicians?




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

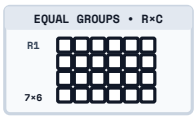
A patio is 9 meters long and 9 meters wide. What is its area in sq m?




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

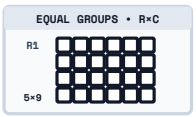
A bookshelf has 7 shelves with 6 encyclopedias per shelf. Total books?




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

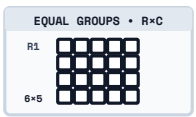
A shipping pallet stacks boxes 5 high and 9 wide. How many boxes?




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

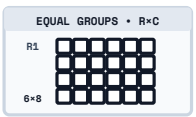
Two garden beds are 6 ft by 5 ft and 4 ft by 5 ft. Find the total area:




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

Spot the Mistake: Leo found the area of a 6 by 8 rug by adding $6 + 8 = 14$. Find true area:




Paper Step: _____ | | SC-6 Ans: _____

# Unit 01 Practice D — Multiplying by Multiples of 10 (10–90)

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Multiply 4 × 30 using Ten-Rods.**

4 groups of 3 tens = 12 tens = 120. Key: [4] [ENTER] [30] [×].

Verified Answer: 120

**EXAMPLE 2** **Multiply 7 × 60 on the stack.**

7 × 6 tens = 42 tens = 420. Key: [7] [ENTER] [60] [×].

Verified Answer: 420

**1** Tier 2 • Detective

Scale by tens: 3 × 40 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×3 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

Scale by tens: 5 × 30 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×5 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

Scale by tens: 6 × 50 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×6 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

Scale by tens: 4 × 70 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×4 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

Scale by tens: 8 × 30 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×8 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

Scale by tens: 7 × 50 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×7 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

Scale by tens: 6 × 80 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×6 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

Scale by tens: 9 × 40 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×9 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

Scale by tens: 8 × 70 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×8 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

Scale by tens: 9 × 60 =

AREA SPLIT • T+O

| | TENS | 1s |
|---|---|---|
| ×9 | | |

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 01 Practice E — Composite Area Models & Distributive Property Splits

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Split 7 × 8 into (7 × 5) + (7 × 3).

35 + 21 = 56. Key: [7] [ENTER] [5] [×] [7] [ENTER] [3] [×] [+].

Verified Answer: 56

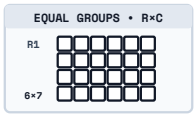
**EXAMPLE 2** Find total area of a 6×10 rectangle and a 6×4 rectangle.

60 + 24 = 84.

Verified Answer: 84

**1** Tier 3 • Challenge

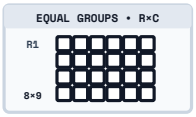
Distributive split for 6 × 7: (6 × 5) + (6 × 2) =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

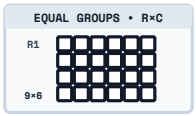
Distributive split for 8 × 9: (8 × 5) + (8 × 4) =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

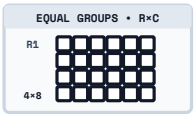
Distributive split for 9 × 6: (5 × 6) + (4 × 6) =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

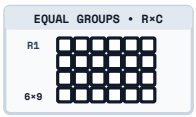
Composite *L*-shape area: (4 × 8) + (3 × 5) =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

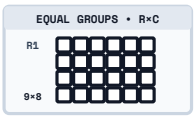
Composite *L*-shape area: (6 × 9) + (4 × 6) =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

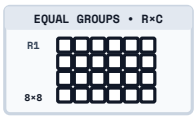
Frame border area (outer minus inner): (9 × 8) − (5 × 4) =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

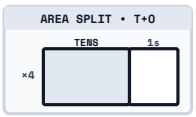
Frame border area (outer minus inner): (8 × 8) − (4 × 4) =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

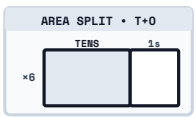
Combine two warehouse bays: (4 × 30) + (5 × 20) =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

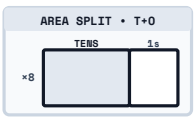
Combine two warehouse bays: (6 × 40) + (3 × 50) =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Difference between two bays: (8 × 50) − (6 × 40) =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 01 Practice F — Multi-Step Area Challenges & Stack Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Three rooms measure 6×4, 5×5, and 7×3 sq m. Total area?

24 + 25 + 21 = 70 sq m.

Verified Answer: 70

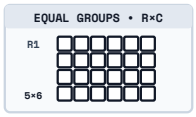
**EXAMPLE 2** Find missing factor: (6 × 8) = 4 × [ ? ].

48 ÷ 4 = 12.

Verified Answer: 12

**1** Tier 3 • Challenge

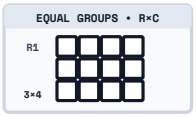
Total area of three rooms: (5 × 6) + (4 × 7) + (3 × 4) =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

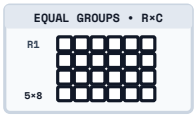
Total crates in 3 trucks, each holding 4 rows of 20 crates: 3 × 4 × 20 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

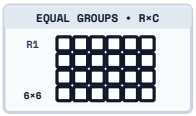
Total tiles in 5 packs of 8×6 grids: 5 × 8 × 6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

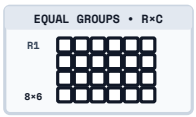
Equal area puzzle: a 6×6 square has the same area as a 4 × [ ? ] rectangle:




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

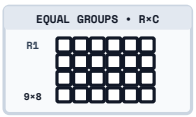
Equal area puzzle: an 8×6 rectangle has the same area as a 12 × [ ? ] rectangle:




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

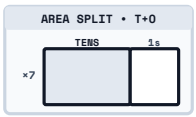
A wall is 9 m by 8 m with a 3 m by 4 m window. Find the paintable area:




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

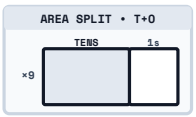
A truck carries 7 pallets of 40 bricks and 6 pallets of 30 bricks. Total bricks:




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

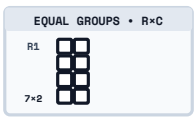
How much larger is 9 × 70 than 8 × 60?




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

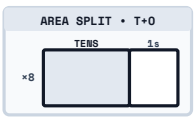
Evaluate on the stack: (7 + 2) × (5 + 3) =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Two-Warehouse Area Challenge: Combine 8 rows of 40 crates and 5 rows of 30 crates on your Place Value Studio:




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 01 Mastery Check: Multiplication & Area Models

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Evaluate area product: 7 × 8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Evaluate area product: 9 × 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Multiply by a multiple of 10: 6 × 40 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Multiply by a multiple of 10: 8 × 70 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Find the missing factor: 9 × [ ? ] = 72**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Evaluate distributive split: (6 × 5) + (6 × 4) =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**A solar roof has 8 rows of 9 panels. How many panels in all?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**Five buses each carry 50 passengers. How many passengers total?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Find the composite area of a 7 m by 8 m room and a 4 m by 6 m porch:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Sam says 7 × 30 = 21. Calculate the true product:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 01 STEM: Warehouse Pallet Storage System

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: WAREHOUSE PALLET STORAGE SYSTEM**

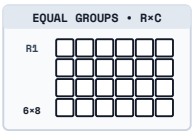
**Logistics managers calculate warehouse floor space in square meters using area models.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Tile Arrays, Skip-Counting & Area Models) • Why 24 Is the Champion Packing Number & 90° Arrays: Beverage crates love 24 because 24 tiles pack into 1×24, 2×12, 3×8, and 4×6 rectangles—and spinning any array 90° preserves its product!

Mathematical Focus: Rectangular Area Grid Diagram & Place Value Studio • Stack Shortcut: [6] [ENTER] [8] [×] = 48

**CASE INVESTIGATION 1** STEM Modeling

**A pallet bay holds 6 rows of 8 pallets. How many pallets fit in the bay?**




1. Model / Expression: _____

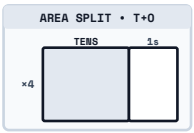
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**If 4 bays each have 30 boxes, how many boxes total?**




1. Model / Expression: _____

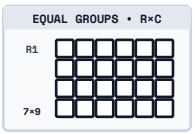
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Rotate a 7 by 9 pallet layout by 90 degrees. Does the area change?**




1. Model / Expression: _____

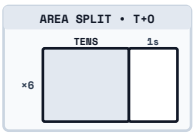
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: A clerk calculates 6 × 80 = 48. Find the true product.**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

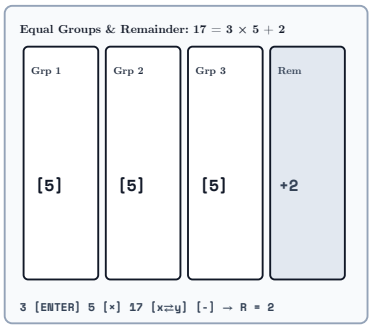
# Division as Fair-Sharing, Grouping & Remainders

Mathematical Focus: Place Value Studio & Prime Factor Studio • Standards: CCSS.MATH.3.OA.A.2, CCSS.MATH.3.OA.B.6

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Fair Sharing on the Stack: 24 ÷ 6 = 4):**

Division inverts multiplication: 24 ÷ 6 = 4 because 6 × 4 = 24.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Tile Arrays, Skip-Counting & Remainders ([÷R])) • Fair-Share Division & The Baker's Remainder Check: Sharing 25 cookies across 4 trays gives 6 per tray with 1 leftover—and rebuilding 6 × 4 + 1 = 25 on the stack proves not a crumb was lost!**




**2. KEY MATHEMATICAL PRINCIPLES: UNDERSTANDING DIVISION AS INVERSE MULTIPLICATION**

- Fair Sharing (Partitive): Partition 24 blocks equally across 6 tray columns → 4 blocks per column.
- Grouping (Quotative): Measure how many groups of 6 blocks fit inside 24 → 4 equal groups.
- On the SC-6: Total Dividend goes to Waiting Shelf Y, Divisor goes to Workbench X, then press [÷].
- Remainders: 25 ÷ 4 gives quotient 6 with 1 leftover block (25 = 4 × 6 + 1).

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Mathematical Setup: Place dividend [2][4] in Row 1 (Waiting Shelf Y) and divisor [6] in Row 0 (Workbench X) of the 6-column Place Value Tray. Question: Distribute 24 equally across the 6 columns of the Place Value Tray. How many per column? (Prediction: 4)

**Verification Keystrokes:** [2] [4] [ENTER] [6] [÷]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [2] [4] | 24 | Total dividend (24) enters Register X |
| [ENTER] | 24 | 24 lifts to Waiting Shelf Register Y |
| [6] | 6 | Number of shares (6) enters Workbench Register X |
| [÷] | 4 | 24 ÷ 6 = 4! Each share receives 4 blocks |

**3. PAPER STEPS & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Mathematical Rule: Fair Sharing (Partitive): Partition 24 blocks equally across 6 tray columns → 4 blocks per column.
- Stack Shortcut & Pattern: Dividend [ENTER] Divisor [÷] = Quotient — Remainders: 25 ÷ 4 gives quotient 6 with 1 leftover block (25 = 4 × 6 + 1).
- 2-Second Paper-to-Screen Audit: Confirm your paper setup and inner values match Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: 36 apples are packed into bags of 4. How many bags can be filled?**

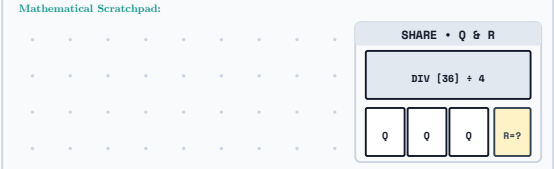
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



# Extension: Division with Remainders: 25 ÷ 4

Deepening Concepts • Standards: CCSS.MATH.3.OA.A.2

**1. EXTENSION CONCEPT & SETUP (Division with Remainders: 25 ÷ 4):**

The whole number part (6) is the quotient; the fractional part (.25 = 1/4) shows remainder 1.

**Extension Setup: Place dividend [2][5] in Row 1 (Waiting Shelf Y) and divisor [4] in Row 0 (Workbench X) of the Place Value Tray. • Question: Partition 25 into 4 equal groups: 4 × 6 = 24, leaving 1 block remaining. (Prediction: 6 per group, 1 leftover)**

**2. EXTENSION STACK SEQUENCE (SC-6):** [2] [5] [ENTER] [4] [÷]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [2] [5] | 25 | 25 enters Register X |
| [ENTER] | 25 | 25 lifts to Waiting Shelf Register Y |
| [4] | 4 | 4 enters Workbench Register X |
| [÷] | 6.25 | 25 ÷ 4 = 6.25 (Whole quotient = 6, 0.25 = 1/4 → Remainder 1) |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS**

**Misconception 1 — Reversing Dividend and Divisor Order on the Stack: Student enters [6] [ENTER] [24] [÷] getting 0.25.**

Mathematical & Stack Remediation: Emphasize that the WHOLE dividend to be shared always waits in Row 1 (Waiting Shelf Y) first!

**Misconception 2 — Treating the Remainder as a Pure Decimal: Seeing 6.25 and thinking the remainder is 25.**

Mathematical & Stack Remediation: 0.25 is one-fourth (1/4); the numerator 1 is the physical leftover block.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: Dividend [ENTER] Divisor [÷] = Quotient**

**Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: A baker puts 29 cookies into boxes of 6. How many full boxes and how many cookies left over?**

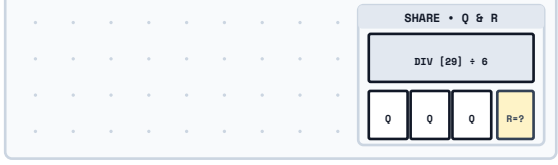
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



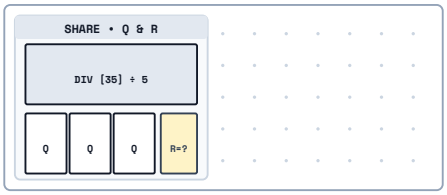
# Unit 02 Lab: Division as Fair-Sharing, Grouping & Remainders

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your place-value / bar model in Box 1. (2) Show your step-by-step paper math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: Divide 35 candies equally into 5 treat bags. How many per bag?**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 2: How many groups of 8 are in 48?**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

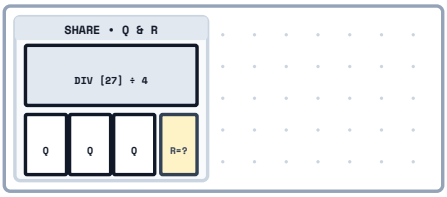
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 3: Share 27 marbles among 4 friends. Find the quotient and remainder:**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

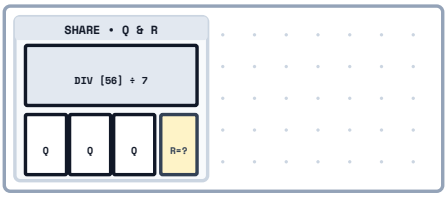
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 4: Divide 56 books equally across 7 classroom shelves. How many per shelf?**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your area grid or expression tree sketch show why your paper answer matched the SC-6 stack?**

# Unit 02 Practice A — Fair-Sharing & Grouping Division Fluency

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Share 24 counters equally into 6 cups.

24 ÷ 6 = 4 per cup. Key: [24] [ENTER] [6] [÷].

Verified Answer: 4

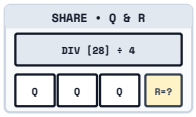
**EXAMPLE 2** How many groups of 8 fit inside 56?

56 ÷ 8 = 7 groups. Key: [56] [ENTER] [8] [÷].

Verified Answer: 7

**1** Tier 1 • Builder

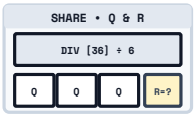
Fair-share 28 blocks among 4 trays: 28 ÷ 4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

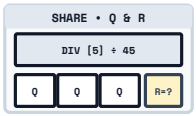
Fair-share 36 blocks among 6 trays: 36 ÷ 6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

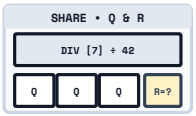
Measure groups of 5 in 45: 45 ÷ 5 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

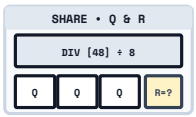
Measure groups of 7 in 42: 42 ÷ 7 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

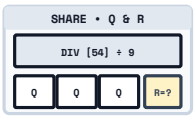
Fair-share 48 blocks among 8 trays: 48 ÷ 8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

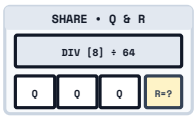
Fair-share 54 blocks among 9 trays: 54 ÷ 9 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

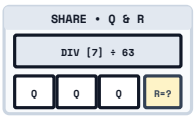
Measure groups of 8 in 64: 64 ÷ 8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

Measure groups of 7 in 63: 63 ÷ 7 =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

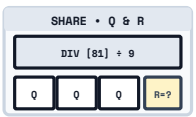
Fair-share 72 blocks among 8 trays: 72 ÷ 8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

Fair-share 81 blocks among 9 trays: 81 ÷ 9 =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 02 Practice B — Inverse Multiplication Fact Families & Unknowns

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **If 6 × 7 = 42, find 42 ÷ 6.**

Division undoes multiplication: 42 ÷ 6 = 7.

Verified Answer: 7

**EXAMPLE 2** **Find the unknown dividend: [ ? ] ÷ 8 = 6.**

Multiply 6 × 8 = 48. Key: [6] [ENTER] [8] [×].

Verified Answer: 48

**1** Tier 1 • Builder

**Since 8 × 7 = 56, evaluate 56 ÷ 7 =**

SHARE • Q & R
DIV [8] ÷ 7
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Since 9 × 6 = 54, evaluate 54 ÷ 6 =**

SHARE • Q & R
DIV [9] ÷ 6
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

**Find the missing divisor: 32 ÷ [ ? ] = 8**

SHARE • Q & R
DIV [32] ÷ 8
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

**Find the missing divisor: 49 ÷ [ ? ] = 7**

SHARE • Q & R
DIV [49] ÷ 7
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

**Find the missing divisor: 72 ÷ [ ? ] = 9**

SHARE • Q & R
DIV [72] ÷ 9
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

**Find the unknown dividend: [ ? ] ÷ 5 = 8**

SHARE • Q & R
DIV [5] ÷ 8
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Find the unknown dividend: [ ? ] ÷ 7 = 9**

SHARE • Q & R
DIV [7] ÷ 9
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Find the unknown dividend: [ ? ] ÷ 9 = 8**

SHARE • Q & R
DIV [9] ÷ 8
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Scale division by tens: 120 ÷ 4 =**

SHARE • Q & R
DIV [120] ÷ 4
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Scale division by tens: 240 ÷ 6 =**

SHARE • Q & R
DIV [240] ÷ 6
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 02 Practice C — Real-World Fair-Sharing & Packaging Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **36 apples are packed into bags of 4. How many bags?**

36 ÷ 4 = 9 bags.

Verified Answer: 9 bags

**EXAMPLE 2** **42 lb of fish is shared among 6 sea otters.**

42 ÷ 6 = 7 lb per otter.

Verified Answer: 7 lb

**1** Tier 2 • Detective

**40 seedlings are planted in 5 equal garden rows. How many per row?**

SHARE • Q & R
DIV [40] ÷ 5
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

**56 granola bars are packed in boxes of 8. How many boxes are filled?**

SHARE • Q & R
DIV [56] ÷ 8
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

**A 45-meter rope is cut into 9-meter lengths. How many pieces?**

SHARE • Q & R
DIV [45] ÷ 9
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

**63 students ride in 7 equal minivans. How many students per van?**

SHARE • Q & R
DIV [63] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

**72 solar cells are wired into panels of 9 cells each. How many panels?**

SHARE • Q & R
DIV [72] ÷ 9
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

**48 liters of water fill 6 identical jugs. How many liters per jug?**

SHARE • Q & R
DIV [48] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

**180 chairs are arranged in 6 equal sections. How many chairs per section?**

SHARE • Q & R
DIV [180] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

**350 tickets are bundled in packs of 7. How many packs?**

SHARE • Q & R
DIV [350] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

**A baker bakes 36 muffins, keeps 4, and packs the rest into boxes of 8. How many boxes?**

SHARE • Q & R
DIV [36] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

**Spot the Mistake: Mia entered [5] [ENTER] [35] [÷] for 35 ÷ 5 and got 0.14. Find 35 ÷ 5:**

SHARE • Q & R
DIV [5] ÷ 35
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

# Unit 02 Practice D — Division with Remainders (Quotient & Leftover)

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Share 25 blocks into 4 cups.

$4 \times 6 = 24$, with 1 leftover → 6 R 1.

Verified Answer: 6 R 1

**EXAMPLE 2** Divide 38 by 5 and state quotient and remainder.

$5 \times 7 = 35$, with 3 left → 7 R 3.

Verified Answer: 7 R 3

**1** Tier 2 • Detective

Find quotient and remainder: 19 ÷ 3 =

SHARE • Q & R
DIV [19] ÷ 3
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

Find quotient and remainder: 23 ÷ 4 =

SHARE • Q & R
DIV [23] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

Find quotient and remainder: 29 ÷ 5 =

SHARE • Q & R
DIV [29] ÷ 5
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

Find quotient and remainder: 34 ÷ 6 =

SHARE • Q & R
DIV [34] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

Find quotient and remainder: 45 ÷ 7 =

SHARE • Q & R
DIV [45] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

Find quotient and remainder: 50 ÷ 8 =

SHARE • Q & R
DIV [50] ÷ 8
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

Find quotient and remainder: 60 ÷ 9 =

SHARE • Q & R
DIV [60] ÷ 9
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

Find quotient and remainder: 53 ÷ 6 =

SHARE • Q & R
DIV [53] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

Find quotient and remainder: 67 ÷ 8 =

SHARE • Q & R
DIV [67] ÷ 8
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

Find quotient and remainder: 76 ÷ 9 =

SHARE • Q & R
DIV [76] ÷ 9
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

# Unit 02 Practice E — Checking Remainders (Q × D + R) & Interpreting Leftovers

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Verify 29 ÷ 6 = 4 R 5 on the stack.**

4 × 6 + 5 = 24 + 5 = 29. Key: [4] [ENTER] [6] [×] [5] [+].

Verified Answer: 29

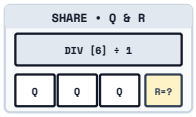
**EXAMPLE 2** **27 campers sleep 4 to a tent. How many FULL tents?**

27 ÷ 4 = 6 R 3 → 6 full tents (3 campers in 7th tent).

Verified Answer: 6 R 3

**1** Tier 3 • Challenge

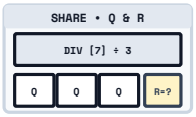
**Verify 6 R 1 for divisor 4: (6 × 4) + 1 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Verify 7 R 3 for divisor 5: (7 × 5) + 3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

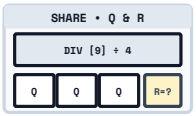
**Verify 8 R 5 for divisor 6: (8 × 6) + 5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

**Verify 9 R 4 for divisor 7: (9 × 7) + 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

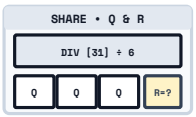
**Verify 8 R 7 for divisor 8: (8 × 8) + 7 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

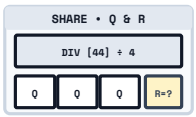
**31 eggs are packed in cartons of 6. Find full cartons and leftover eggs:**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

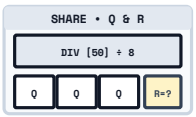
**44 wheels are used to build 4-wheel carts. How many carts can be built?**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

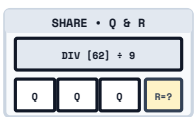
**50 markers are put in boxes of 8. How many markers are left over?**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

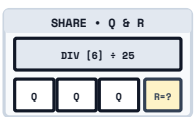
**62 beads make necklaces of 9 beads each. How many beads are left over?**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**How many 6-seat boats are needed to carry ALL 25 hikers (including any remainder)?**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 02 Practice F — Multi-Step Division & Multiplication Pipelines

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **4 boxes of 9 apples are shared among 6 friends.**

$(4 \times 9) \div 6 = 36 \div 6 = 6$ apples each.

Verified Answer: 6

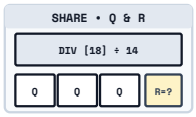
**EXAMPLE 2** **Subtract 12 from 60, then divide by 8.**

$(60 - 12) \div 8 = 48 \div 8 = 6$.

Verified Answer: 6

**1** Tier 3 • Challenge

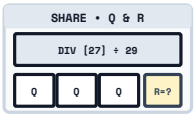
**Combine and share: $(18 + 14) \div 4 =$**




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 3 • Challenge

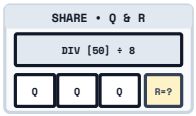
**Combine and share: $(27 + 29) \div 7 =$**




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 3 • Challenge

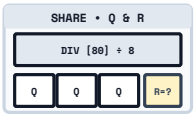
**Remove damaged items, then pack: $(50 - 8) \div 6 =$**




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 3 • Challenge

**Remove damaged items, then pack: $(80 - 8) \div 9 =$**




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 3 • Challenge

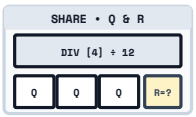
**Multiply packs, then repack: $(6 \times 6) \div 4 =$**




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 3 • Challenge

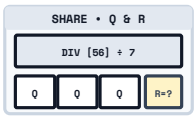
**Multiply packs, then repack: $(4 \times 12) \div 8 =$**




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 3 • Challenge

**Compare two group sizes: $(56 \div 7) + (45 \div 9) =$**




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 3 • Challenge

**Difference between two shares: $(72 \div 8) - (30 \div 6) =$**




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 3 • Challenge

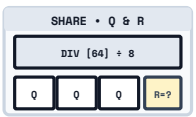
**Find mystery number: when divided by 7, quotient is 8 and remainder is 5:**




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 3 • Challenge

**Two-Branch Packing Puzzle: Share 64 blocks equally among 8 trays, then add 3 packs of 5 blocks to Tray 1. Total in Tray 1?**




Paper Step: _____ | | SC-6 Ans: _____

# Unit 02 Mastery Check: Division Concepts & Remainders

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Evaluate exact quotient: 54 ÷ 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Evaluate exact quotient: 63 ÷ 9 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Find the missing divisor: 48 ÷ [ ? ] = 6**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Find quotient and remainder: 29 ÷ 4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Find quotient and remainder: 47 ÷ 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Verify division using $(Q \times D) + R$: $(8 \times 7) + 4 =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**72 books are placed equally onto 8 library shelves. How many books per shelf?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**38 tennis balls are packed in cans of 3. Find full cans and leftover balls:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Evaluate two-step pipeline: (42 + 14) ÷ 8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Ben says 37 ÷ 5 = 6 R 7 because 5×6+7=37. Find the true quotient and remainder:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 02 STEM: Wildlife Rescue Meal Distribution

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: WILDLIFE RESCUE MEAL DISTRIBUTION**

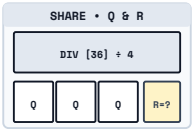
**Marine biologists divide daily fish rations and vitamin packs among orphaned seal pups and sea otters.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Tile Arrays, Skip-Counting & Remainders ([÷R])) • Fair-Fair Division & The Baker's Remainder Check: Sharing 25 cookies across 4 trays gives 6 per tray with 1 leftover—and rebuilding 6 × 4 + 1 = 25 on the stack proves not a crumb was lost!

Mathematical Focus: Place Value Studio & Prime Factor Studio • Stack Shortcut: Dividend [ENTER] Divisor [÷] = Quotient

**CASE INVESTIGATION 1** STEM Modeling

**Share 36 kg of herring equally among 4 seal pups. How many kg per pup?**




1. Model / Expression: _____

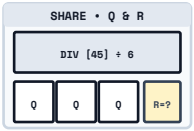
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Divide 45 vitamin tablets into packs of 6. Find quotient and remainder.**




1. Model / Expression: _____

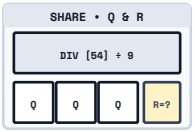
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**If 54 fish are shared among 9 otters, how many fish does each otter receive?**




1. Model / Expression: _____

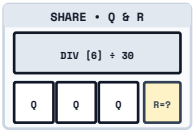
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: Sam enters 6 [ENTER] 30 [÷] and gets 0.2. What should he do?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

UNIT 03 CORE CONCEPT LAB (PAGE 1 OF 11) • GRADE 3 – 4 (SC-6)

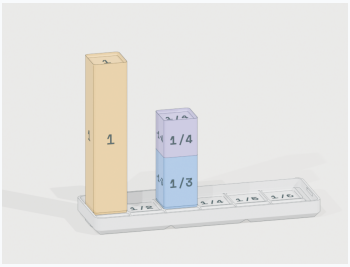
# Unit Fractions, Denominator Strips & Comparator Heights

Hands-On Block Studio: Fraction Studio (84 mm Proportional Towers) • Standards: CCSS.MATH.3.NF.A.1, CCSS.MATH.3.NF.A.2

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Unit Fractions on the Comparator: 1/3 vs 1/4):**

The denominator determines how many equal parts make the whole; more parts mean smaller pieces.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fractions, Decimals & 84 mm Fraction Towers) • The Pharaoh's Bakery & Why Fraction Towers Stand 84 mm: Ancient Egyptian scribes shared loaves with unit fractions (1/2, 1/3, 1/4), and our 84 mm Whole divides into exact millimeter heights (42, 28, 21, 14, 7 mm)!**




Fraction Studio • 84 mm Comparator Well

**2. HANDS-ON BLOCKS & CONCEPT RULES: HOW FRACTIONS REPRESENT EXACT PHYSICAL PROPORTIONS**

- 1 Whole stands 84.0 mm tall; every fraction tower has an exact proportional height.
- Halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), twelfths (7 mm).
- Larger Denominator = Smaller Pieces: 1/3 (28 mm) is physically taller than 1/4 (21 mm)!
- On the SC-6: Type two-dot fractions using [.]: 1/2 is [.] [1] [.] [2].

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Block Setup: Place the 1/3 tower (28 mm) and 1/4 tower (21 mm) side-by-side in the Fraction Studio comparator well. Question: Which tower is physically taller? What is the height difference? (Prediction: 1/3 is 7 mm taller = 1/12)

**Verification Keystrokes:** [•] [1] [•] [3] [ENTER] [•] [1] [•] [4] [-]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [•] [1] [•] [3] | 1/3 | 1/3 enters Register X (28 mm height) |
| [ENTER] | 1/3 | 1/3 lifts to Waiting Shelf Register Y |
| [•] [1] [•] [4] | 1/4 | 1/4 enters Workbench Register X (21 mm height) |
| [-] | 1/12 | 1/3 − 1/4 = 1/12 (7 mm height difference) |

**3. 3D BLOCK STUDIO & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Physical Block Studio Rule: 1 Whole stands 84.0 mm tall; every fraction tower has an exact proportional height.
- Stack Shortcut & Pattern: [.] [1] [.] [2] = 1/2 (42.0 mm height) — On the SC-6: Type two-dot fractions using [.]: 1/2 is [.] [1] [.] [2].
- 2-Second Tray-to-Screen Audit: Confirm your physical 3D Block Studio setup matches Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: Place two 1/4 blocks into the comparator. What fraction height do they match?**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Ordering Fractions with Like Numerators: 1/2, 1/3, 1/4, 1/6

Deepening Concepts • Standards: CCSS.MATH.3.NF.A.2

**1. EXTENSION CONCEPT & SETUP (Ordering Fractions with Like Numerators: 1/2, 1/3, 1/4, 1/6):**

Because the whole is fixed at 84 mm, dividing into fewer parts produces larger individual parts.

**Hands-On Extension Setup: Line up the 1/2, 1/3, 1/4, and 1/6 blocks from tallest to shortest in the Fraction Studio. • Question: How much taller is 1/2 (42 mm) than 1/6 (14 mm)? (Prediction: 28 mm = 1/3)**

**2. EXTENSION STACK SEQUENCE (SC-6):** [•] [1] [•] [2] [ENTER] [•] [1] [•] [6] [-]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [•] [1] [•] [2] | 1/2 | 1/2 enters Register X |
| [ENTER] | 1/2 | 1/2 lifts to Waiting Shelf Register Y |
| [•] [1] [•] [6] | 1/6 | 1/6 enters Workbench Register X |
| [-] | 1/3 | 1/2 − 1/6 = 1/3 (28 mm difference) |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1 — Believing Larger Denominators Mean Larger Fractions: Student claims 1/8 is larger than 1/4 because 8 > 4.**

Hands-On Block Studio Remediation: Stand the 10.5 mm (1/8) and 21 mm (1/4) blocks side-by-side in the comparator well.

**Misconception 2 — Adding Numerators and Denominators Directly: Student claims 1/2 + 1/3 = 2/5.**

Hands-On Block Studio Remediation: Stack 1/2 and 1/3; show the combined tower stands at 70 mm (5/6), not 2/5 (33.6 mm).

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [.] [1] [.] [2] = 1/2 (42.0 mm height)**

**Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Compare 1/4 and 1/8. How many 1/8 blocks equal 1/4?**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



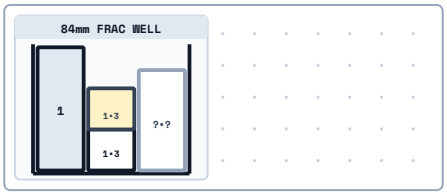
# Unit 03 Lab: Unit Fractions, Denominator Strips & Comparator Heights

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your fraction bar or ratio model in Box 1. (2) Show your fraction/decimal math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

### Investigation 1: Combine two 1/3 fraction blocks: 1/3 + 1/3 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

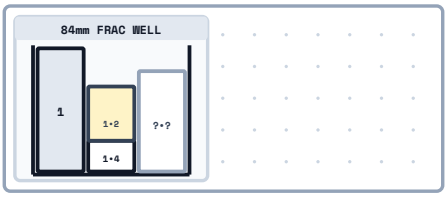
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 2: Subtract 1/4 from 1/2: 1/2 − 1/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

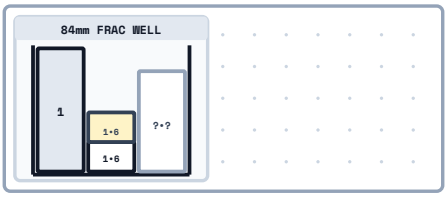
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 3: Find the height of three 1/6 blocks: 1/6 + 1/6 + 1/6 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

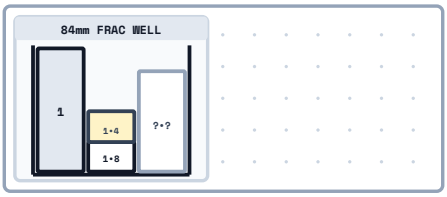
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 4: Subtract 1/8 from 1/4: 1/4 − 1/8 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 03 Practice A — Unit Fractions & Like-Denominator Sums in the 84 mm Well

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Combine two 1/3 blocks in the comparator well.

1/3 + 1/3 = 2/3. Key: [.1.3] [ENTER] [.1.3] [+].

Verified Answer: 2/3

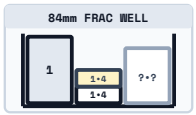
**EXAMPLE 2** Subtract 1/8 from 5/8 on the stack.

5/8 − 1/8 = 4/8 = 1/2. Key: [.5.8] [ENTER] [.1.8] [-].

Verified Answer: 1/2

**1** Tier 1 • Builder

Combine two 1/4 fraction blocks: 1/4 + 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

Combine three 1/4 fraction blocks: 1/4 + 2/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

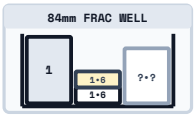
Add sixths in the comparator: 1/6 + 4/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

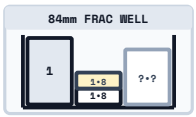
Add sixths and simplify: 1/6 + 1/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

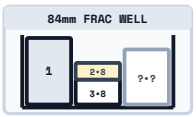
Add eighths and simplify: 1/8 + 1/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

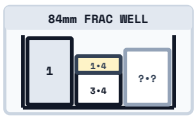
Add eighths in the comparator: 3/8 + 2/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

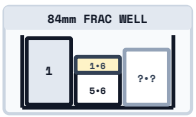
Subtract fourths: 3/4 − 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

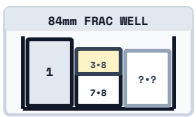
Subtract sixths: 5/6 − 1/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

Subtract eighths: 7/8 − 3/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

Subtract twelfths: 11/12 − 5/12 =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Practice B — Comparing Unit Fractions & Comparator Height Differences

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Which is taller: 1/3 or 1/4? Find 1/3 − 1/4.**

1/3 (28 mm) is taller than 1/4 (21 mm) by 1/12 (7 mm).

Verified Answer: 1/12

**EXAMPLE 2** **Compare 1/2 and 1/6: find 1/2 − 1/6.**

3/6 − 1/6 = 2/6 = 1/3.

Verified Answer: 1/3

**1** Tier 1 • Builder

**Find the height difference: 1/2 − 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Find the height difference: 1/3 − 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

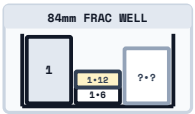
**Find the height difference: 1/4 − 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

**Find the height difference: 1/6 − 1/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

**Find the height difference: 1/2 − 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

**Find the height difference: 1/2 − 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Find the height difference: 1/3 − 1/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Find the height difference: 1/4 − 1/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Find the height difference: 1/2 − 1/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Find the height difference: 2/3 − 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Practice C — Physical 84 mm Fraction Tower Heights & Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **1 Whole stands 84 mm tall. How tall is 1/3?**

84 ÷ 3 = 28 mm. Key: [84] [ENTER] [3] [÷].

Verified Answer: 28 mm

**EXAMPLE 2** **How tall is a tower of three 1/4 blocks (21 mm each)?**

21 × 3 = 63 mm. Key: [21] [ENTER] [3] [×].

Verified Answer: 63 mm

**1** Tier 2 • Detective

**1 Whole is 84 mm. Find the height of the 1/2 block in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**1 Whole is 84 mm. Find the height of the 1/4 block in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**1 Whole is 84 mm. Find the height of the 1/6 block in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

**1 Whole is 84 mm. Find the height of the 1/12 block in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**1 Whole is 84 mm. Find the height of the 1/8 block in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

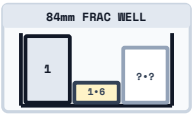
**1 Whole is 84 mm. Stack two 1/3 blocks and find the total height in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

**1 Whole is 84 mm. Stack five 1/6 blocks and find the total height in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

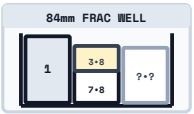
**A glazier cuts 1/4 m and 2/4 m of glass. What is the total fraction of a meter?**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

**A ribbon is 7/8 yard long. You use 3/8 yard. How much ribbon remains?**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Spot the Mistake: Sam says 1/8 is taller than 1/4 because 8 > 4. Find 1/4 — 1/8:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Practice D — Completing 1 Whole & Complements to the Whole

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** How much more to make 1 Whole from 1/3?

1 − 1/3 = 2/3. Key: [1] [ENTER] [.1.3] [-].

Verified Answer: 2/3

**EXAMPLE 2** Stack three 1/3 blocks together: 1/3 + 1/3 + 1/3.

3/3 = 1 Whole. Key: [.1.3] [ENTER] [.1.3] [+] [.1.3] [+].

Verified Answer: 1

**1** Tier 2 • Detective

Complement to 1 Whole: 1 − 1/2 =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

Complement to 1 Whole: 1 − 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

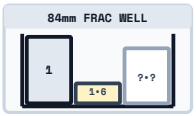
Complement to 1 Whole: 1 − 3/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

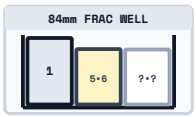
Complement to 1 Whole: 1 − 1/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

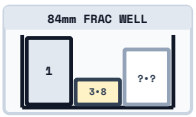
Complement to 1 Whole: 1 − 5/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

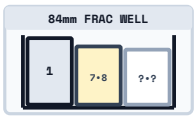
Complement to 1 Whole: 1 − 3/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

Complement to 1 Whole: 1 − 7/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

Complement to 1 Whole: 1 − 5/12 =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

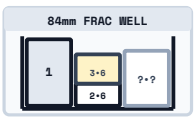
Build 1 Whole from fourths: 1/4 + 1/4 + 1/4 + 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

Find the missing piece to reach 1 Whole: 2/6 + 3/6 + [ ? ] = 1




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Practice E — Chained Unit-Fraction Pipelines on the Stack

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Stack 1/6 + 1/6 + 1/3 in the comparator well.**

1/6 + 1/6 = 1/3; 1/3 + 1/3 = 2/3.

Verified Answer: 2/3

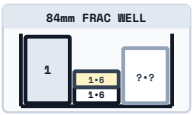
**EXAMPLE 2** **Stack 1/8 + 1/8 + 1/4 on the workbench.**

1/8 + 1/8 = 1/4; 1/4 + 1/4 = 1/2.

Verified Answer: 1/2

**1** Tier 3 • Challenge

**Chained fraction tower: 1/6 + 1/6 + 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Chained fraction tower: 1/8 + 1/8 + 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

**Chained fraction tower: 1/8 + 1/8 + 2/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

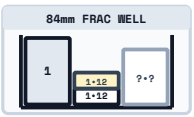
**Chained fraction tower: 1/12 + 1/12 + 1/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

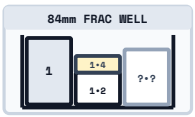
**Chained fraction tower: 1/12 + 1/12 + 2/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

**Combine halves and fourths: 1/2 + 1/4 + 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

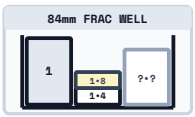
**Combine thirds and sixths: 1/3 + 1/6 + 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

**Combine fourths and eighths: 1/4 + 1/8 + 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

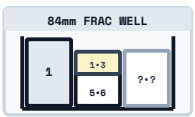
**Step up and down: 3/4 + 1/8 − 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Step up and down: 5/6 − 1/3 + 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Practice F — Comparator Challenges, Number Lines & Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** How many 1/12 blocks match one 1/3 block?

28 mm ÷ 7 mm = 4 twelfths (or 1/3 ÷ 1/12 = 4).

Verified Answer: 4

**EXAMPLE 2** Find the midpoint between 1/4 and 3/4.

2/4 simplifies to 1/2.

Verified Answer: 1/2

**1** Tier 3 • Challenge

How many 1/8 blocks stack to match 1/2?




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

How many 1/6 blocks stack to match 2/3?




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

How many 1/12 blocks stack to match 3/4?




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

Difference between 7/8 and 1/4: 7/8 − 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

Difference between 5/6 and 1/3: 5/6 − 1/3 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

Difference between 11/12 and 1/4: 11/12 − 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

Find the missing fraction: 1/4 + [ ? ] = 7/8




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

Find the missing fraction: 5/6 − [ ? ] = 1/2




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

In the 84 mm comparator, how many mm taller is the 1/2 block than the 1/3 block?




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Flush Comparator Puzzle: Stack a 1/2 tower (42 mm), a 1/6 tower (14 mm), and a 1/3 tower (28 mm). What Whole is formed?




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 03 Mastery Check: Unit Fractions & Proportional Heights

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Add unit fractions and simplify: 1/4 + 1/4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Subtract like fractions and simplify: 5/6 − 1/6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Find the comparator difference: 1/3 ◯ 1/4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Find the comparator difference: 1/2 ◯ 1/6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Find the complement to 1 Whole: 1 − 3/8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Evaluate chained sum: 1/6 + 1/6 + 1/6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**If 1 Whole is 84 mm tall, how tall is a 1/3 block in mm?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**How many 1/8 blocks are needed to match 3/4?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**A carpenter glues a 3/8 in strip to a 1/8 in strip. Total thickness?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Noah added 1/4 + 1/4 = 2/8. Simplify 1/4 + 1/4 correctly:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 03 STEM: Architectural Glass Window Panes

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: ARCHITECTURAL GLASS WINDOW PANES**

**An architect proportions structural glass windows using unit fraction ratios.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fractions, Decimals & 84 mm Fraction Towers) • The Pharaoh's Bakery & Why Fraction Towers Stand 84 mm: Ancient Egyptian scribes shared loaves with unit fractions (1/2, 1/3, 1/4), and our 84 mm Whole divides into exact millimeter heights (42, 28, 21, 14, 7 mm)!

Block Studio: Fraction Studio (84 mm Proportional Towers) • Stack Shortcut: [.] [1] [.] [2] = 1/2 (42.0 mm height)

**CASE INVESTIGATION 1** STEM Modeling

**Clerestory vs. Transom Pane: An upper clerestory pane is 1/3 of the frame height and a transom pane is 1/4. Compare your 84 mm towers to find the fraction difference:**




1. Model / Expression: _____

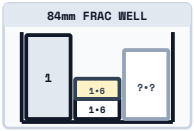
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Modular Louver Stack: Each glass louver vent is 1/6 of a meter tall. How many 1/6 louver blocks stack up in the comparator well to fill a 1/2-meter window opening?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Compound Mullion Frame: The architect stacks a 1/4-meter tinted pane above a 1/8-meter acoustic sill. What total fraction of a meter is the combined assembly?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: A junior drafter claims a 1/8-meter glass strip is taller than a 1/4-meter strip because 8 > 4. Use the 84 mm Fraction Comparator to explain why this is false:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

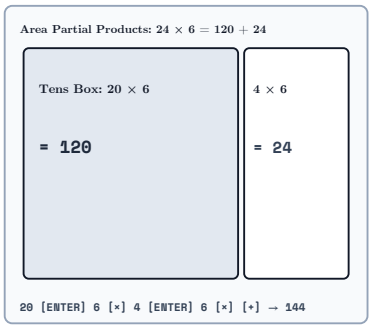
# Multi-Digit Multiplication & Partial Products

Mathematical Focus: Partial-Product Area Grid Diagram & Place Value Studio • Standards: CCSS.MATH.4.NBT.B.5

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Place Value Partial Products: 24 × 6 = 144):**

The distributive property allows us to multiply tens and ones separately and sum the results.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Multi-Digit Area Models & Partial Products) • Slicing Wide Tile Grids Along the Ten-Lines: Instead of blind carry rules, slice 32 × 15 along its ten-lines into four visible area rectangles (300 + 150 + 20 + 10 = 480) on the 4-shelf stack!**




**2. KEY MATHEMATICAL PRINCIPLES: DECOMPOSING NUMBERS INTO PARTIAL PRODUCTS**

- To multiply 24 × 6: break 24 into 20 + 4. Partial products: (20 × 6) + (4 × 6) = 120 + 24 = 144.
- For 2-digit by 2-digit (32 × 15): draw a 2×2 area grid for (30×10) + (30×5) + (2×10) + (2×5) = 480.
- Stack lift preserves partial products in upper registers as you accumulate.
- On the SC-6: Type [2] [4] [ENTER] [6] [×] to verify the complete product instantly.

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Mathematical Setup: Decompose 24 into (20 + 4) along the top of an area rectangle (or [2][4] on the Place Value Tray) and multiply each part by 6. Question: Tens product: 20 × 6 = 120. Ones product: 4 × 6 = 24. Combine partial products: 120 + 24 = 144. (Prediction: 144)

**Verification Keystrokes:** [2] [4] [ENTER] [6] [×]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [2] [4] | 24 | 24 enters Register X |
| [ENTER] | 24 | 24 lifts to Waiting Shelf Register Y |
| [6] | 6 | 6 enters Workbench Register X |
| [×] | 144 | 24 × 6 = 144! Partial products combine |

**3. PAPER STEPS & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Mathematical Rule: To multiply 24 × 6: break 24 into 20 + 4. Partial products: (20 × 6) + (4 × 6) = 120 + 24 = 144.
- Stack Shortcut & Pattern: [2] [4] [ENTER] [6] [×] = 144 — On the SC-6: Type [2] [4] [ENTER] [6] [×] to verify the complete product instantly.
- 2-Second Paper-to-Screen Audit: Confirm your paper setup and inner values match Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: Use partial products to find 38 × 7 on paper, then check on the stack.**

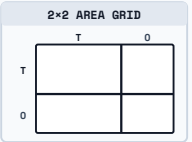
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



# Extension: Two-Digit by Two-Digit Multiplication: 32 × 15

Deepening Concepts • Standards: CCSS.MATH.4.NBT.B.5

**1. EXTENSION CONCEPT & SETUP (Two-Digit by Two-Digit Multiplication: 32 × 15):**

Every term in the first binomial multiplies every term in the second: (30+2)(10+5) = 480.

**Extension Setup: Draw a 2×2 area grid on paper splitting 32 into (30 + 2) along the top and 15 into (10 + 5) along the side. • Question: 30×10=300, 30×5=150, 2×10=20, 2×5=10. Sum: 300+150+20+10 = 480. (Prediction: 480)**

**2. EXTENSION STACK SEQUENCE (SC-6):** 3 2 ENTER 1 5 ×

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| 3 2 | 32 | 32 enters Register X |
| ENTER | 32 | 32 lifts to Waiting Shelf Register Y |
| 1 5 | 15 | 15 enters Workbench Register X |
| × | 480 | 32 × 15 = 480! Four partial products verified |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS**

**Misconception 1 — Multiplying Only Tens-by-Tens and Ones-by-Ones: In 32 × 15, student computes 30×10=300 and 2×5=10 → 310, missing cross-terms.**

Mathematical & Stack Remediation: Use a 2×2 partial-product grid showing all 4 sub-products.

**Misconception 2 — Dropping Place Value Zeroes: Computing 30 × 5 = 15 instead of 150.**

Mathematical & Stack Remediation: On the Place Value Tray, 3 Tens × 5 = 15 Tens = 150.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [2] [4] [ENTER] [6] [×] = 144**

**Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Multiply 43 × 26 using an area grid model on paper, then check on the stack.**

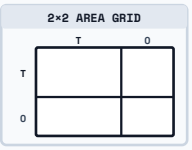
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



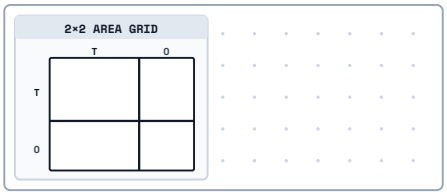
# Unit 04 Lab: Multi-Digit Multiplication & Partial Products

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your place-value / bar model in Box 1. (2) Show your step-by-step paper math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: Multiply using partial products: 52 × 8 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

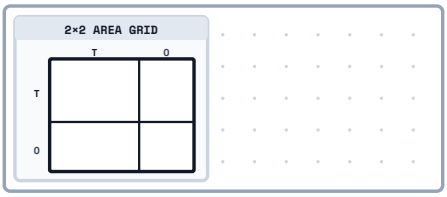
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 2: Multiply two 2-digit numbers: 28 × 14 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

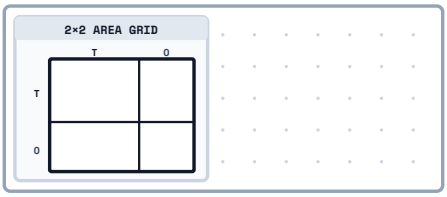
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 3: Multiply using area decomposition: 64 × 7 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

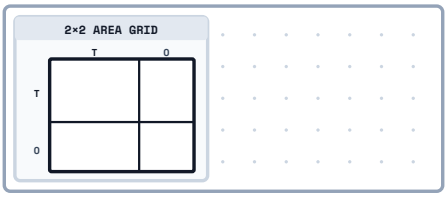
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 4: Multiply two 2-digit numbers: 45 × 32 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your area grid or expression tree sketch show why your paper answer matched the SC-6 stack?**

# Unit 04 Practice A — 2-Digit by 1-Digit Multiplication Fluency

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Multiply 24 × 6 using partial products.**

(20 × 6) + (4 × 6) = 120 + 24 = 144. Key: [24] [ENTER] [6] [×].

Verified Answer: 144

**EXAMPLE 2** **Multiply 38 × 7 on the stack.**

(30 × 7) + (8 × 7) = 210 + 56 = 266.

Verified Answer: 266

**1** Tier 1 • Builder

Area split (20 + 3) × 4: 23 × 4 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

Area split (30 + 4) × 5: 34 × 5 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

Area split (40 + 6) × 3: 46 × 3 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

Area split (20 + 7) × 6: 27 × 6 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

Area split (50 + 2) × 8: 52 × 8 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

Area split (60 + 4) × 7: 64 × 7 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

Area split (70 + 5) × 6: 75 × 6 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

Area split (40 + 8) × 9: 48 × 9 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

Area split (80 + 3) × 7: 83 × 7 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

Area split (90 + 6) × 8: 96 × 8 =

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 04 Practice B — Explicit Partial Products on the 4-Register Stack

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Decompose 36 × 4 into (30 × 4) + (6 × 4).**

120 + 24 = 144. Key: [30] [ENTER] [4] [×] [6] [ENTER] [4] [×] [+].

Verified Answer: 144

**EXAMPLE 2** **Decompose 57 × 8 into (50 × 8) + (7 × 8).**

400 + 56 = 456.

Verified Answer: 456

**1** Tier 1 • Builder

**Partial products for 26 × 5: (20 × 5) + (6 × 5) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Partial products for 43 × 7: (40 × 7) + (3 × 7) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

**Partial products for 58 × 6: (50 × 6) + (8 × 6) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

**Partial products for 67 × 4: (60 × 4) + (7 × 4) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

**Partial products for 74 × 8: (70 × 8) + (4 × 8) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

**Partial products for 85 × 9: (80 × 9) + (5 × 9) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Partial products for 93 × 6: (90 × 6) + (3 × 6) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Compensation strategy for 29 × 6: (30 × 6) − (1 × 6) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Compensation strategy for 49 × 8: (50 × 8) − (1 × 8) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Compensation strategy for 99 × 7: (100 × 7) − (1 × 7) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 04 Practice C — 3-Digit by 1-Digit & Real-World Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** A theater has 6 sections of 125 seats.

125 × 6 = 750 seats.

Verified Answer: 750 seats

**EXAMPLE 2** 4 pallets each hold 235 bricks.

235 × 4 = 940 bricks.

Verified Answer: 940 bricks

**1** Tier 2 • Detective

**Multiply 3-digit by 1-digit: 124 × 3 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

**Multiply 3-digit by 1-digit: 145 × 6 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

**Multiply 3-digit by 1-digit: 218 × 4 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

**Multiply 3-digit by 1-digit: 306 × 5 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

**A solar farm has 8 rows of 135 panels. How many panels in all?**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

**A jet flies 460 miles per hour for 4 hours. Total distance?**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

**Seven crates each weigh 245 lb. What is the total cargo weight?**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

**A school orders 9 boxes of 160 pencils. How many pencils total?**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

**3-part decomposition of 134 × 5: (100 × 5) + (30 × 5) + (4 × 5) =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

**Spot the Mistake: For 208 × 4, Leo forgot the 0 tens and wrote 832 vs 112. Verify 208 × 4:**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ | | SC-6 Ans: _____

# Unit 04 Practice D — 2-Digit by 2-Digit Area Model Multiplication

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Multiply 32 × 15 using a 2×2 area grid.**

300 + 150 + 20 + 10 = 480. Key: [32] [ENTER] [15] [×].

Verified Answer: 480

**EXAMPLE 2** **Multiply 43 × 26 on the stack.**

800 + 240 + 60 + 18 = 1118.

Verified Answer: 1118

**1** Tier 2 • Detective

**Multiply two 2-digit numbers: 24 × 12 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**Multiply two 2-digit numbers: 28 × 14 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**Multiply two 2-digit numbers: 35 × 16 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

**Multiply two 2-digit numbers: 42 × 23 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**Multiply two 2-digit numbers: 45 × 32 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

**Multiply two 2-digit numbers: 54 × 25 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

**Multiply two 2-digit numbers: 62 × 18 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**Multiply two 2-digit numbers: 37 × 24 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

**Multiply two 2-digit numbers: 56 × 34 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Multiply two 2-digit numbers: 75 × 28 =**

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 04 Practice E — 4-Box Area Sub-Rectangles & Engineering Tables

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Sum the 4 boxes of 23 × 14: 200 + 80 + 30 + 12.**

200 + 80 + 30 + 12 = 322.

Verified Answer: 322

**EXAMPLE 2** **Find the missing cross-term in 34 × 12 = 300 + 60 + [ ? ] + 8.**

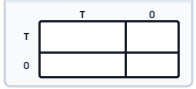
4 × 10 = 40 (total 408).

Verified Answer: 40

**1** Tier 3 • Challenge

**Sum the 4 area boxes for 25 × 13: 200 + 60 + 50 + 15 =**

2×2 AREA GRID

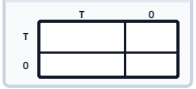



Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Sum the 4 area boxes for 34 × 22: 600 + 60 + 80 + 8 =**

2×2 AREA GRID

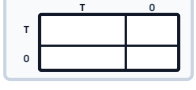



Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

**Sum the 4 area boxes for 46 × 15: 400 + 200 + 60 + 30 =**

2×2 AREA GRID

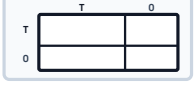



Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

**Find missing box in 27 × 16: (27 × 16) − 200 − 120 − 42 =**

2×2 AREA GRID

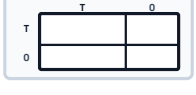



Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

**A battery bank has 24 racks of 16 cells. Find the total number of cells:**

2×2 AREA GRID

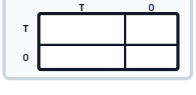



Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

**An auditorium has 28 rows of 35 seats. Find the total number of seats:**

2×2 AREA GRID

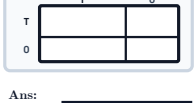



Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

**A greenhouse has 36 trays with 24 seedlings each. Find the total seedlings:**

2×2 AREA GRID

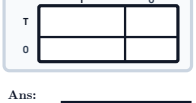



Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

**Compare two solar arrays: (25 × 18) + (15 × 18) =**

2×2 AREA GRID

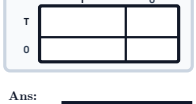



Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

**Difference between two storage yards: (45 × 24) − (32 × 20) =**

2×2 AREA GRID




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Error Analysis: In 32 × 15, Sam added 30×10 + 2×5 = 310. By how much was his answer too low?**

2×2 AREA GRID




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 04 Practice F — Multi-Step Multiplication Pipelines & Challenge

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Evaluate $(18 + 14) \times (20 - 5)$ on the stack.

$32 \times 15 = 480$.

Verified Answer: 480

**EXAMPLE 2** Find the area of a border: $(30 \times 20) - (24 \times 14)$.

$600 - 336 = 264$.

Verified Answer: 264

**1** Tier 3 • Challenge

Two-branch product: $(16 + 12) \times (10 + 5) =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

Two-branch product: $(40 - 15) \times (20 + 4) =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

Courtyard minus fountain area: $(35 \times 24) - (12 \times 10) =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

Combined shipment weight: $(18 \times 25) + (22 \times 25) =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

Three-factor volume of a box 12 cm by 15 cm by 5 cm: $12 \times 15 \times 5 =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

Three-factor volume of a crate 25 in by 16 in by 4 in: $25 \times 16 \times 4 =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

Find the missing factor: $24 \times [\ ?\ ] = 360$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

Find the missing factor: $[\ ?\ ] \times 18 = 450$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

Square of 25 on the stack: $25 \times 25 =$

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Composite Area Challenge: Combine a $34 \times 18$ solar array and a $26 \times 12$ solar array. Find the total cells:

2×2 AREA GRID

| | T | O |
|---|---|---|
| T | | |
| O | | |

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 04 Mastery Check: Multi-Digit Multiplication & Partial Products

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Multiply 2-digit by 1-digit: 47 × 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Multiply 2-digit by 1-digit: 68 × 7 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Multiply 3-digit by 1-digit: 135 × 4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Evaluate partial products: (40 × 8) + (6 × 8) =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Multiply 2-digit by 2-digit: 32 × 15 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Multiply 2-digit by 2-digit: 45 × 24 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**A battery array has 26 modules with 15 cells each. Total cells?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**Find the area of a 38 m by 22 m field minus a 10 m by 8 m shed:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Find the missing factor: 16 × [ ? ] = 400**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Mia wrote 24 × 13 = (20×10) + (4×3) = 212. Find the true product 24 × 13:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 04 STEM: Solar Battery Storage Bank

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: SOLAR BATTERY STORAGE BANK**

**Electrical engineers calculate total energy cells across multi-tier battery racks.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Multi-Digit Area Models & Partial Products) • Slicing Wide Tile Grids Along the Ten-Lines: Instead of blind carry rules, slice 32 × 15 along its ten-lines into four visible area rectangles (300 + 150 + 20 + 10 = 480) on the 4-shelf stack!

Mathematical Focus: Partial-Product Area Grid Diagram & Place Value Studio • Stack Shortcut: [2] [4] [ENTER] [6] [×] = 144

**CASE INVESTIGATION 1** STEM Modeling

**A residential battery rack holds 24 rows of 6 lithium-iron cells. Decompose 24 into 20 + 4 on your Place Value Tray to find total cells:**

2×2 AREA GRID

| | T | 0 |
|---|---|---|
| T | | |
| 0 | | |

1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**A commercial microgrid bank has 32 modules with 15 supercapacitors each. Use a 2×2 area model to find the total capacitors:**

2×2 AREA GRID

| | T | 0 |
|---|---|---|
| T | | |
| 0 | | |

1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Backup Rack Expansion: 7 inverter trays each hold 43 battery cells. Split 43 into 40 + 3 partial products to find the total cells:**

2×2 AREA GRID

| | T | 0 |
|---|---|---|
| T | | |
| 0 | | |

1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: To size a 25-row by 14-column solar array, an apprentice computed (20×10) + (5×4) = 220, forgetting the two cross-rectangles. Find the true cell count:**

2×2 AREA GRID

| | T | 0 |
|---|---|---|
| T | | |
| 0 | | |

1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

# Long Division, Quotients & Remainders

Hands-On Block Studio: Place Value Studio (6×2 Whole Number Tray) • Standards: CCSS.MATH.4.NBT.B.6

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Multi-Digit Division with Remainder: 87 ÷ 4 = 21 R 3):**

87 ÷ 4 yields a whole quotient of 21 and a remainder of 3, since 4 × 21 = 84 and 84 + 3 = 87.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Multi-Digit Division & Place-Value Unbundling ([÷R])) • Unpacking Multi-Story Crates & The Q × *D* + *R* Audit: Long division unpacks place-value crates from Tens to Ones: 87 ÷ 4 gives 21 full rows with 3 left over, verified by 21 × 4 + 3 = 87!**




Place Value Studio • Multi-Digit Quotient Tray

**2. HANDS-ON BLOCKS & CONCEPT RULES: STEP-BY-STEP MULTI-DIGIT DIVISION**

- Long division decomposes the dividend from greatest place value to least: Divide, Multiply, Subtract, Bring Down.
- In 87 ÷ 4 on the Place Value Tray: 8 Tens ÷ 4 = 2 Tens (80). 7 Ones ÷ 4 = 1 One with 3 remaining → 21 R 3!
- Stack verification formula: Quotient × Divisor + Remainder = Dividend.
- On the SC-6: [2] [1] [ENTER] [4] [×] [3] [+] = 87 proves your division 100% correct.

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Block Setup: Place [8] in the Tens socket and [7] in the Ones socket of Row 1 (Waiting Shelf Y) on the Place Value Tray, and divisor [4] in Row 0 (Workbench X). Question: Share 8 Tens into 4 groups = 2 Tens each. Share 7 Ones into 4 groups = 1 One each with 3 left. (Prediction: 21 R 3)

**Verification Keystrokes:** [8] [7] [ENTER] [4] [÷]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [8] [7] | 87 | 87 enters Register X |
| [ENTER] | 87 | 87 lifts to Waiting Shelf Register Y |
| [4] | 4 | 4 enters Workbench Register X |
| [÷] | 21.75 | 87 ÷ 4 = 21.75 (Quotient = 21, 0.75 = 3/4 → Remainder 3) |

**3. 3D BLOCK STUDIO & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Physical Block Studio Rule: Long division decomposes the dividend from greatest place value to least: Divide, Multiply, Subtract, Bring Down.
- Stack Shortcut & Pattern: [21] [ENTER] [4] [×] [3] [+] = 87 (Verification Formula) — On the SC-6: [2] [1] [ENTER] [4] [×] [3] [+] = 87 proves your division 100% correct.
- 2-Second Tray-to-Screen Audit: Confirm your physical 3D Block Studio setup matches Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

**WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Core Investigation: Divide 95 ÷ 6 on paper. State quotient and remainder, then verify on stack.**

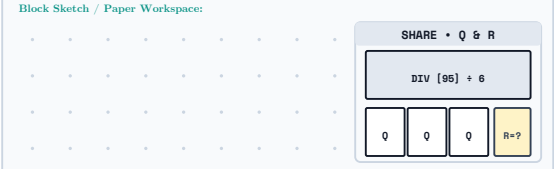
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Rebuilding the Dividend on the Stack: Q × D + R

Deepening Concepts • Standards: CCSS.MATH.4.NBT.B.6

**1. EXTENSION CONCEPT & SETUP (Rebuilding the Dividend on the Stack: Q × D + R):**

The division verification pipeline confirms both the quotient and the remainder without hidden errors.

**Hands-On Extension Setup: Place Quotient [2]|[1] in Row 1 (Waiting Shelf Y) and Divisor [4] in Row 0 (Workbench X) of the Place Value Tray, with Remainder [3] ready to add. • Question: Does 21 × 4 + 3 return exactly 87? (Prediction: 84 + 3 = 87)**

**2. EXTENSION STACK SEQUENCE (SC-6):** [2] [1] [ENTER] [4] [×] [3] [+]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [2] [1] | 21 | Quotient (21) enters Register X |
| [ENTER] | 21 | 21 lifts to Waiting Shelf Register Y |
| [4] | 4 | Divisor (4) enters Workbench Register X |
| [×] | 84 | 21 × 4 = 84! Subtotal in Register X |
| [3] | 3 | Remainder (3) enters Register X |
| [+] | 87 | 84 + 3 = 87! Exact match to original dividend |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1 — Allowing a Remainder Larger than the Divisor: Student claims 87 ÷ 4 = 20 R 7.**

Hands-On Block Studio Remediation: Point out that remainder 7 can make another full group of 4, leaving remainder 3!

**Misconception 2 — Dropping Zeroes in the Middle of Quotients: In 816 ÷ 4, answering 24 instead of 204.**

Hands-On Block Studio Remediation: Place Value Trays show 1 Ten cannot be divided into 4 Tens, so a [0] Digit Block must go in the Tens socket.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [21] [ENTER] [4] [×] [3] [+] = 87 (Verification Formula)**

**Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: If 134 ÷ 5 = 26 R 4, verify by evaluating 26 × 5 + 4 on the stack.**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



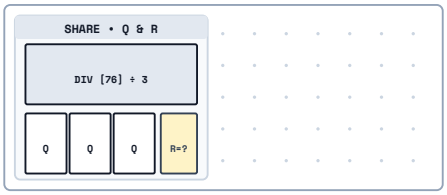
# Unit 05 Lab: Long Division, Quotients & Remainders

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your place-value / bar model in Box 1. (2) Show your step-by-step paper math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: Divide 76 ÷ 3 on paper: state quotient and remainder =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

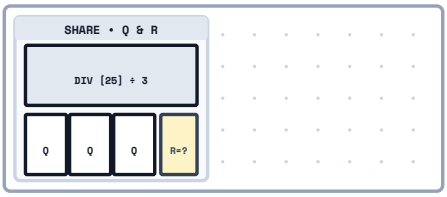
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 2: Verify division on stack: 25 × 3 + 1 =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

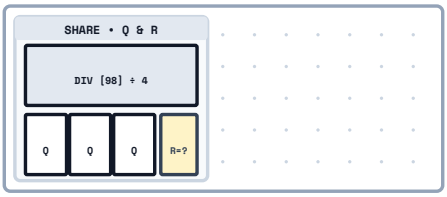
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 3: Divide 98 ÷ 4 on paper: state quotient and remainder =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 4: Divide 125 ÷ 6 on paper: state quotient and remainder =**

**1. Place-Value / Bar Model Sketch:**




**2. Step-by-Step Paper Work:**

Step 1 (First Group / Product):

Step 2 (Combine / Divide):

Equation & Solution:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 05 Practice A — Exact Multi-Digit Division Fluency

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Divide 84 by 4 using place value.

8 tens ÷ 4 = 2 tens; 4 ones ÷ 4 = 1 one → 21.

Verified Answer: 21

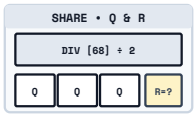
**EXAMPLE 2** Divide 135 by 5 on the stack.

100 ÷ 5 = 20; 35 ÷ 5 = 7 → 27.

Verified Answer: 27

**1** Tier 1 • Builder

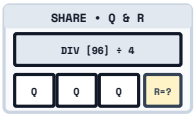
Share tens & ones: 68 ÷ 2 =




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 1 • Builder

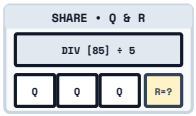
Unbundle tens to share: 96 ÷ 4 =




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 1 • Builder

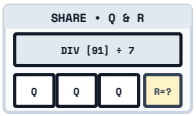
Unbundle tens to share: 85 ÷ 5 =




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 1 • Builder

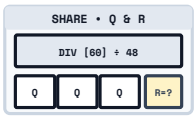
Unbundle tens to share: 91 ÷ 7 =




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 1 • Builder

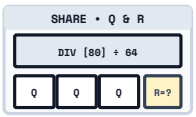
Decompose (60 + 48) ÷ 6: 108 ÷ 6 =




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 1 • Builder

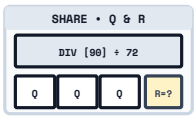
Decompose (80 + 64) ÷ 8: 144 ÷ 8 =




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 1 • Builder

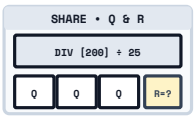
Decompose (90 + 72) ÷ 9: 162 ÷ 9 =




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 1 • Builder

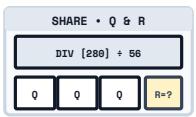
Decompose (200 + 25) ÷ 5: 225 ÷ 5 =




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 1 • Builder

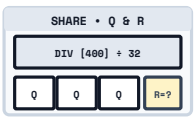
Decompose (280 + 56) ÷ 7: 336 ÷ 7 =




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 1 • Builder

Decompose (400 + 32) ÷ 8: 432 ÷ 8 =




Paper Step: _____ | | SC-6 Ans: _____

# Unit 05 Practice B — Multi-Digit Long Division with Remainders

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Divide 87 ÷ 4 and find quotient and remainder.

87 = 4 × 21 + 3 → 21 R 3.

Verified Answer: 21 R 3

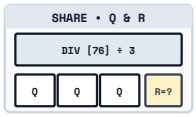
**EXAMPLE 2** Divide 134 ÷ 5 and find quotient and remainder.

134 = 5 × 26 + 4 → 26 R 4.

Verified Answer: 26 R 4

**1** Tier 1 • Builder

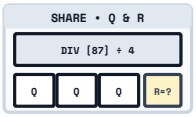
Find quotient and remainder: 76 ÷ 3 =




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 1 • Builder

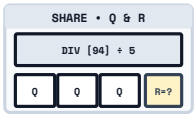
Find quotient and remainder: 87 ÷ 4 =




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 1 • Builder

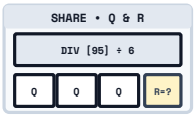
Find quotient and remainder: 94 ÷ 5 =




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 1 • Builder

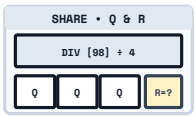
Find quotient and remainder: 95 ÷ 6 =




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 1 • Builder

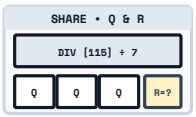
Find quotient and remainder: 98 ÷ 4 =




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 1 • Builder

Find quotient and remainder: 115 ÷ 7 =




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 1 • Builder

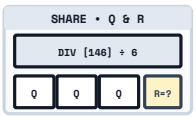
Find quotient and remainder: 125 ÷ 6 =




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 1 • Builder

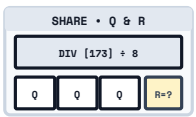
Find quotient and remainder: 146 ÷ 6 =




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 1 • Builder

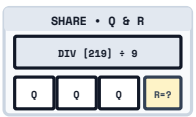
Find quotient and remainder: 173 ÷ 8 =




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 1 • Builder

Find quotient and remainder: 219 ÷ 9 =




Paper Step: _____ | | SC-6 Ans: _____

# Unit 05 Practice C — Rebuilding the Dividend ($Q \times D + R =$ Dividend)

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Verify $87 \div 4 = 21$ R 3 on the stack.**

$21 \times 4 + 3 = 84 + 3 = 87$. Key: [21] [ENTER] [4] [×] [3] [+].

Verified Answer: 87

**EXAMPLE 2** **Find the dividend when $Q = 26$, $D = 5$, $R = 4$.**

$26 \times 5 + 4 = 134$.

Verified Answer: 134

**1** Tier 2 • Detective

**Verify division: $(25 \times 3) + 1 =$**

SHARE • Q & R
DIV [25] ÷ 3
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**Verify division: $(18 \times 5) + 4 =$**

SHARE • Q & R
DIV [18] ÷ 5
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**Verify division: $(15 \times 6) + 5 =$**

SHARE • Q & R
DIV [15] ÷ 6
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

**Verify division: $(24 \times 4) + 2 =$**

SHARE • Q & R
DIV [24] ÷ 4
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**Verify division: $(16 \times 7) + 3 =$**

SHARE • Q & R
DIV [16] ÷ 7
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

**Verify division: $(20 \times 6) + 5 =$**

SHARE • Q & R
DIV [20] ÷ 6
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

**Verify division: $(24 \times 6) + 2 =$**

SHARE • Q & R
DIV [24] ÷ 6
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**Verify division: $(21 \times 8) + 5 =$**

SHARE • Q & R
DIV [21] ÷ 8
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

**Verify division: $(32 \times 7) + 6 =$**

SHARE • Q & R
DIV [32] ÷ 7
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Given quotient 18, find the remainder when 150 is divided by 8:**

SHARE • Q & R
DIV [18] ÷ 150
Q Q Q R=?

Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 05 Practice D — Zeroes in the Quotient & Place Value Decomposition

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Divide 816 ÷ 4 carefully.

800 ÷ 4 = 200; 16 ÷ 4 = 4 → 204 (not 24!).

Verified Answer: 204

**EXAMPLE 2** Decompose 168 ÷ 6 into (120 ÷ 6) + (48 ÷ 6).

20 + 8 = 28.

Verified Answer: 28

**1** Tier 2 • Detective

**Zero in the tens place: 408 ÷ 4 =**

SHARE • Q & R
DIV [408] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

**Zero in the tens place: 615 ÷ 3 =**

SHARE • Q & R
DIV [615] ÷ 3
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

**Zero in the tens place: 816 ÷ 4 =**

SHARE • Q & R
DIV [816] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

**Zero in the tens place: 535 ÷ 5 =**

SHARE • Q & R
DIV [535] ÷ 5
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

**Zero in the tens place: 642 ÷ 6 =**

SHARE • Q & R
DIV [642] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

**Zero in the tens place: 728 ÷ 7 =**

SHARE • Q & R
DIV [728] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

**Zero in the ones place: 480 ÷ 4 =**

SHARE • Q & R
DIV [480] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

**Zero in the ones place: 750 ÷ 5 =**

SHARE • Q & R
DIV [750] ÷ 5
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

**Partial quotients for 156 ÷ 6: (120 ÷ 6) + (36 ÷ 6) =**

SHARE • Q & R
DIV [156] ÷ 6
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

**Partial quotients for 252 ÷ 7: (210 ÷ 7) + (42 ÷ 7) =**

SHARE • Q & R
DIV [252] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

# Unit 05 Practice E — Cargo Logistics, Remainder Decisions & Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** 87 shipping containers load in rows of 4. Full rows and remainder?

87 ÷ 4 = 21 R 3.

Verified Answer: 21 R 3

**EXAMPLE 2** How many 6-pallet trucks are needed to move ALL 95 pallets?

95 ÷ 6 = 15 R 5 → 15 + 1 = 16 trucks.

Verified Answer: 16 trucks

**1** Tier 3 • Challenge

168 solar panels are packed 6 to a crate. How many crates?

SHARE • Q & R | DIV [168] ÷ 6 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 3 • Challenge

270 chairs are set up in 9 equal rows. How many chairs per row?

SHARE • Q & R | DIV [270] ÷ 9 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 3 • Challenge

134 loaves of bread are packed 5 to a tray. Find full trays and leftover loaves:

SHARE • Q & R | DIV [134] ÷ 5 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 3 • Challenge

215 meters of cable are cut into 8-meter spools. Find full spools and leftover:

SHARE • Q & R | DIV [215] ÷ 8 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 3 • Challenge

324 apples are packed in boxes of 12. How many boxes are filled?

SHARE • Q & R | DIV [324] ÷ 12 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 3 • Challenge

450 gallons of water fill 15-gallon drums. How many drums?

SHARE • Q & R | DIV [450] ÷ 15 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 3 • Challenge

528 tiles cover a hall 16 tiles wide. How many tiles long is the hall?

SHARE • Q & R | DIV [528] ÷ 16 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 3 • Challenge

A baker has 100 cookies. After eating 4, she packs the rest in boxes of 8. How many boxes?

SHARE • Q & R | DIV [100] ÷ 4 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 3 • Challenge

Two warehouses ship 140 and 116 boxes in trucks that hold 8 boxes each. Total trucks?

SHARE • Q & R | DIV [140] ÷ 116 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 3 • Challenge

Spot the Mistake: For 624 ÷ 6, Ben wrote 14 instead of 104. By how much was his quotient off?

SHARE • Q & R | DIV [624] ÷ 6 | Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

# Unit 05 Practice F — Multi-Step Division & Stack Verification Challenges

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find average of 3flights: (120 + 135 + 150) ÷ 3.**

405 ÷ 3 = 135.

Verified Answer: 135

**EXAMPLE 2** **Divide 576 by 12, then divide by 4.**

576 ÷ 12 = 48; 48 ÷ 4 = 12.

Verified Answer: 12

**1** Tier 3 • Challenge

**Find the average of 3 scores: 84, 92, and 88:**

SHARE • Q & R
DIV [3] ÷ 84
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 3 • Challenge

**Find the average of 4 weights: 42, 48, 54, and 56:**

SHARE • Q & R
DIV [4] ÷ 42
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 3 • Challenge

**Two-step division: 480 ÷ 8 ÷ 5 =**

SHARE • Q & R
DIV [480] ÷ 8
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 3 • Challenge

**Two-step division: 648 ÷ 9 ÷ 6 =**

SHARE • Q & R
DIV [648] ÷ 9
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 3 • Challenge

**Multiply then divide: (24 × 15) ÷ 9 =**

SHARE • Q & R
DIV [24] ÷ 15
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 3 • Challenge

**Multiply then divide: (36 × 14) ÷ 8 =**

SHARE • Q & R
DIV [36] ÷ 14
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 3 • Challenge

**Difference of two quotients: (288 ÷ 8) − (156 ÷ 6) =**

SHARE • Q & R
DIV [288] ÷ 8
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 3 • Challenge

**Sum of two quotients: (315 ÷ 7) + (225 ÷ 9) =**

SHARE • Q & R
DIV [315] ÷ 7
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 3 • Challenge

**Mystery dividend: when divided by 9, quotient is 34 and remainder is 7:**

SHARE • Q & R
DIV [9] ÷ 34
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 3 • Challenge

**Zero-Placeholder Verification: Divide 816 blocks equally among 4 trays, then subtract 204. What remains?**

SHARE • Q & R
DIV [816] ÷ 4
Q Q Q R=?

Paper Step: _____ | | SC-6 Ans: _____

# Unit 05 Mastery Check: Long Division & Remainders

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Evaluate exact quotient: 156 ÷ 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Evaluate exact quotient with zero in tens: 816 ÷ 4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Evaluate 2-digit divisor quotient: 360 ÷ 15 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Find quotient and remainder: 87 ÷ 4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Find quotient and remainder: 146 ÷ 6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Verify division on stack: (26 × 5) + 4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**252 books are packed into 7 equal cartons. How many books per carton?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**95 containers are stacked 6 high. Find full stacks and leftover containers:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Find the average of 3 container loads: 115, 125, and 135:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: A student says 98 ÷ 4 = 23 R 6. Find the true quotient and remainder:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 05 STEM: Cargo Ship Container Loading

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: CARGO SHIP CONTAINER LOADING**

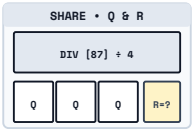
**Port logistics engineers pack intermodal shipping containers into uniform vessel bays and verify remainders.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 4: Multi-Digit Division & Place-Value Unbundling ([÷R])) • Unpacking Multi-Story Crates & The Q × D + R Audit: Long division unpacks place-value crates from Tens to Ones: 87 ÷ 4 gives 21 full rows with 3 left over, verified by 21 × 4 + 3 = 87!

Block Studio: Place Value Studio (6×2 Whole Number Tray) • Stack Shortcut: [21] [ENTER] [4] [×] [3] [+] = 87 (Verification Formula)

**CASE INVESTIGATION 1** STEM Modeling

**87 intermodal containers are loaded into vessel bays of 4 containers each. Find the full bays and remaining containers with [÷R]:**




1. Model / Expression: _____

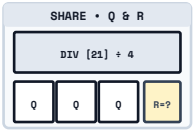
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Manifest Audit: Rebuild the total container count on the stack from 21 full bays of 4 plus 3 deck containers (21 × 4 + 3):**




1. Model / Expression: _____

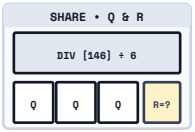
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Bulk Cement Hold: 146 export pallets are crane-lifted in slings of 6 pallets per lift. Find full lifts and leftover pallets:**




1. Model / Expression: _____

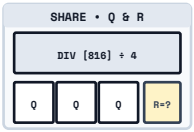
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: Distributing 816 tons of ballast equally across 4 hull tanks (816 ÷ 4), a clerk wrote 24 tons per tank. Find the true load:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

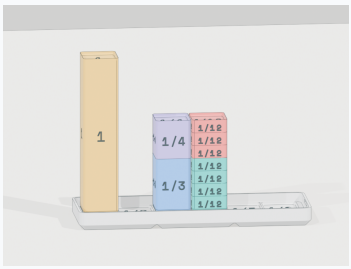
# Fraction Equivalence & Common Denominators

Hands-On Block Studio: Fraction Studio (Comparator Well & Proportional Towers) • Standards: CCSS.MATH.4.NF.A.1, CCSS.MATH.4.NF.A.2

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Equivalent Fractions on the Comparator: 2/4 = 1/2):**

Multiplying or dividing numerator and denominator by the same number scales the piece count without changing total height.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Equivalent Fractions & Common Denominators ([SIMP])) • Hot Dogs vs. Buns, Drum Polyrhythms & The 84 mm Well: Why do 3-against-4 drum beats and 1/3 + 1/4 glass strips meet at 12ths? In the 84 mm well, 1/3 (28 mm) + 1/4 (21 mm) = seven 7 mm twelfths (49 mm)!**




**Common Denominator: 1/3 + 1/4 = 7/12**

**2. HANDS-ON BLOCKS & CONCEPT RULES: WHY EQUIVALENT FRACTIONS STAND AT EQUAL HEIGHTS**

- Equivalent fractions have equal values and identical heights: 1/2 = 2/4 = 3/6 = 4/8 (42.0 mm).
- To add fractions with unlike denominators, find a Common Denominator: 1/3 (4/12) + 1/4 (3/12) = 7/12.
- The SC-6 automatically simplifies fractions to lowest terms: [.] [2] [.] [4] displays as 1/2.
- Physical verification: stack 4/12 and 3/12 into the comparator well; they stand flush with 7/12 (49.0 mm)!

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Block Setup: Place the 1/2 tower (42 mm) in the comparator well. Beside it, stack two 1/4 blocks (21+21 = 42 mm). Question: Do two 1/4 blocks stand at the exact same height as 1/2? (Prediction: Yes, 42.0 mm flush!)

**Verification Keystrokes:** [•] [2] [•] [4]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [•] [2] [•] [4] | 1/2 | Enter 2/4: calculator automatically reduces to 1/2 |

**3. 3D BLOCK STUDIO & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Physical Block Studio Rule: Equivalent fractions have equal values and identical heights: 1/2 = 2/4 = 3/6 = 4/8 (42.0 mm).
- Stack Shortcut & Pattern: [.] [1] [.] [3] [ENTER] [.] [1] [.] [4] [+] = 7/12 — Physical verification: stack 4/12 and 3/12 into the comparator well; they stand flush with 7/12 (49.0 mm)!
- 2-Second Tray-to-Screen Audit: Confirm your physical 3D Block Studio setup matches Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: Test 3/6 and 4/8 in the comparator well. What single simplified fraction do both equal?**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Building Common Denominators: 1/3 + 1/4 = 7/12

Deepening Concepts • Standards: CCSS.MATH.4.NF.A.2

**1. EXTENSION CONCEPT & SETUP (Building Common Denominators: 1/3 + 1/4 = 7/12):**

Twelfths are the common unit because 12 is a multiple of both 3 and 4.

**Hands-On Extension Setup: Place 1/3 (28 mm = four 1/12 blocks) and 1/4 (21 mm = three 1/12 blocks) together in the comparator well. • Question: Total twelfths: 4/12 + 3/12 = 7/12. Height = 49.0 mm. (Prediction: 7/12)**

**2. EXTENSION STACK SEQUENCE (SC-6):** [ . ] [ 1 ] [ . ] [ 3 ] [ ENTER ] [ . ] [ 1 ] [ . ] [ 4 ] [ + ]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [ . ] [ 1 ] [ . ] [ 3 ] | 1/3 | 1/3 enters Register X |
| [ ENTER ] | 1/3 | 1/3 lifts to Waiting Shelf Register Y |
| [ . ] [ 1 ] [ . ] [ 4 ] | 1/4 | 1/4 enters Workbench Register X |
| [ + ] | 7/12 | 1/3 + 1/4 = 7/12! Common denominator 12 |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1 — Adding Denominators Across: Student adds 1/3 + 1/4 = 2/7.**

Hands-On Block Studio Remediation: Compare 7/12 (49 mm) against 2/7 (24 mm) in the comparator well.

**Misconception 2 — Scaling Only the Numerator: Claiming 1/3 is equivalent to 2/3 instead of 2/6.**

Hands-On Block Studio Remediation: Show that if you double the pieces, each piece must be half as tall.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [.] [1] [.] [3] [ENTER] [.] [1] [.] [4] [+] = 7/12**

**Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Find the common denominator and sum of 1/2 + 1/3 on paper, then check on the stack.**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



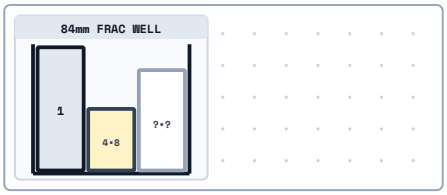
# Unit 06 Lab: Fraction Equivalence & Common Denominators

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your fraction bar or ratio model in Box 1. (2) Show your fraction/decimal math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: Find equivalent fraction: 4/8 simplifies to =**

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

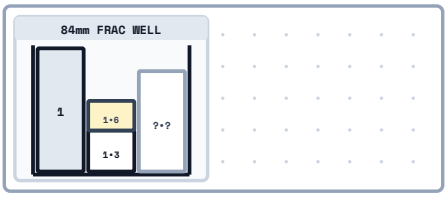
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 2: Add unlike fractions: 1/3 + 1/6 =**

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

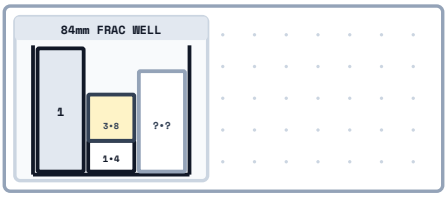
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 3: Add unlike fractions: 1/4 + 3/8 =**

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

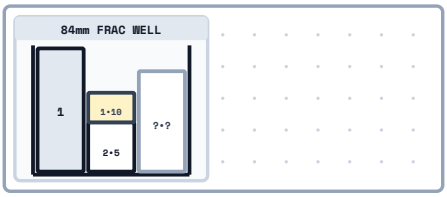
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 4: Find common denominator and sum: 2/5 + 1/10 =**

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 06 Practice A — Fraction Equivalence & Automatic Simplification

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Enter 2/4 on the SC-6 ([.2.4]).**

Two 1/4 blocks (42 mm) match one 1/2 block (42 mm) → 1/2.

Verified Answer: 1/2

**EXAMPLE 2** **Enter 8/12 on the SC-6 ([.8.12]).**

Divide top and bottom by GCF 4 → 2/3.

Verified Answer: 2/3

**1** Tier 1 • Builder

**Simplify equivalent fraction: 2/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Simplify equivalent fraction: 3/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

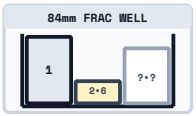
**Simplify equivalent fraction: 4/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

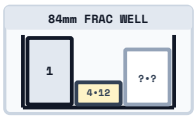
**Simplify equivalent fraction: 2/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

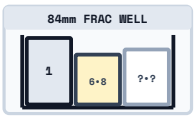
**Simplify equivalent fraction: 4/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

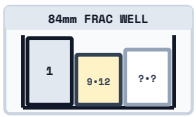
**Simplify equivalent fraction: 6/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Simplify equivalent fraction: 9/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Simplify equivalent fraction: 4/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

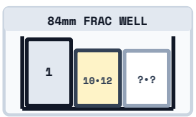
**Simplify equivalent fraction: 8/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Simplify equivalent fraction: 10/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Practice B — Adding & Subtracting with Related Denominators

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Add 1/2 + 1/4 using fourths.

1/2 = 2/4; 2/4 + 1/4 = 3/4. Key: [.1.2] [ENTER] [.1.4] [+].

Verified Answer: 3/4

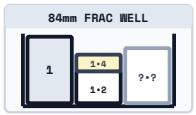
**EXAMPLE 2** Add 1/3 + 1/6 using sixths.

1/3 = 2/6; 2/6 + 1/6 = 3/6 = 1/2.

Verified Answer: 1/2

**1** Tier 1 • Builder

Add related fractions: 1/2 + 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

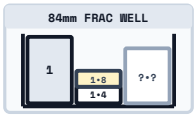
Add related fractions: 1/3 + 1/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

Add related fractions: 1/4 + 1/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

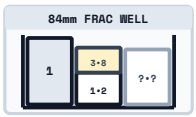
Add related fractions: 1/4 + 3/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

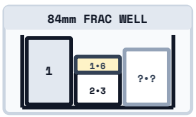
Add related fractions: 1/2 + 3/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

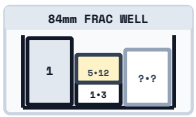
Add related fractions: 2/3 + 1/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

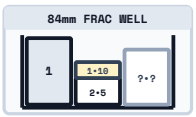
Add related fractions: 1/3 + 5/12 =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

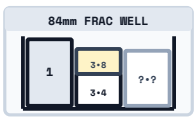
Add related fractions: 2/5 + 1/10 =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

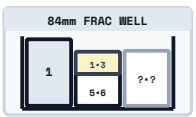
Subtract related fractions: 3/4 − 3/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

Subtract related fractions: 5/6 − 1/3 =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Practice C — Common Denominators for Unlike Fractions (1/3 & 1/4)

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Add 1/3 + 1/4 using common denominator 12.**

4/12 + 3/12 = 7/12 (49 mm). Key: [.1.3] [ENTER] [.1.4] [+].

Verified Answer: 7/12

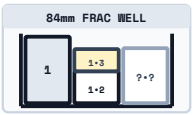
**EXAMPLE 2** **Subtract 2/3 − 1/4 using twelfths.**

8/12 − 3/12 = 5/12.

Verified Answer: 5/12

**1** Tier 2 • Detective

**Add unlike fractions: 1/2 + 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**Add unlike fractions: 1/3 + 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**Add unlike fractions: 2/3 + 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

**Add unlike fractions: 1/3 + 3/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**Add unlike fractions: 1/2 + 2/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

**Subtract unlike fractions: 1/2 − 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

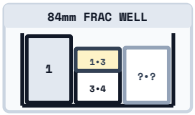
**Subtract unlike fractions: 1/3 − 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**Subtract unlike fractions: 3/4 − 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

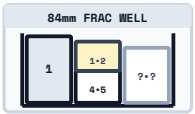
**Subtract unlike fractions: 2/3 − 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Subtract unlike fractions: 4/5 − 1/2 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Practice D — Finding Missing Equivalent Numerators & Heights

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find the missing numerator: 1/3 = [ ? ] / 12.**

Multiply by 12: (1/3) × 12 = 4.

Verified Answer: 4

**EXAMPLE 2** **Find the missing numerator: 3/4 = [ ? ] / 12.**

Multiply by 12: (3/4) × 12 = 9.

Verified Answer: 9

**1** Tier 2 • Detective

**How many twelfths equal 1/2?**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**How many twelfths equal 1/3?**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**How many twelfths equal 2/3?**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

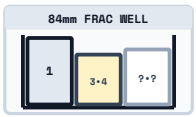
**How many twelfths equal 1/4?**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**How many twelfths equal 3/4?**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

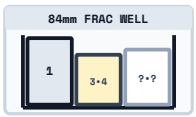
**How many twelfths equal 5/6?**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

**How many eighths equal 3/4?**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**How many tenths equal 3/5?**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

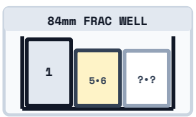
**Find the height of 7/12 in the 84 mm comparator in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Find the height of 5/6 in the 84 mm comparator in mm:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Practice E — Stained Glass, Carpentry & Recipe Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** A stained glass frame joins 1/3 m red and 1/4 m blue glass.

1/3 + 1/4 = 4/12 + 3/12 = 7/12 m.

Verified Answer: 7/12 m

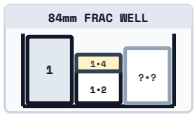
**EXAMPLE 2** A chef has 3/4 cup olive oil and uses 1/3 cup.

3/4 − 1/3 = 9/12 − 4/12 = 5/12 cup.

Verified Answer: 5/12 cup

**1** Tier 3 • Challenge

A baker mixes 1/2 cup sugar with 1/4 cup brown sugar. Total sugar?




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

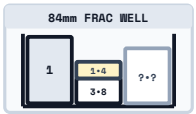
A trail is 1/3 mile gravel and 1/2 mile boardwalk. Total trail length?




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

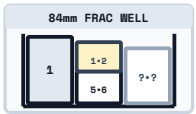
A carpenter glues a 3/8 inch plywood sheet to a 1/4 inch veneer. Total thickness?




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

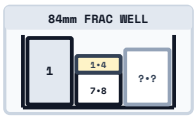
A pitcher holds 5/6 liter of juice. You pour out 1/2 liter. How much is left?




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

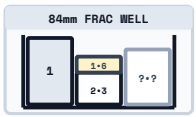
A bolt is 7/8 inch long. A nut covers 1/4 inch. How much bolt thread shows?




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

Sam walks 2/3 mile before lunch and 1/6 mile after lunch. Total distance?




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

Maya paints 1/3 of a wall on Monday and 1/4 on Tuesday. What fraction is painted?




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

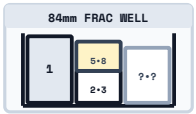
In Problem 7, what fraction of the wall remains unpainted?




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

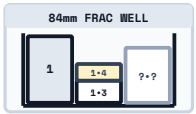
Which is longer: 2/3 m or 5/8 m? Find the difference 2/3 − 5/8 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Spot the Mistake: Leo added 1/3 + 1/4 = 2/7. Find the true sum 1/3 + 1/4:




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Practice F — Three-Fraction Pipelines & Equivalence Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Evaluate $1/2 + 1/3 + 1/6$ on the stack.

$3/6 + 2/6 + 1/6 = 6/6 = 1$ Whole.

Verified Answer: 1

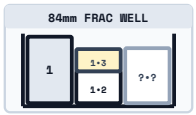
**EXAMPLE 2** Find the missing fraction: $1/4 + 1/3 + [\ ?\ ] = 1$.

$1 - 7/12 = 5/12$.

Verified Answer: 5/12

**1** Tier 3 • Challenge

**Three-fraction sum: $1/2 + 1/3 + 1/6 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Three-fraction sum: $1/2 + 1/4 + 1/8 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

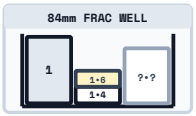
**Three-fraction sum: $1/3 + 1/4 + 1/6 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

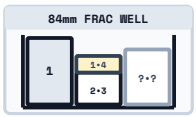
**Three-fraction sum: $1/4 + 1/6 + 1/12 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

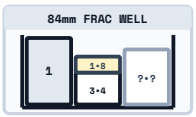
**Add then subtract: $2/3 + 1/4 - 1/6 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

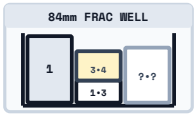
**Add then subtract: $3/4 + 1/8 - 1/2 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

**Find the missing fraction: $1/3 + [\ ?\ ] = 3/4$**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

**Find the missing fraction: $5/6 - [\ ?\ ] = 1/4$**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

**Complement of two parts: $1 - (1/3 + 1/4) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Three-Tower Flush Challenge: Stack a 2/3 tower, a 1/4 tower, and a 1/12 tower in your 84 mm Comparator Well:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 06 Mastery Check: Fraction Equivalence & Denominators

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Simplify to lowest terms: 6/8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Simplify to lowest terms: 8/12 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Add related fractions: 1/3 + 1/6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Add related fractions: 1/4 + 3/8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Add unlike fractions: 1/2 + 1/3 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Add unlike fractions: 1/3 + 1/4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**Subtract unlike fractions: 3/4 − 1/3 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**How many twelfths are equivalent to 2/3?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**A stained-glass border combines 1/2 m, 1/4 m, and 1/8 m strips. Total length?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: A student adds 1/2 + 1/3 = 2/5. Find the true sum:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 06 STEM: Stained Glass Window Assembly

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: STAINED GLASS WINDOW ASSEMBLY**

**Architectural glass artisans cut proportional strips and match lead-came joints using common denominators.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Equivalent Fractions & Common Denominators ([SIMP])) • Hot Dogs vs. Buns, Drum Polyrhythms & The 84 mm Well: Why do 3-against-4 drum beats and 1/3 + 1/4 glass strips meet at 12ths? In the 84 mm well, 1/3 (28 mm) + 1/4 (21 mm) = seven 7 mm twelfths (49 mm)!

Block Studio: Fraction Studio (Comparator Well & Proportional Towers) • Stack Shortcut: [.] [1] [.] [3] [ENTER] [.] [1] [.] [4] [+] = 7/12

**CASE INVESTIGATION 1** STEM Modeling

**Cathedral Medallion Check: Two 1/4-meter ruby glass panels are soldered side-by-side. Compare them in the 84 mm well to simplify 2/4:**




1. Model / Expression: _____

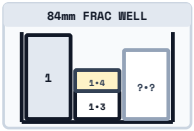
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Lead-Came Border Joint: An artisan joins a 1/3-meter cobalt strip and a 1/4-meter amber strip. Trade for twelfths in the comparator well to find total length:**




1. Model / Expression: _____

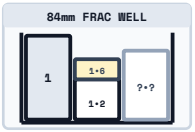
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Rose Window Arch: Combine a 1/2-meter central panel with a 1/6-meter border arch and simplify to lowest terms:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: An apprentice glazier added $1/2 + 1/3 = 2/5$ m of glass and cut the frame too short! Find the true combined length:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

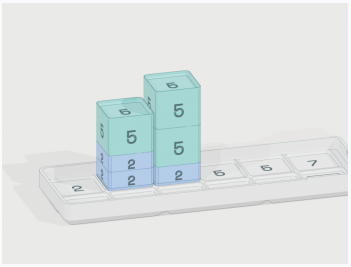
# Prime Factorization, GCF/LCM & Factor Trees

Hands-On Block Studio: Prime Factor Studio (Logarithmic Prime Blocks 2, 3, 5, 7) • Standards: CCSS.MATH.4.OA.B.4, CCSS.MATH.6.NS.B.4

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Building Composite Numbers from Prime Factor Blocks: 2 × 2 × 3 = 12):**

Because Prime Factor Studio block heights are proportional to log(p), stacking prime blocks physically adds heights to multiply factors.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 7: Prime Numbers, Factor Towers, [GCF] & [LCM]) • 17-Year Prime Cicadas & Hunting-Tooth Bicycle Gears: Magicicada broods emerge every 13 or 17 prime years to beat predator LCM cycles, while engineers choose prime gear teeth so wear spreads evenly!**




Prime Factor Studio • Log Prime Towers

**2. HANDS-ON BLOCKS & CONCEPT RULES: HOW PRIME BLOCKS BUILD EVERY COMPOSITE NUMBER**

- Prime numbers have exactly two factors: 1 and itself (2, 3, 5, 7, 11...). 1 is NOT prime.
- Fundamental Theorem of Arithmetic: every integer > 1 has a unique prime factorization.
- Prime Factor Studio Blocks: equal products stand at the exact same physical height (e.g. 2×2×3 = 12 matches 3×4).
- GCF is the shared prime intersection; LCM is the complete prime union.

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Block Setup: Stack two '2' prime blocks and one '3' prime block in the Prime Factor Studio comparator tower. Question: Multiply the prime chain: 2 × 2 = 4, then 4 × 3 = 12. Does the stack match the '12' mark? (Prediction: 12)

**Verification Keystrokes:** [2] [ENTER] [2] [×] [3] [×]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [2] | 2 | First prime factor (2) enters Register X |
| [ENTER] | 2 | 2 lifts to Waiting Shelf Register Y |
| [2] | 2 | Second prime factor (2) enters Register X |
| [×] | 4 | 2 × 2 = 4 in Register X |
| [3] | 3 | Third prime factor (3) enters Register X |
| [×] | 12 | 4 × 3 = 12! Unique prime factorization verified |

**STACK SHORTCUT & BLOCK STUDIO TIP (SC-6):**

[12] [ENTER] [18] [×] [6] [÷] = 36 (LCM via GCF Formula) • Prime numbers have exactly two factors: 1 and itself (2, 3, 5, 7, 11...). 1 is NOT prime.

## WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: Find the prime factorization of 18 and verify the product on your stack.**

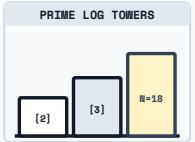
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Finding GCF and LCM via Prime Factor Intersection & Union

Deepening Concepts • Standards: CCSS.MATH.6.NS.B.4

**1. EXTENSION CONCEPT & SETUP (Finding GCF and LCM via Prime Factor Intersection & Union):**

Every pair of numbers satisfies $a \times b = \text{GCF}(a, b) \times \text{LCM}(a, b)$.

**Hands-On Extension Setup: Build Tower 12 (2 × 2 × 3) and Tower 18 (2 × 3 × 3) side-by-side in the Prime Factor Studio. • Question: Shared prime blocks (GCF): one '2' and one '3' → 2 × 3 = 6. LCM = (12 × 18) ÷ 6 = 36.**

**2. EXTENSION STACK SEQUENCE (SC-6):** [1] [2] [ENTER] [1] [8] [×] [6] [÷]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [1] [2] | 12 | 12 enters Register X |
| [ENTER] | 12 | 12 lifts to Register Y |
| [1] [8] | 18 | 18 enters Register X |
| [×] | 216 | 12 × 18 = 216 in Register X |
| [6] | 6 | GCF (6) enters Register X |
| [÷] | 36 | 216 ÷ 6 = 36! LCM = 36 verified |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1 — Stopping Factor Trees at Composite Numbers: Stopping a factor tree of 24 at 4 × 6 instead of factoring all the way to primes 2 × 2 × 2 × 3.**

Hands-On Block Studio Remediation: Only Prime Factor Studio blocks (2, 3, 5, 7) exist in the tray—4 and 6 must be split into prime blocks.

**Misconception 2 — Confusing GCF Intersection with LCM Union: Giving a multiple (like 36) when asked for the GCF of 12 and 18 (which is 6).**

Hands-On Block Studio Remediation: GCF is the shared prime sub-tower inside both stacks; LCM is the taller combined tower that both divide into.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [12] [ENTER] [18] [×] [6] [÷] = 36 (LCM via GCF Formula)**

**Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Given GCF(8, 12) = 4, find LCM(8, 12) on the stack.**

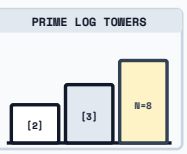
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



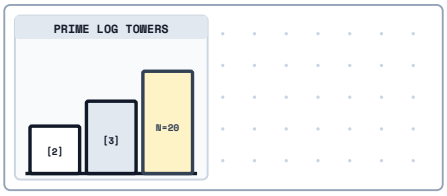
# Unit 07 Lab: Prime Factorization, GCF/LCM & Factor Trees

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your factor tree or grouping diagram in Box 1. (2) Show your inside-out steps in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

## Investigation 1: Multiply prime factorization of 20: 2 × 2 × 5 =

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

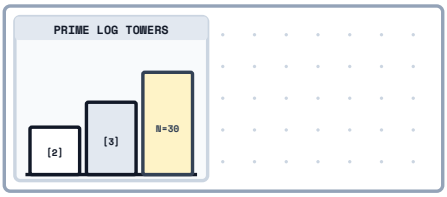
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

## Investigation 2: Multiply prime factorization of 30: 2 × 3 × 5 =

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

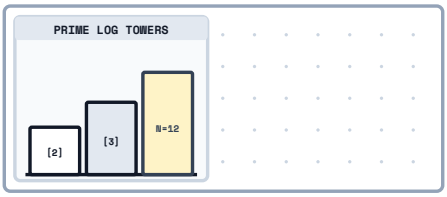
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

## Investigation 3: Find the Greatest Common Factor (GCF) of 12 and 18:

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

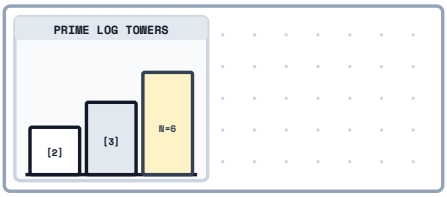
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

## Investigation 4: Find the Least Common Multiple (LCM) of 6 and 8 given GCF(6, 8) = 2:

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 07 Practice A — Prime Factorization, Exponents, GCF & LCM

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find the prime factorization of 36.**

$36 = 4 \times 9 = 2^2 \times 3^2$. Key: [36] [FACT].

Verified Answer: $2^2 \times 3^2$

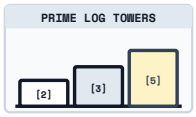
**EXAMPLE 2** **Find GCF(18, 24) and LCM(18, 24).**

$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ → Shared GCF = $2 \times 3 = 6$; Union LCM = $2^3 \times 3^2 = 72$.

Verified Answer: GCF=6, LCM=72

**1** Tier 1 • Builder

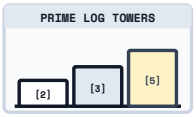
**Evaluate the power: $2^5$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

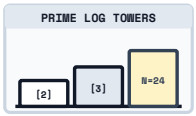
**Evaluate the power: $3^4$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

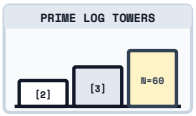
**Write the prime factorization of 24 using exponents:**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

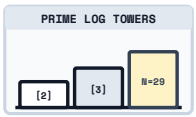
**Write the prime factorization of 60 using exponents:**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

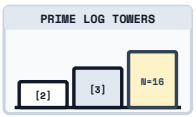
**Test if 29 has only 1 and 29 as factors: 29 ÷ 1 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

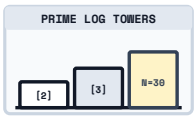
**Find the Greatest Common Factor: GCF(16, 24) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

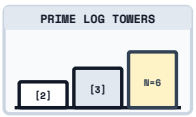
**Find the Greatest Common Factor: GCF(30, 45) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

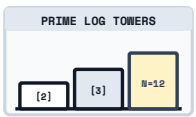
**Find the Least Common Multiple: LCM(6, 8) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

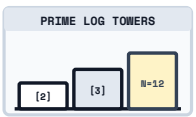
**Find the Least Common Multiple: LCM(12, 15) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Use GCF × LCM = $a \times b$: If GCF(12, 18) = 6, find LCM(12, 18):**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 07 Practice B — Factor Trees, Prime Factor Towers & Divisibility Tests

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Stack the Prime Factor Tower [2] × [2] × [3] × [5] and find the composite number.**

2 × 2 = 4, 4 × 3 = 12, 12 × 5 = 60. Key: [2] [ENTER] [2] [×] [3] [×] [5] [×].

Verified Answer: 60

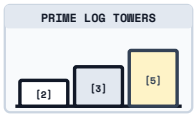
**EXAMPLE 2** **Use the digit-sum rule to test if 126 is divisible by 9, then compute 126 ÷ 9.**

Digit sum: 1 + 2 + 6 = 9 (divisible by 9!). 126 ÷ 9 = 14.

Verified Answer: 14

**1** Tier 1 • Builder

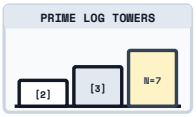
**Build from Prime Factor blocks: 2 × 3 × 5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

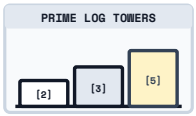
**Build from Prime Factor blocks: $2^2 \times 3 \times 7$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

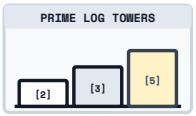
**Build from Prime Factor blocks: $2^3 \times 5$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

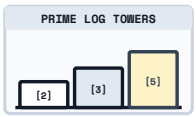
**Build from Prime Factor blocks: $3^2 \times 5$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

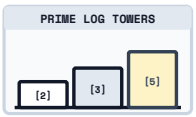
**Build from Prime Factor blocks: $2 \times 5^2$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

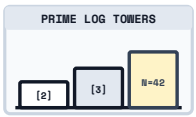
**Build from Prime Factor blocks: $2^3 \times 3^2$ =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

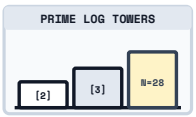
**Find the missing prime factor block: 2 × 3 × [ ? ] = 42**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

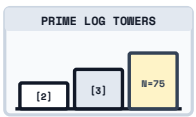
**Find the missing prime factor block: $2^2 \times$ [ ? ] = 28**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

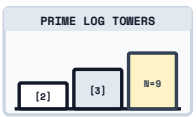
**Find the missing prime factor block: 3 × 5 × [ ? ] = 75**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Divisibility by 9 check: Since 1+3+5 = 9, evaluate 135 ÷ 9 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 07 Practice C — Hunting-Tooth Bicycle Gears, 17-Year Cicadas & GCF/LCM Stories

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Two beacons flash every 8 sec and 12 sec. When do they flash together again?

LCM(8, 12) = 24 seconds. Key: [8] [ENTER] [12] [LCM].

Verified Answer: 24 seconds

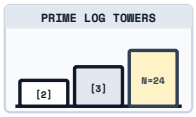
**EXAMPLE 2** Spot the Mistake: For GCF(12, 18), Leo multiplied 12 × 18 = 216.

GCF is the largest shared factor (6), while LCM is 36!

Verified Answer: 6

**1** Tier 2 • Detective

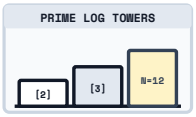
A florist has 24 roses and 36 tulips to make identical bouquets with none left over. Max bouquets (GCF)?




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

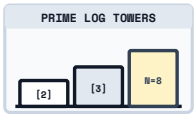
Gear A has 12 teeth and Gear B has 16 teeth. After how many tooth-meshes do the same teeth meet (LCM)?




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

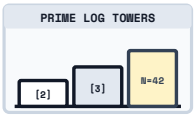
Hot dog buns come in packs of 8 and sausages in packs of 6. Min number of each to match evenly?




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

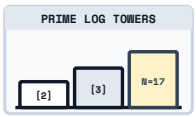
Hunting-Tooth Bicycle Gear: A 42-tooth chainring drives a 14-tooth cog. Find GCF(42, 14):




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

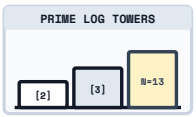
17-Year Prime Cicadas emerge every 17 years; a predator cycles every 3 years. Next overlap (LCM)?




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

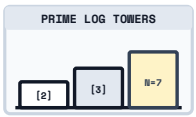
A 13-year cicada brood and a 17-year cicada brood emerge together today. Years until next overlap?




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

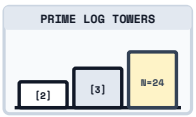
Three prime factor blocks [2], [3], and [7] are stacked into one tower. What product do they represent?




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

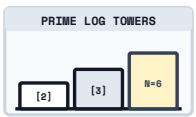
How many total positive factors does $24 = 2^3 \times 3^1$ have?




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

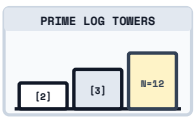
Polyrhythm Drums: Drum A beats every 6 counts and Drum B every 9 counts. When do they align (LCM)?




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

Spot the Mistake: Sam evaluated $4^3$ as $4 \times 3 = 12$. Compute the true value of $4^3$!




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 07 Practice D — Divisibility Rules, Factor Counts & Square Roots

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** How many positive factors does $18 = 2^1 \times 3^2$ have?

Add 1 to each prime exponent and multiply: $(1 + 1)(2 + 1) = 2 \times 3 = 6$ factors.

Verified Answer: 6 factors

**EXAMPLE 2** Evaluate $\sqrt{3^2 + 4^2}$ from the inside out.

$3^2 + 4^2 = 9 + 16 = 25$, then $\sqrt{25} = 5$.

Verified Answer: 5

**1** Tier 2 • Detective

Evaluate exact square root: $\sqrt{64}$ =

PRIME LOG TOWERS [2] [3] N=64

Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

Evaluate exact square root: $\sqrt{144}$ =

PRIME LOG TOWERS [2] [3] N=144

Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

Evaluate exact cube root: $\sqrt[3]{27}$ =

PRIME LOG TOWERS [2] [3] N=27

Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

Evaluate inside-out radical: $\sqrt{6^2 + 8^2}$ =

PRIME LOG TOWERS [2] [3] [5]

Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

Evaluate inside-out radical: $\sqrt{10^2 - 6^2}$ =

PRIME LOG TOWERS [2] [3] [5]

Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

How many positive factors does $12 = 2^2 \times 3^1$ have?

PRIME LOG TOWERS [2] [3] N=12

Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

How many positive factors does $36 = 2^2 \times 3^2$ have?

PRIME LOG TOWERS [2] [3] N=36

Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

What is the sum of the distinct prime factors of 30?

PRIME LOG TOWERS [2] [3] N=30

Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

Find GCF of three numbers: GCF(12, 18, 30) =

PRIME LOG TOWERS [2] [3] N=12

Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

Find LCM of three numbers: LCM(4, 6, 8) =

PRIME LOG TOWERS [2] [3] N=6

Paper Step: _____ | | SC-6 Ans: _____

# Unit 07 Practice E — The GCF × LCM Symmetry Law & Sieve of Eratosthenes

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Verify GCF(12, 18) × LCM(12, 18) = 12 × 18.**

GCF = 6, LCM = 36 → 6 × 36 = 216, and 12 × 18 = 216!

Verified Answer: 216

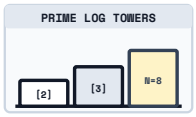
**EXAMPLE 2** **Sieve of Eratosthenes: Find the sum of the first four prime numbers (2, 3, 5, 7).**

2 + 3 + 5 + 7 = 17 (which is also prime!).

Verified Answer: 17

**1** Tier 3 • Challenge

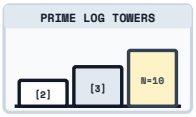
**Symmetry Check: Evaluate GCF(8, 12) × LCM(8, 12) =**




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 3 • Challenge

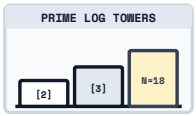
**Symmetry Check: Evaluate GCF(10, 15) × LCM(10, 15) =**




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 3 • Challenge

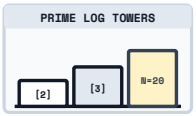
**Given GCF($a$, 18) = 6 and LCM($a$, 18) = 36, find $a$ = (6 × 36) ÷ 18:**




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 3 • Challenge

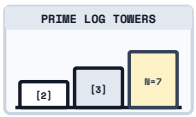
**Given GCF($a$, 20) = 4 and LCM($a$, 20) = 60, find $a$ = (4 × 60) ÷ 20:**




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 3 • Challenge

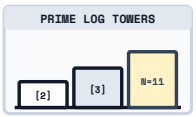
**Sieve of Eratosthenes: Sum the first five prime numbers (2 + 3 + 5 + 7 + 11):**




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 3 • Challenge

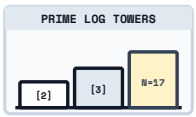
**Twin Primes differ by 2: Find the sum of the twin primes 11 + 13 =**




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 3 • Challenge

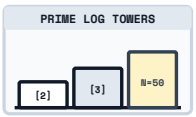
**Twin Primes differ by 2: Find the sum of the twin primes 17 + 19 =**




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 3 • Challenge

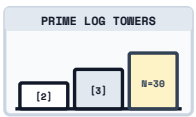
**What is the largest prime number less than 50?**




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 3 • Challenge

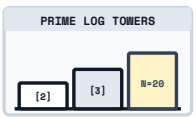
**Goldbach Conjecture: 30 = 13 + [ ? ] (find the missing prime partner):**




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 3 • Challenge

**How many prime numbers are there between 1 and 20 (2, 3, 5, 7, 11, 13, 17, 19)?**




Paper Step: _____ | | SC-6 Ans: _____

# Unit 07 Practice F — Perfect Squares from Prime Towers, GCF/LCM Puzzles & Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** What is the smallest prime block needed to turn $18 = 2^1 \times 3^2$ into a perfect square?

Multiply by [2] so every prime exponent is even: $2^2 \times 3^2 = 36 = 6^2$!

Verified Answer: 2 (makes 36)

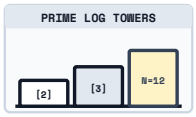
**EXAMPLE 2** Two numbers have GCF = 6 and LCM = 36. If one number is 12, find the other.

$a \times b = \text{GCF} \times \text{LCM} = 6 \times 36 = 216 \rightarrow 216 \div 12 = 18$.

Verified Answer: 18

**1** Tier 3 • Challenge

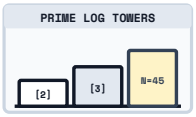
Smallest multiplier to make $12 = 2^2 \times 3^1$ a perfect square:




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

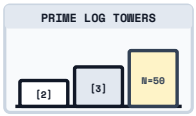
Smallest multiplier to make $45 = 3^2 \times 5^1$ a perfect square:




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

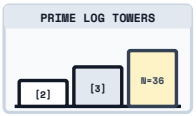
Smallest multiplier to make $50 = 2^1 \times 5^2$ a perfect square:




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

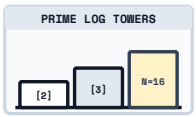
Find GCF(36, 48) using shared prime towers:




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

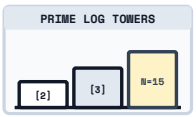
Find LCM(16, 20) using union of prime towers:




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

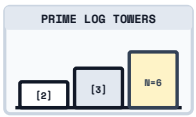
GCF × LCM identity: GCF($a$, 15) = 5 and LCM($a$, 15) = 30. Find a:




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

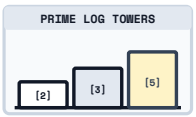
Smallest positive integer divisible by 2, 3, 4, 5, and 6 (LCM):




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

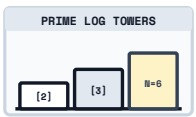
Sum of cubes (Nicomachus Theorem): $1^3 + 2^3 + 3^3 = (1 + 2 + 3)^2 =$




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

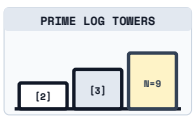
Three-planet alignment: Moons orbit in 4, 6, and 10 days. Days until all 3 align (LCM)?




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Pythagorean Tower Challenge: Find the hypotenuse of a right triangle with legs 9 and 12: $\sqrt{9^2 + 12^2} =$




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 07 Mastery Check: Prime Factors, GCF & LCM

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Write the prime factorization of 45:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Find GCF(24, 36) using shared prime factors:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Find LCM(8, 10) using union of prime factors:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Stack the Prime Factor blocks: $2^2 \times 3 \times 5 =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**How many positive factors does $18 = 2^1 \times 3^2$ have?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Given GCF($a$, 12) = 4 and LCM($a$, 12) = 24, find a:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**Bus A arrives every 10 min and Bus B every 15 min. Both arrive at 9:00 AM. Minutes until next overlap (LCM)?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**32 sixth-graders and 40 seventh-graders form equal-sized trivia teams with no mixed grades. Max team size?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**17-Year Prime Cicadas emerge every 17 years; a predator cycles every 2 years. Years until both overlap?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Sam claims 51 is prime. Divide 51 ÷ 3 on your stack to find its hidden prime factor!**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 07 STEM: Gear Train Planetary Mechanism

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: GEAR TRAIN PLANETARY MECHANISM**

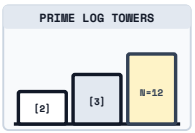
**Mechanical engineers design clockwork gear trains using prime tooth counts (2, 3, 5, 7) so wear distributes evenly across every rotation cycle.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 7: Prime Numbers, Factor Towers, [GCF] & [LCM]) • 17-Year Prime Cicadas & Hunting-Tooth Bicycle Gears: Magicicada broods emerge every 13 or 17 prime years to beat predator LCM cycles, while engineers choose prime gear teeth so wear spreads evenly!

Block Studio: Prime Factor Studio (Logarithmic Prime Blocks 2, 3, 5, 7) • Stack Shortcut: [12] [ENTER] [18] [×] [6] [÷] = 36 (LCM via GCF Formula)

**CASE INVESTIGATION 1** STEM Modeling

**Pinion Tooth Decomposition: A 12-tooth brass pinion gear is machined for a clockwork train. Stack your Prime Factor blocks to build 12:**




1. Model / Expression: _____

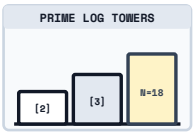
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Shared Gear Pitch (GCF): An 18-tooth drive gear meshes with a 24-tooth idler gear. Find the largest shared tooth grouping GCF(18, 24):**




1. Model / Expression: _____

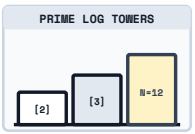
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Full Mesh Alignment Cycle (LCM): A 12-tooth gear drives an 18-tooth gear. After how many tooth-meshes do the exact same two marked teeth meet again?**




1. Model / Expression: _____

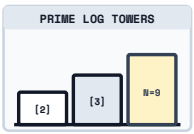
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: An apprentice claims a 9-tooth cog has a prime number of teeth so it cannot share factors. Divide 9 ÷ 3 to prove 9 is composite:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

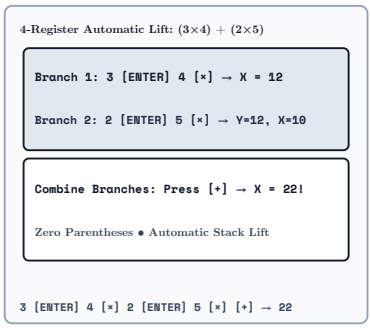
# The 4-Register Stack: Four Operations & Automatic Lift

Mathematical Focus: Expression Tree Diagram & 4-Register Stack • Standards: CCSS.MATH.5.OA.A.1, CCSS.MATH.5.OA.A.2

### 1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Multi-Step Word Problems Without Parentheses):

In a multi-step expression like (3 × 4) + (2 × 5), each product is a self-contained quantity that must be evaluated before the two groups can be added. On the 4-level stack, the first product (12) waits safely on Shelf Y while the second product (10) is formed on X, requiring zero parentheses.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 2: Stacking Numbers Just Like Blocks (Expression Trees)) • Beating the Equals-Sign Trap & The 4-Floor Elevator: Infix equations trap intermediate values inside parentheses, but the 4-story stack elevator (X, Y, Z, T) solves from the inside out—just like building with blocks!**




### 2. KEY MATHEMATICAL PRINCIPLES: EVALUATING EXPRESSIONS WITHOUT PARENTHESES

- Evaluate inside-out: always compute the deepest parenthetical sub-expression first.
- Register Y acts as the Waiting Shelf: in (15 − 6) ÷ 3, compute 15 − 6 = 9 first, then bring in 3 and divide.
- Two-branch expressions: (8 + 2) × (9 − 4) lifts Branch 1 (10) to Shelf Y while you compute Branch 2 (5).
- No parentheses keys needed: automatic stack lift manages mathematical precedence transparently.

### 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Mathematical Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display. Question: Problem: (8 - 3) × 2. Predict: 8 - 3 = 5, then 5 × 2 = 10.

**Verification Keystrokes:** [8] [ENTER] [3] [-] [2] [×]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [8] | 8 | 8 in X |
| [ENTER] | 8 | 8 copied to Y |
| [3] | 3 | 3 entered into X |
| [-] | 5 | 8 - 3 = 5 in X |
| [2] | 2 | Automatic lift! 5 lifts to Y; 2 enters X |
| [×] | 10 | 5 × 2 = 10 in X! |

**STACK SHORTCUT & PAPER STRATEGY TIP (SC-6):**

[15] [ENTER] [6] [-] [3] [÷] = 3 • Evaluate inside-out: always compute the deepest parenthetical sub-expression first.

### WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!

**Core Investigation: A baker bakes 15 cookies, sells 6 of them, and packs the remaining cookies equally into 3 bags. How many cookies go in each bag?**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



# Extension: Automatic Stack Lift & Register Retention

Deepening Concepts • Standards: CCSS.MATH.4.OA.A.3

**1. EXTENSION CONCEPT & SETUP (Automatic Stack Lift & Register Retention):**

When an inner operation finishes, its result becomes the starting quantity for the next step. On the stack, that intermediate result automatically lifts to Waiting Shelf Y as soon as the next number is typed—allowing multi-step math like (12 ÷ 4) + 7 to flow continuously from the inside out.

**Extension Setup: Observe the 4-line stack screen as values automatically lift into register Y: (12 / 4) + 7. • Question: 12 / 4 = 3. Entering 7 will lift 3 to Y. 3 + 7 = 10.**

**2. EXTENSION STACK SEQUENCE (SC-6):** [1] [2] [ENTER] [4] [÷] [7] [+]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [1] | 1 | First digit of 12 enters X |
| [2] | 12 | 12 in register X |
| [ENTER] | 12 | 12 copied to Y |
| [4] | 4 | 4 enters X |
| [÷] | 3 | 12 / 4 = 3 in X |
| [7] | 7 | 7 entered: 3 lifts to Y automatically! |
| [+] | 10 | 3 + 7 = 10! |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & STACK REMEDIATIONS**

**Misconception 1: Pressing [ENTER] after an operator (+, −, ×, ÷) before typing the next number in a chained calculation, wasting a stack register.**

Mathematical & Stack Remediation: Demonstrate Automatic Stack Lift: after [18] [ENTER] [6] [−] produces 12, typing [3] automatically lifts 12 to Register Y—no [ENTER] needed before [3] [×]!

**Misconception 2: Reversing operand order in subtraction or division (computing *X* − *Y* instead of *Y* − *X*).**

Mathematical & Stack Remediation: Remind students that Register Y is always the starting quantity on top (Row 1) and Register X is the operator's action (Row 0); if entered backward, press [*x* swap *y*] before [−] or [÷].

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [15] [ENTER] [6] [-] [3] [÷] = 3**

**Explain-Why Reflection: How does your paper diagram or stack trace prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Evaluate (24 ÷ 6) + 15 by solving the division group first and then adding 15 to that result.**

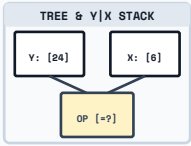
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Mathematical Scratchpad:



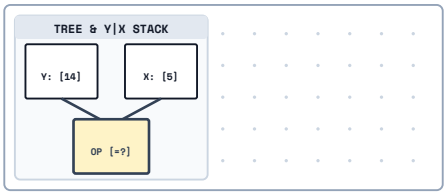
# Unit 08 Lab: The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your factor tree or grouping diagram in Box 1. (2) Show your inside-out steps in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

**Investigation 1: Solve (14 - 5) × 3 by finding the difference inside the parentheses first, then multiplying by 3.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

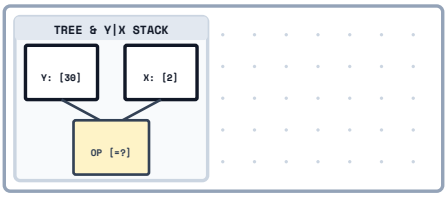
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 2: Solve 30 ÷ (2 + 3) by adding the divisor group (2 + 3) first, then dividing 30 by that sum.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

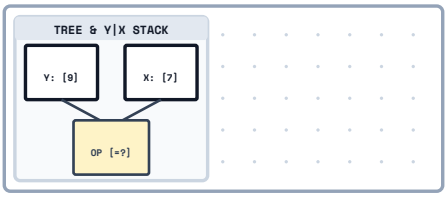
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 3: Evaluate (9 + 7) ÷ (8 - 4) by finding the numerator sum and denominator difference first, then dividing.**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

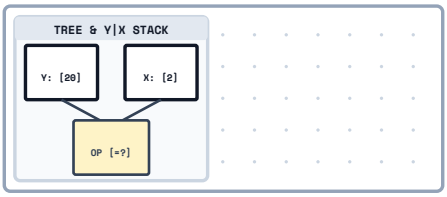
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**Investigation 4: A farmer collects 20 eggs, sets aside 2 cracked eggs, and divides the rest equally among 6 cartons. How many per carton?**

**1. Factor Tree / Grouping Sketch:**




**2. Inside-Out Evaluation Steps:**

Step 1 (Inner Group / Power):

Step 2 (Next Operation):

Final Value / Prime Factors:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your area grid or expression tree sketch show why your paper answer matched the SC-6 stack?**

# Unit 08 Practice A — Order of Operations (PEMDAS) & Inside-Out Evaluation

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Evaluate 30 - 4 × 6 on the 4-Register Stack.**

Multiply inner bundle first: 4 × 6 = 24, then 30 - 24 = 6. Key: [30] [ENTER] [4] [ENTER] [6] [×] [-].

Verified Answer: 6

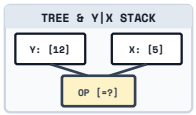
**EXAMPLE 2** **Evaluate (18 + 6) ÷ 4 using Automatic Stack Lift.**

Add parenthesis first: 18 + 6 = 24, then [4] [÷] → 6.

Verified Answer: 6

**1** Tier 1 • Builder

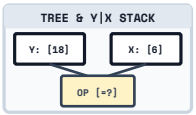
**Multiply before adding: 12 + 5 × 4 =**




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 1 • Builder

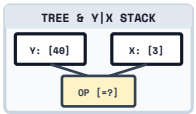
**Multiply before adding: 18 + 6 × 7 =**




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 1 • Builder

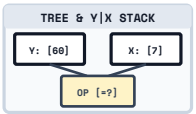
**Multiply before subtracting: 40 - 3 × 8 =**




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 1 • Builder

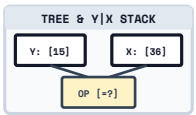
**Multiply before subtracting: 60 - 7 × 6 =**




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 1 • Builder

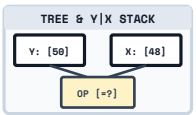
**Divide before adding: 15 + 36 ÷ 4 =**




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 1 • Builder

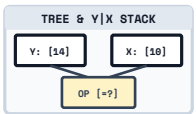
**Divide before subtracting: 50 - 48 ÷ 6 =**




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 1 • Builder

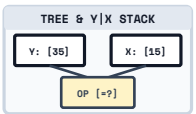
**Inside-out parentheses: (14 + 10) ÷ 3 =**




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 1 • Builder

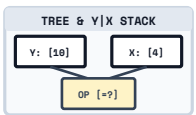
**Inside-out parentheses: (35 - 15) × 4 =**




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 1 • Builder

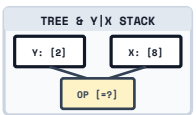
**Power before adding: 10 + $3^2$ × 4 =**




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 1 • Builder

**Power before subtracting: $5^2$ - 2 × 8 =**




Paper Step: _____ | | SC-6 Ans: _____

# Unit 08 Practice B — Automatic Stack Lift & Two-Branch Expression Trees

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Evaluate (18 - 6) × 3 using Automatic Stack Lift.**

Step 1: 18 - 6 = 12. Step 2: Type [3] [×] → 12 × 3 = 36 (no extra ENTER!).

Verified Answer: 36

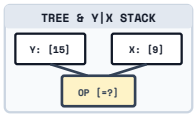
**EXAMPLE 2** **Evaluate (7 + 5) × (11 - 6) using Register Y and Register X.**

Left branch: 7 + 5 = 12 (lifts to Y). Right branch: 11 - 6 = 5 (in X). Press [×] → 60.

Verified Answer: 60

**1** Tier 1 • Builder

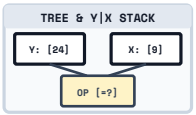
**Chained calculation: (15 + 9) ÷ 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

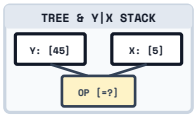
**Chained calculation: (24 - 9) × 3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

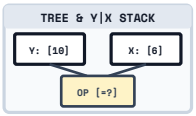
**Chained calculation: (45 ÷ 5) + 18 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

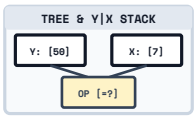
**Inside-out order of operations: 10 + 6 × 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

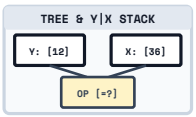
**Inside-out order of operations: 50 - 7 × 5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

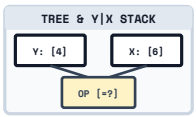
**Inside-out order of operations: 12 + 36 ÷ 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

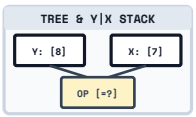
**Two-branch sum: (4 × 6) + (3 × 5) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

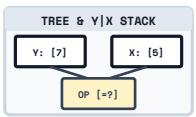
**Two-branch difference: (8 × 7) - (6 × 4) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

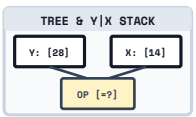
**Two-branch product: (7 + 5) × (11 - 6) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Two-branch quotient: (28 + 14) ÷ (13 - 6) =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 08 Practice C — Multi-Step Bar-Model Word Problems & Operand Order Check

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** A school buys 4 boxes of 12 markers and 15 loose markers. Total markers?

$4 \times 12 = 48$, plus $15 = 63$ markers.

Verified Answer: 63 markers

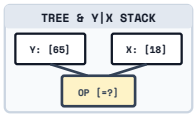
**EXAMPLE 2** Spot the Mistake: For 40 - 12, Eli typed |12| |ENTER| |40| |-| and got -28.

In *Y - X*, the starting total 40 must be in Register Y! Press $[x \rightleftarrows y]$ $[-] \rightarrow 28$.

Verified Answer: 28

**1** Tier 2 • Detective

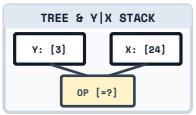
A library has 65 books on a cart. Students return 18 more, then 25 are shelved. How many left?




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

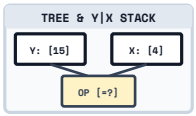
3 buses each carry 24 students, and 8 teachers ride in a van. Total people on the trip?




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

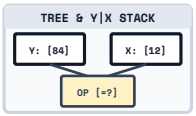
Maya saves $15 per week for 4 weeks, then spends $22 on a book. How much money left?




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

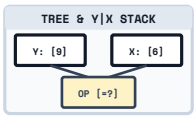
A baker bakes 84 cookies, keeps 12 for display, and packs the rest into boxes of 8. How many boxes?




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

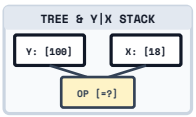
Tickets cost $9 for adults and $6 for kids. Find the cost for 3 adults and 4 kids.




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

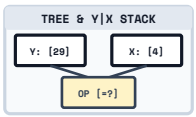
A 100-meter spool of wire has three 18-meter pieces cut from it. How much wire remains?




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

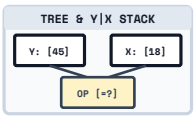
29 students are placed into lab teams of 4. How many full teams and how many left over?




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

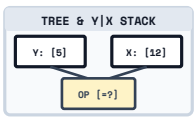
A drone climbs to 45 m, rises 18 m, then descends 27 m. What is its altitude now?




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

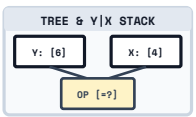
A theater has 5 rows of 12 seats in the balcony and 6 rows of 15 on the floor. Total seats?




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

Spot the Mistake: A basic phone calculator says $6 + 4 \times 5 = 50$. Find the true mathematical answer!




Paper Step: _____ | | SC-6 Ans: _____

# Unit 08 Practice D — Nested Brackets, Fraction Bars & Postfix (RPN) Translation

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Evaluate $(4^2 + 8) \div (9 - 3)$ on the 4-Register Stack.

Numerator: 16 + 8 = 24 (in Y). Denominator: 9 - 3 = 6 (in X). Divide: 24 ÷ 6 = 4.

Verified Answer: 4

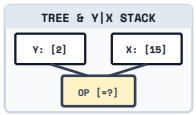
**EXAMPLE 2** Translate RPN sequence 9 [ENTER] 3 [+] 4 [ENTER] 2 [-] [×] to infix and evaluate.

(9 + 3) × (4 - 2) = 12 × 2 = 24.

Verified Answer: 24

**1** Tier 2 • Detective

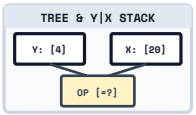
Nested parentheses: $2 \times [15 - (4 + 3)] =$




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

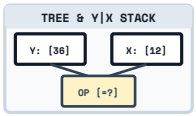
Nested parentheses: $4 \times [20 - (3 \times 5)] =$




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

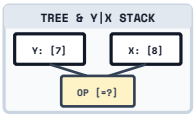
Horizontal fraction bar: $(36 - 12) \div (2 + 4) =$




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

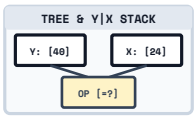
Horizontal fraction bar: $(7 \times 8 - 6) \div (2 + 3) =$




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

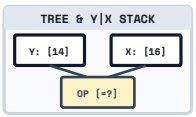
Horizontal fraction bar: $(40 + 24) \div (5^2 - 9) =$




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

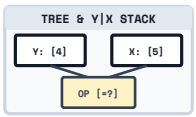
Horizontal fraction bar: $(14 + 16) \div (2^3 - 3) =$




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

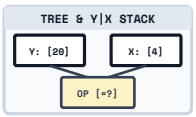
Three-branch expression: $(4 \times 5) + (6 \times 3) - 8 =$




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

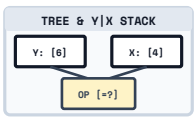
Evaluate RPN sequence: 20 [ENTER] 4 [÷] 7 [+] =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

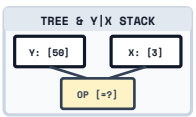
Evaluate RPN sequence: 6 [ENTER] 4 [+] 5 [ENTER] 2 [-] [×] =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

Evaluate RPN sequence: 50 [ENTER] 3 [ENTER] 6 [×] [-] =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 08 Practice E — Galileo's Penguin Huddle: 3D Surface Area & Volume on the Stack

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find the surface area of a 3×4×5 box: 2 × (3×4 + 3×5 + 4×5).**

Pairs of faces: 12 + 15 + 20 = 47; double for all 6 faces → 94 sq units!

Verified Answer: 94

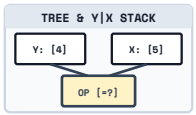
**EXAMPLE 2** **Galileo Penguin Huddle: Compare surface area of 1 solo cube (6 faces) vs 2 huddled cubes.**

2 solo cubes = 12 faces; huddled side-by-side hides 2 touching faces → 10 faces!

Verified Answer: 10 faces

**1** Tier 3 • Challenge

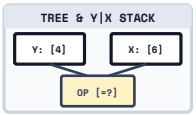
**Volume of a rectangular prism (4 cm × 5 cm × 6 cm):**




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 3 • Challenge

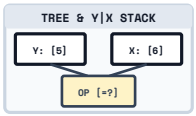
**Surface area of a cube with side 4 cm ($6 \times 4^2$):**




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 3 • Challenge

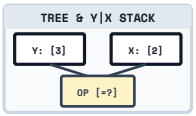
**Surface area of a cube with side 5 cm ($6 \times 5^2$):**




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 3 • Challenge

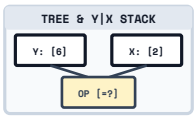
**Sum of 3 distinct face areas of a 2×3×4 box (2×3 + 2×4 + 3×4):**




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 3 • Challenge

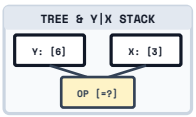
**Total 6-face surface area of a 2×3×4 box: 2 × (2×3 + 2×4 + 3×4) =**




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 3 • Challenge

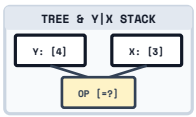
**Total 6-face surface area of a 3×4×5 box: 2 × (3×4 + 3×5 + 4×5) =**




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 3 • Challenge

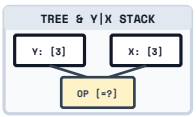
**Combined volume of a 4-cm cube and a 3-cm cube ($4^3 + 3^3$):**




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 3 • Challenge

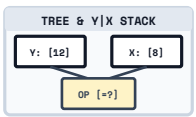
**Penguin Huddle: 3 unit cubes in a row have 3 × 6 - 4 touching faces =**




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 3 • Challenge

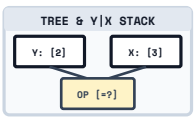
**Difference of squares on the stack: ($12^2$ - $8^2$) ÷ (12 - 8) =**




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 3 • Challenge

**Four-branch sum: (2 × 3) + (4 × 5) + (6 × 7) =**




Paper Step: _____ | | SC-6 Ans: _____

# Unit 08 Practice F — 4-Register Stack Pipelines, Non-Commutative Swap [$x \rightleftarrows y$] & Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Evaluate $120 \div (14 + 16)$ using [$x \rightleftarrows y$] on the stack.**

Add denominator first: $14 + 16 = 30$. Type 120, press [$x \rightleftarrows y$] [÷] → 4!

Verified Answer: 4

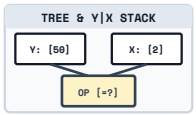
**EXAMPLE 2** **Evaluate $3 \times [20 - (2^3 + 4)]$ from the deepest bracket outward.**

Inner: $2^3 + 4 = 12$; Middle: $20 - 12 = 8$; Outer: $3 \times 8 = 24$.

Verified Answer: 24

**1** Tier 3 • Challenge

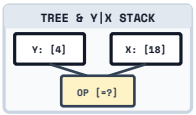
**Deepest-first bracket: $50 - 2 \times (3^2 + 6) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

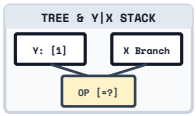
**Deepest-first bracket: $4 \times [18 - (10 - 4) \times 2] =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

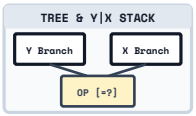
**Fraction bar with powers: $(5^2 + 3^2) \div (4^2 + 1) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

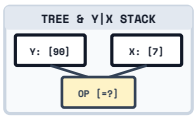
**Fraction bar with powers: $(6^2 - 2^2) \div (2^3) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

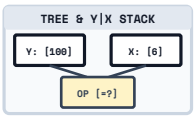
**Denominator-first division with [$x \rightleftarrows y$]: $90 \div (7 + 8) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

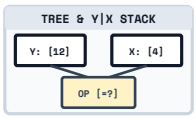
**Inner-product-first subtraction with [$x \rightleftarrows y$]: $100 - (6 \times 9) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

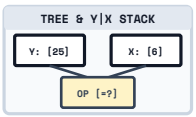
**Two-branch sum: $(12 \times 4) + (13 \times 4) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

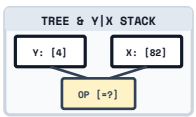
**Two-branch difference: $(25 \times 6) - (15 \times 4) =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

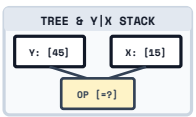
**Average of 4 test scores: $(82 + 86 + 90 + 94) \div 4 =$**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Two-Branch Expression Tree Puzzle: Evaluate $(45 + 15) \div (18 - 6)$ from the inside out on the 4-level stack:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 08 Mastery Check: Order of Operations & The Stack

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Multiply before adding: $14 + 5 \times 6 =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Divide & power order: $24 - 16 \div 4 + 2^3 =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Inside-out parentheses: $(16 + 8) \div 3 =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Two-branch expression tree: $(9 - 4) \times (6 + 2) =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Two-branch expression tree: $(12 + 8) \times (15 - 9) =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Horizontal fraction bar: $(5^2 + 11) \div (4 + 2) =$**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**A crate holds 6 rows of 8 apples, plus 12 pears. How many pieces of fruit in all?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**Noah has $50. He buys 4 notebooks at $7 each. How much change does he receive?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**A maker lab orders 4 kits at $18 each and 3 sensors at $9 each. Total cost?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Why does $(9 + 3) \div (6 - 2)$ need NO parentheses on a 4-register RPN stack?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 08 STEM: Space Elevator Cargo Transfer

Name: _____ Date: _____ Score: _____ / 4

## REAL-WORLD STEM APPLICATION & WORKSHOP: SPACE ELEVATOR CARGO TRANSFER

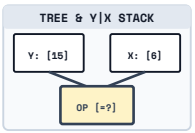
**Astrodynamicists balance orbital payload pods on a 4-Register Stack Stage without parentheses.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 2: Stacking Numbers Just Like Blocks (Expression Trees)) • Beating the Equals-Sign Trap & The 4-Floor Elevator: Infix equations trap intermediate values inside parentheses, but the 4-story stack elevator (X, Y, Z, T) solves from the inside out—just like building with blocks!

Mathematical Focus: Expression Tree Diagram & 4-Register Stack • Stack Shortcut: [15] [ENTER] [6] [-] [3] [÷] = 3

### CASE INVESTIGATION 1 — STEM Modeling

**Pod A carries 15 tons and transfers 6 tons to the station; the remaining cargo is split across 3 shuttles: (15 − 6) ÷ 3 =**




1. Model / Expression: _____

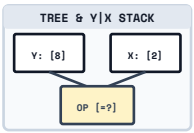
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 2 — STEM Modeling

**Two thruster banks fire together: (8 + 2) pods each burning (9 − 4) kg of fuel. Evaluate (8 + 2) × (9 − 4):**




1. Model / Expression: _____

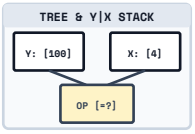
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 3 — Applied Calculation

**From a 100-ton counterweight reserve, subtract 4 racks of 5 tons plus a 10-ton tether clamp: 100 − (4 × 5 + 10) =**




1. Model / Expression: _____

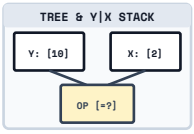
2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 4 — Error Analysis Workshop

**Error Analysis: Student executes left-to-right on 10 + 2 × 5 and gets 60.**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

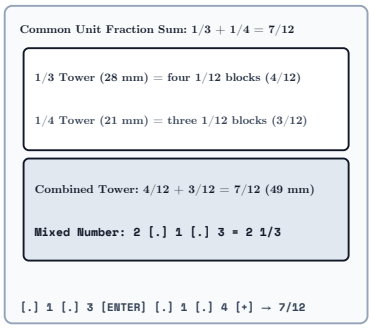
# Adding & Subtracting Fractions & Mixed Numbers

Hands-On Block Studio: Fraction Studio (84 mm Proportional Towers) • Standards: CCSS.MATH.5.NF.A.1, CCSS.MATH.5.NF.A.2

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Adding Mixed Numbers: 2 1/3 + 1 1/2 = 3 5/6):**

Both whole number and fractional components combine into an exact mixed number result.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Mixed Numbers, Improper Fractions & Regrouping) • Overflowing the 84 mm Well & Artisan Bakery Scaling: When 3/4 (63 mm) and 2/3 (56 mm) stack to 119 mm in the comparator well, they rise 35 mm (5/12) above the 84 mm Whole to show 1 5/12 instantly!**




**2. HANDS-ON BLOCKS & CONCEPT RULES: COMBINING MIXED NUMBERS ON THE STACK**

- To add mixed numbers: combine whole numbers and fractions: 2 1/3 + 1 1/2 = (2+1) + (1/3+1/2) = 3 + 5/6 = 3 5/6.
- When subtracting requires regrouping: trade 1 whole for fractional parts: 3 1/4 − 1 1/2 = 2 5/4 − 1 2/4 = 1 3/4.
- On the SC-6: Enter mixed numbers using two dots: [2] [.] [1] [.] [3] represents 2 1/3.
- Verification: Stack heights match paper arithmetic with zero rounding errors.

**3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6**

Hands-On Block Setup: Assemble Tower A: two 1-whole blocks + one 1/3 block. Tower B: one 1-whole block + one 1/2 block. Question: Wholes: 2 + 1 = 3. Fractions: 1/3 (2/6) + 1/2 (3/6) = 5/6. Total? (Prediction: 3 5/6)

**Verification Keystrokes:** [2] [•] [1] [•] [3] [ENTER] [1] [•] [1] [•] [2] [+]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [2] [•] [1] [•] [3] | 2 1/3 | 2 1/3 enters Register X |
| [ENTER] | 2 1/3 | 2 1/3 lifts to Waiting Shelf Register Y |
| [1] [•] [1] [•] [2] | 1 1/2 | 1 1/2 enters Workbench Register X |
| [+] | 3 5/6 | 2 1/3 + 1 1/2 = 3 5/6! Exact mixed number sum |

**3. 3D BLOCK STUDIO & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Physical Block Studio Rule: To add mixed numbers: combine whole numbers and fractions: 2 1/3 + 1 1/2 = (2+1) + (1/3+1/2) = 3 + 5/6 = 3 5/6.
- Stack Shortcut & Pattern: [2] [.] [1] [.] [3] [ENTER] [1] [.] [1] [.] [2] [+] = 3 5/6 — Verification: Stack heights match paper arithmetic with zero rounding errors.
- 2-Second Tray-to-Screen Audit: Confirm your physical 3D Block Studio setup matches Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

**WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Core Investigation: Add 1 3/4 + 2 1/8 on paper and verify on your calculator.**

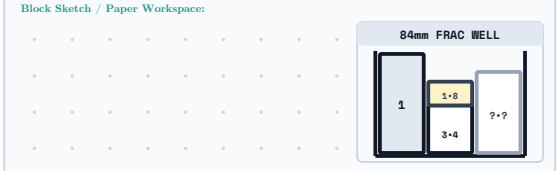
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Subtracting Mixed Numbers with Regrouping: 3 1/4 − 1 1/2

Deepening Concepts • Standards: CCSS.MATH.5.NF.A.1

**1. EXTENSION CONCEPT & SETUP (Subtracting Mixed Numbers with Regrouping: 3 1/4 − 1 1/2):**

Regrouping 1 whole into 4 fourths allows subtraction when the minuend fraction is smaller than the subtrahend.

Hands-On Extension Setup: Build Tower 3 1/4 (three wholes and one 1/4 block). • Question: 1/4 is smaller than 1/2 (2/4). Unbundle 1 whole into 4/4 → 2 5/4. 2 5/4 − 1 2/4 = 1 3/4. (Prediction: 1 3/4)

**2. EXTENSION STACK SEQUENCE (SC-6):** 3 • 1 • 4 ENTER 1 • 1 • 2 -

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| 3 • 1 • 4 | 3 1/4 | 3 1/4 enters Register X |
| ENTER | 3 1/4 | 3 1/4 lifts to Waiting Shelf Register Y |
| 1 • 1 • 2 | 1 1/2 | 1 1/2 enters Workbench Register X |
| - | 1 3/4 | 3 1/4 − 1 1/2 = 1 3/4! Regrouping verified |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1 — Subtracting Smaller Fraction from Larger Fraction Regardless of Position: In 3 1/4 − 1 1/2, student computes 1/2 − 1/4 = 1/4 and 3 − 1 = 2 → 2 1/4.**

Hands-On Block Studio Remediation: Point out that 1/4 is on top; you must regroup 1 whole from 3 to make 2 5/4!

**Misconception 2: Adding Wholes but Forgetting Common Denominators for Fractions**

Hands-On Block Studio Remediation: Always find the common denominator (like 4 or 6) before adding or subtracting.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

Calculator Shortcut: [2] [.] [1] [.] [3] [ENTER] [1] [.] [1] [.] [2] [+] = 3 5/6

Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Solve 4 1/3 − 2 1/2 on paper with regrouping and check on the stack.**

1. First Quantity / Inner Step: _____

2. Second Quantity / Operation: _____

3. My Paper Prediction: _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



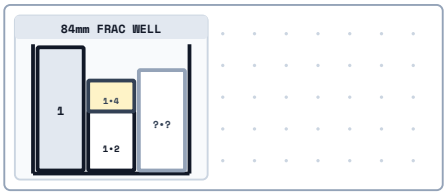
# Unit 09 Lab: Adding & Subtracting Fractions & Mixed Numbers

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your fraction bar or ratio model in Box 1. (2) Show your fraction/decimal math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

### Investigation 1: Add mixed numbers: 1 1/2 + 2 1/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 2: Subtract mixed numbers: 3 1/2 − 1 1/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

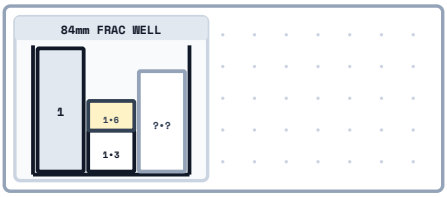
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 3: Add mixed numbers with unlike denominators: 2 1/3 + 1 1/6 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

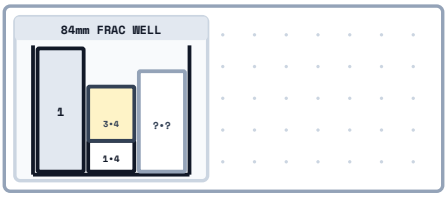
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 4: Subtract with regrouping: 4 1/4 − 1 3/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 09 Practice A — Adding Mixed Numbers (Like & Related Denominators)

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Add 1 1/4 + 2 1/4 on the stack.**

Wholes: 1+2=3; Fractions: 1/4+1/4=1/2 → 3 1/2.

Verified Answer: 3 1/2

**EXAMPLE 2** **Add 2 1/2 + 1 1/4 on the stack.**

2 2/4 + 1 1/4 = 3 3/4. Key: [2.1.2] [ENTER] [1.1.4] [+].

Verified Answer: 3 3/4

**1** Tier 1 • Builder

**Add mixed numbers: 1 1/3 + 2 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Add mixed numbers: 1 1/4 + 2 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

**Add mixed numbers: 2 1/6 + 1 3/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

**Add mixed numbers: 1 1/2 + 2 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

**Add mixed numbers: 2 1/3 + 1 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

**Add mixed numbers: 3 1/4 + 1 3/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Add mixed numbers: 1 3/4 + 2 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

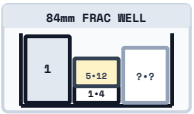
**Add mixed numbers: 2 1/2 + 3 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Add mixed numbers: 4 1/4 + 1 5/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Add mixed numbers: 2 3/10 + 1 1/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 09 Practice B — Adding Mixed Numbers (Unlike Denominators & Regrouping)

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Add 2 1/3 + 1 1/2 on the stack.

2 2/6 + 1 3/6 = 3 5/6. Key: [2.1.3] [ENTER] [1.1.2] [+].

Verified Answer: 3 5/6

**EXAMPLE 2** Add 1 3/4 + 2 1/2 (fractions exceed 1 Whole).

1 3/4 + 2 2/4 = 3 5/4 = 4 1/4.

Verified Answer: 4 1/4

**1** Tier 1 • Builder

Add unlike mixed numbers: 2 1/3 + 1 1/2 =




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

Add unlike mixed numbers: 1 1/3 + 2 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

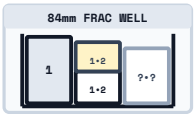
Add unlike mixed numbers: 3 2/3 + 1 1/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

Add with fraction carry: 1 1/2 + 2 1/2 =




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

Add with fraction carry: 1 3/4 + 2 1/2 =




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

Add with fraction carry: 2 2/3 + 1 1/2 =




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

Add with fraction carry: 2 2/3 + 1 3/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

Add with fraction carry: 3 5/6 + 1 1/3 =




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

Add with fraction carry: 2 5/8 + 1 3/4 =




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

Add with fraction carry: 4 3/4 + 2 5/6 =




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 09 Practice C — Subtracting Mixed Numbers (No Regrouping Needed)

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Subtract 3 1/2 − 1 1/4.**

3 2/4 − 1 1/4 = 2 1/4. Key: [3.1.2] [ENTER] [1.1.4] [-].

Verified Answer: 2 1/4

**EXAMPLE 2** **Subtract 4 2/3 − 1 1/4.**

4 8/12 − 1 3/12 = 3 5/12.

Verified Answer: 3 5/12

**1** Tier 2 • Detective

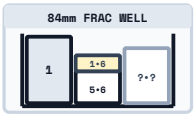
**Subtract mixed numbers: 3 3/4 − 1 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**Subtract mixed numbers: 4 5/6 − 2 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**Subtract mixed numbers: 3 1/2 − 1 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

**Subtract mixed numbers: 4 5/6 − 1 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**Subtract mixed numbers: 5 7/8 − 2 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

**Subtract mixed numbers: 3 1/2 − 1 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

**Subtract mixed numbers: 4 2/3 − 1 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**Subtract mixed numbers: 5 3/4 − 2 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

**Subtract mixed numbers: 6 5/6 − 3 1/2 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Subtract mixed numbers: 4 4/5 − 1 1/2 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 09 Practice D — Subtracting Mixed Numbers WITH Regrouping (Unbundling 1 Whole)

Name: _____ Date: _____ Score: _____ / 10

TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST

**EXAMPLE 1** Subtract 3 1/4 − 1 1/2 by unbundling 1 Whole.

3 1/4 = 2 5/4; 2 5/4 − 1 2/4 = 1 3/4.

Verified Answer: 1 3/4

**EXAMPLE 2** Subtract 4 1/3 − 2 1/2 with regrouping.

4 2/6 = 3 8/6; 3 8/6 − 2 3/6 = 1 5/6.

Verified Answer: 1 5/6

**1** Tier 2 • Detective

Subtract from whole number: 3 − 1 1/4 =




Paper Step: _____ | | SC-6 Ans: _____

**2** Tier 2 • Detective

Subtract from whole number: 5 − 2 2/3 =




Paper Step: _____ | | SC-6 Ans: _____

**3** Tier 2 • Detective

Subtract with regrouping: 3 1/4 − 1 3/4 =




Paper Step: _____ | | SC-6 Ans: _____

**4** Tier 2 • Detective

Subtract with regrouping: 4 1/3 − 1 2/3 =




Paper Step: _____ | | SC-6 Ans: _____

**5** Tier 2 • Detective

Subtract with regrouping: 3 1/4 − 1 1/2 =




Paper Step: _____ | | SC-6 Ans: _____

**6** Tier 2 • Detective

Subtract with regrouping: 4 1/3 − 2 1/2 =




Paper Step: _____ | | SC-6 Ans: _____

**7** Tier 2 • Detective

Subtract with regrouping: 5 1/6 − 2 1/2 =




Paper Step: _____ | | SC-6 Ans: _____

**8** Tier 2 • Detective

Subtract with regrouping: 4 1/4 − 1 2/3 =




Paper Step: _____ | | SC-6 Ans: _____

**9** Tier 2 • Detective

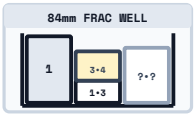
Subtract with regrouping: 6 1/8 − 2 1/2 =




Paper Step: _____ | | SC-6 Ans: _____

**10** Tier 2 • Detective

Subtract with regrouping: 5 1/3 − 2 3/4 =




Paper Step: _____ | | SC-6 Ans: _____

# Unit 09 Practice E — Artisan Bakery, Carpentry & Measurement Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** A chef mixes 2 1/3 cups rye and 1 1/2 cups wheat flour.

2 1/3 + 1 1/2 = 3 5/6 cups.

Verified Answer: 3 5/6 cups

**EXAMPLE 2** A 5-foot board has 2 3/8 feet sawed off.

5 − 2 3/8 = 2 5/8 feet.

Verified Answer: 2 5/8 ft

**1** Tier 3 • Challenge

A bread recipe uses 2 1/4 cups bread flour and 1 1/2 cups whole wheat. Total flour?




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

A hiker walks 3 1/3 miles in the morning and 2 1/2 miles in the afternoon. Total miles?




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

A carpenter has a 6-foot oak plank and cuts off 2 3/4 feet. How much plank remains?




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

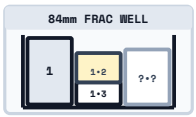
A puppy weighed 4 1/4 lb in May and gained 2 3/8 lb in June. New weight?




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

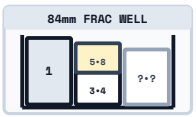
A water jug holds 4 1/3 liters. Athletes drink 1 1/2 liters. How much remains?




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

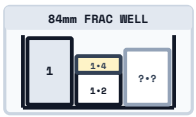
Two rainstorms drop 1 3/4 inches and 2 5/8 inches of rain. Total rainfall?




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

A picture frame needs two sides of 3 1/2 ft and two of 2 1/4 ft. Find its perimeter:




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

How much taller is a 5 1/4 ft fence than a 3 2/3 ft gate?




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

A spool has 8 yards of ribbon. After using 2 1/3 yd and 1 1/2 yd, how much is left?




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

Spot the Mistake: For 3 1/4 − 1 1/2, Ben wrote 2 1/4 instead of regrouping. Find 3 1/4 − 1 1/2:




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 09 Practice F — Multi-Step Mixed Number Pipelines & Mastery

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Evaluate 2 1/2 + 1 3/4 − 1 1/3 on the stack.

2 1/2 + 1 3/4 = 4 1/4; 4 1/4 − 1 1/3 = 2 11/12.

Verified Answer: 2 11/12

**EXAMPLE 2** Find the missing mixed number: 1 2/3 + [ ? ] = 4 1/4.

4 1/4 − 1 2/3 = 2 7/12.

Verified Answer: 2 7/12

**1** Tier 3 • Challenge

**Multi-step pipeline: 1 1/2 + 2 1/4 + 1 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Multi-step pipeline: 2 1/3 + 1 1/2 + 1 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

**Add then subtract: 3 1/2 + 1 1/4 − 2 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

**Add then subtract: 2 1/2 + 1 3/4 − 1 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

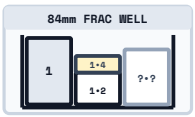
**Subtract then add: 5 1/4 − 1 1/2 + 2 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

**Two-branch difference: (3 1/2 + 1 1/4) − (1 1/2 + 3/4) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

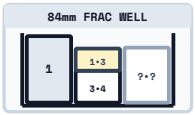
**Find the missing addend: 2 1/3 + [ ? ] = 5 1/6**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

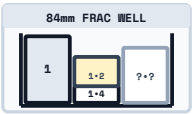
**Find the missing addend: 1 3/4 + [ ? ] = 4 1/3**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

**Find the missing subtrahend: 6 1/4 − [ ? ] = 2 1/2**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Mixed-Number Tower Transfer: Start with 4 1/3 towers, remove 1 1/2 towers, then add 2 1/6 towers:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 09 Mastery Check: Adding & Subtracting Fractions

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Add mixed numbers: 1 1/2 + 2 1/4 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Add unlike mixed numbers: 2 1/3 + 1 1/2 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Add with fraction carry: 1 3/4 + 2 1/2 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Subtract mixed numbers: 4 5/6 − 1 1/3 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Subtract from a whole number: 4 − 1 3/8 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Subtract with regrouping: 3 1/4 − 1 1/2 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**Subtract with regrouping: 4 1/3 − 2 1/2 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**A baker combines 2 2/3 cups oats and 1 3/4 cups flour. Total dry ingredients?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Evaluate multi-step pipeline: 3 1/2 + 1 1/3 − 2 1/6 =**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Analysis: Mia wrote 4 1/4 − 1 3/4 = 3 2/4. Find the true difference:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Unit 09 STEM: The Chef's Artisan Bread Recipe

Name: _____ Date: _____ Score: _____ / 4

## REAL-WORLD STEM APPLICATION & WORKSHOP: THE CHEF'S ARTISAN BREAD RECIPE

**A master baker combines sourdough flours across multiple dough troughs.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Mixed Numbers, Improper Fractions & Regrouping) • Overflowing the 84 mm Well & Artisan Bakery Scaling: When 3/4 (63 mm) and 2/3 (56 mm) stack to 119 mm in the comparator well, they rise 35 mm (5/12) above the 84 mm Whole to show 1 5/12 instantly!

Block Studio: Fraction Studio (84 mm Proportional Towers) • Stack Shortcut: [2] [.] [1] [.] [3] [ENTER] [1] [.] [1] [.] [2] [+] = 3 5/6

### CASE INVESTIGATION 1

STEM Modeling

**Multi-Grain Trough Blend: The baker blends 2 1/3 cups of dark rye flour with 1 1/2 cups of whole wheat flour. Find the combined flour volume:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 2

STEM Modeling

**Levain Starter Portioning: A proofing tub holds 3 1/4 cups of sourdough starter. After scooping out 1 1/2 cups for baguettes, how much starter remains?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 3

Applied Calculation

**Brioche Hydration Mix: Combine 1 3/8 cups of warm buttermilk and 2 1/4 cups of whole milk in the mixer bowl. What is the total liquid volume?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### CASE INVESTIGATION 4

Error Analysis Workshop

**Error Analysis: Subtracting 1 1/2 cups of flour from 3 1/4 cups, an apprentice subtracted 1/2 − 1/4 backward and wrote 2 1/4 cups. Trade 1 Whole for 4/4 to fix it:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

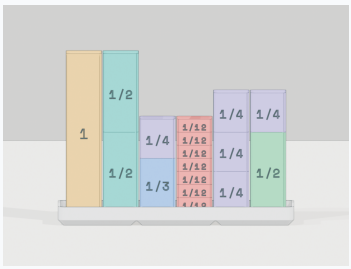
# Fraction Studio: Proportions, Multiplication & Tower Division

Hands-On Block Studio: Fraction Studio (84 mm Proportional Towers) • Standards: CCSS.MATH.5.NF.B.4, CCSS.MATH.5.NF.B.7

**1. MATHEMATICAL CONCEPT & INSIDE-OUT APPROACH (Scaling a Fraction Tower: 1/2 of 3/4 = 3/8):**

Multiplying numerators and denominators (1×3)/(2×4) = 3/8 scales the 3/4 tower to half its height.

**Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fraction Scaling & Split-and-Rebuild Division) • Pythagoras's Musical Strings & Split-and-Rebuild Towers: Pythagoras found that 1/2, 2/3, and 3/4 string lengths create pure musical chords—and splitting and rebuilding 84 mm towers proves fraction division by hand!**




Fraction Studio • 84 mm Flush Towers

**2. HANDS-ON BLOCKS & CONCEPT RULES: SCALING AND DIVIDING WITH PHYSICAL FRACTION TOWERS**

- Multiplying fractions scales a quantity: 1/2 of 3/4 means taking half of the 3/4 tower → 3/8.
- Dividing by a Whole Number: 1/2 ÷ 3 splits the 1/2 block (42 mm) into 3 equal pieces → each piece is 1/6 (14 mm).
- Dividing by a Fraction (Split & Rebuild 1 Full Tower): Just as A ÷ 3 splits tower A into 3 parts to find 1 full tower, A ÷ (c/d) splits tower A into c equal parts (÷ c) and stacks d of them (× d) to build 1 full mystery tower (pooling blocks across a 4-kit table pod when needed).
- On the SC-6: Type [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷] = 2/3 (56.0 mm tower height).

## 3. STEP-BY-STEP VERIFICATION WALKTHROUGH ON THE SC-6

Hands-On Block Setup: Stand three 1/4 blocks (63 mm = 3/4) in the comparator well. Find half of that height (31.5 mm). Question: Half of each 1/4 block is 1/8; three 1/8 blocks = 3/8 (31.5 mm). (Prediction: 3/8)

**Verification Keystrokes:** [•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [×]

| Button Pressed | Screen (X) | How the 4-Level Stack Mirrors Your Mathematical Steps (T, Z, Y, X) |
|---|---|---|
| [•] [1] [•] [2] | 1/2 | 1/2 enters Register X |
| [ENTER] | 1/2 | 1/2 lifts to Waiting Shelf Register Y |
| [•] [3] [•] [4] | 3/4 | 3/4 enters Workbench Register X |
| [×] | 3/8 | 1/2 × 3/4 = 3/8! (31.5 mm height verified) |

**3. 3D BLOCK STUDIO & SC-6 STACK LEARNING GUIDE & SELF-CHECK**

- Physical Block Studio Rule: Multiplying fractions scales a quantity: 1/2 of 3/4 means taking half of the 3/4 tower → 3/8.
- Stack Shortcut & Pattern: [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷] = 2/3 (56 mm tower) — On the SC-6: Type [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷] = 2/3 (56.0 mm tower height).
- 2-Second Tray-to-Screen Audit: Confirm your physical 3D Block Studio setup matches Register Y (Waiting Shelf) and Register X (Workbench) before pressing the operator key.

**WARM-UP CHECK A: SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Core Investigation: Find 2/3 of 3/4 by multiplying (2/3) × (3/4) and simplifying to lowest terms.**

**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



# Extension: Dividing by a Fraction: Split & Rebuild 1 Full Tower (1/2 ÷ 3/4 = 2/3)

Deepening Concepts • Standards: CCSS.MATH.5.NF.B.7

**1. EXTENSION CONCEPT & SETUP (Dividing by a Fraction: Split & Rebuild 1 Full Tower (1/2 ÷ 3/4 = 2/3)):**

Dividing by 3/4 splits the dividend by 3 (÷3) and scales up by 4 (×4), proving physically why ÷(3/4) = ×(4/3)!

**Hands-On Extension Setup: Pool Fraction Studio blocks across a 2-kit pair. Place the 1/2 block (42 mm) as 3/4 of a mystery tower. • Question: Split 1/2 (42 mm) into 3 equal parts (÷3) → each is 1/6 (14 mm). Stack 4 of those 1/6 blocks (×4) → 4/6 = 2/3 (56 mm)!**

**2. EXTENSION STACK SEQUENCE (SC-6):** [•] [1] [•] [2] [ENTER] [•] [3] [•] [4] [÷]

| Button Pressed | Screen (X) | Extension Stack Dynamics (T, Z, Y, X) |
|---|---|---|
| [•] [1] [•] [2] | 1/2 | 1/2 enters Register X (42 mm) |
| [ENTER] | 1/2 | 1/2 lifts to Waiting Shelf Register Y |
| [•] [3] [•] [4] | 3/4 | Divisor 3/4 enters Workbench Register X |
| [÷] | 2/3 | 1/2 ÷ 3/4 = 2/3! (56 mm full mystery tower) |

**3. PARENT & TEACHER COACHING: TARGET MISCONCEPTIONS & 3D BLOCK STUDIO REMEDIATIONS**

**Misconception 1: Adding numerators and denominators directly (e.g., 1/2 + 1/4 = 2/6) or treating 'invert and multiply' for fraction division as an arbitrary rule.**

Hands-On Block Studio Remediation: In the comparator lid, 1/2 (42 mm) + 1/4 (21 mm) matches three 1/4 blocks (63 mm), not two 1/6 blocks (28 mm). For division (e.g., 1/2 ÷ 3/4 = 2/3), have a 4-kit table pod combine blocks to Split & Rebuild 1 Full Tower: split 1/2 (42 mm) into three 1/6 blocks (÷3 → 14 mm is 1/4 of the mystery tower), then stack four 1/6 blocks (×4 → 56 mm = two 1/3 blocks) to prove ÷(3/4) = ×(4/3).

**Misconception 2: Entering a proper fraction like 3/4 as [3] [.] [4] (which is decimal 3.4) instead of using Two-Dot fraction notation [.][3][.][4].**

Hands-On Block Studio Remediation: Teach the rule: 'First dot = whole number zero, second dot = fraction bar.' Typing [.][3][.][4] displays exact 3/4 on the stack.

**4. STACK SHORTCUT & PARTNER EXPLAIN-WHY CHALLENGE (SC-6)**

**Calculator Shortcut: [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷] = 2/3 (56 mm tower)**

**Explain-Why Reflection: How do your Place Value, Fraction, or Prime blocks prove why Misconception 1 is wrong?**

**WARM-UP CHECK B (EXTENSION): SOLVE ON PAPER FIRST, THEN VERIFY ON YOUR CALCULATOR!**

**Extension Challenge: Use Split & Rebuild to evaluate (3/4) ÷ (3/8) on paper and check on your SC-6.**

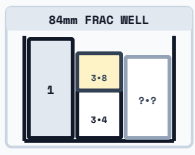
**1. First Quantity / Inner Step:** _____

**2. Second Quantity / Operation:** _____

**3. My Paper Prediction:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

Block Sketch / Paper Workspace:



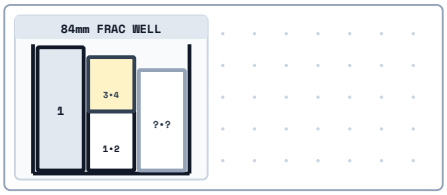
# Unit 10 Lab: Fraction Studio: Proportions, Multiplication & Tower Division

Name: _____ Date: _____ Score: _____ / 4

**DIRECTIONS: (1) Sketch your fraction bar or ratio model in Box 1. (2) Show your fraction/decimal math in Box 2.**
(3) Write your Prediction BEFORE pressing keys, then verify your answer on the SC-6!

### Investigation 1: Multiply fraction scaling: 1/2 × 3/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

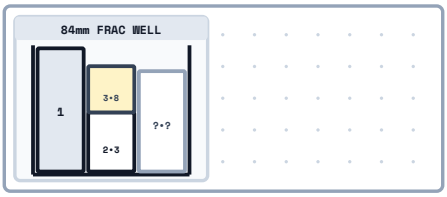
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 2: Multiply fractions and simplify: 2/3 × 3/8 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

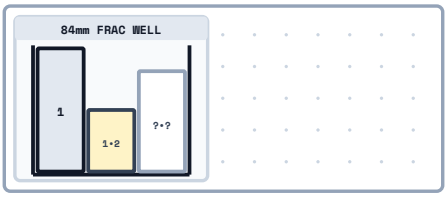
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 3: Divide by a whole number: 1/2 ÷ 3 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

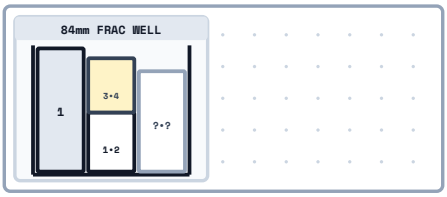
**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

### Investigation 4: Divide by a fraction (Split & Rebuild): 1/2 ÷ 3/4 =

**1. Fraction Bar / Ratio Sketch:**




**2. Equivalent Fraction / Decimal Work:**

Common Unit / Denominator:

Operation Step:

Simplified Answer:

**3. My Prediction:**

**Final Verified Answer:**

**[ ] Verified on SC-6 Stack**

**EXIT REFLECTION: How did your Place Value, Fraction, or Prime blocks show why your paper answer matched the SC-6 stack?**

# Unit 10 Practice A — Multiplying Fractions by Whole Numbers & Fractions of a Set

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE A (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Stack 3 blocks of 1/4 in the 84 mm Well: 3 × 1/4.**

Three 21 mm blocks reach 63 mm = 3/4. Key: [3] [ENTER] [.1.4] [×].

Verified Answer: 3/4

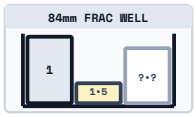
**EXAMPLE 2** **Find 3/4 of a 20-student choir using a 4-part Bar Model.**

1 unit = 20 ÷ 4 = 5 students; 3 units = 3 × 5 = 15 students!

Verified Answer: 15

**1** Tier 1 • Builder

**Stack unit fractions: 3 × 1/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Stack unit fractions: 5 × 1/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

**Multiply whole number by fraction: 4 × 2/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

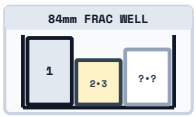
**Multiply whole number by fraction: 6 × 3/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

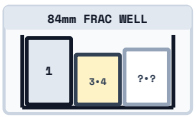
**Find the fractional part of a bar model: 2/3 of 18 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

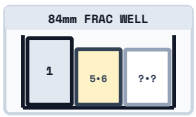
**Find the fractional part of a bar model: 3/4 of 28 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Find the fractional part of a bar model: 5/6 of 42 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

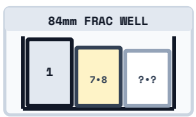
**Find the fractional part of a bar model: 3/5 of 45 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Find the fractional part of a bar model: 7/8 of 32 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Find the fractional part of a bar model: 4/9 of 36 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Practice B — Fraction × Fraction Scaling & Tower Division (Split & Rebuild)

Name: _____ Date: _____ Score: _____ / 10

**TIER 1 • LEVEL 1: BUILDER FOUNDATION • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE B (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find 1/2 of a 2/3 tower in the 84 mm Well: 1/2 × 2/3.**

Half of two 1/3 blocks is one 1/3 block (28 mm)! Key: [.1.2] [ENTER] [.2.3] [×].

Verified Answer: 1/3

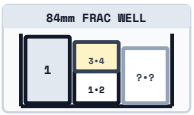
**EXAMPLE 2** **Divide 1/2 ÷ 3/4 (Split & Rebuild 1 Full Tower in the 84 mm Well).**

1/2 (42mm) is 3/4 of tower: split ÷3 → 1/6 (14mm), stack ×4 → 4/6 = 2/3 (56mm)!

Verified Answer: 2/3

**1** Tier 1 • Builder

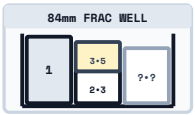
**Fraction of a fraction: 1/2 × 3/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 1 • Builder

**Fraction of a fraction: 2/3 × 3/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 1 • Builder

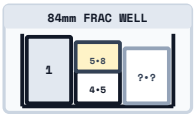
**Fraction of a fraction: 3/4 × 2/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 1 • Builder

**Fraction of a fraction: 4/5 × 5/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 1 • Builder

**Fraction of a fraction: 3/4 × 4/9 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 1 • Builder

**Divide whole number by unit fraction: 3 ÷ 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 1 • Builder

**Divide whole number by unit fraction: 4 ÷ 1/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 1 • Builder

**Split a unit fraction into equal parts: 1/3 ÷ 2 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 1 • Builder

**Tower Division in the 84 mm Well: 3/4 ÷ 3/8 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 1 • Builder

**Tower Division in the 84 mm Well: 5/6 ÷ 5/12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Practice C — Pythagoras's Musical Strings & Fractional Bar-Model Word Problems

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE C (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** Pythagoras stops a 36 cm harp string at 2/3 of its length (Perfect Fifth). Length?

2/3 × 36 = 24 cm. Key: [.2.3] [ENTER] [36] [×].

Verified Answer: 24 cm

**EXAMPLE 2** Spot the Mistake: Sam computed 2/3 × 3/4 by adding 2+3=5 and 3+4=7 to get 5/7.

Multiply numerators and denominators (or cancel 3s): (2×3)/(3×4) = 6/12 = 1/2!

Verified Answer: 1/2

**1** Tier 2 • Detective

**Pythagoras's Octave: Stop a 48 cm monochord string at 1/2 of its length. Vibrating length?**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

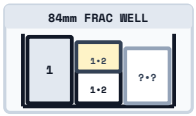
**Pythagoras's Perfect Fourth: Stop a 36 cm string at 3/4 of its length. Vibrating length?**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

**Two-Octave Halving: What fraction of a string vibrates if you take 1/2 of 1/2?**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

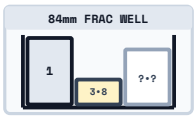
**A 6-cup pitcher of lemonade is poured into 3/4-cup glasses. How many full glasses?**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

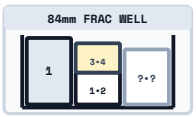
**In a class of 24 students, 3/8 play a string instrument. How many play strings?**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

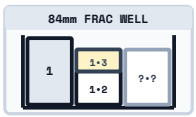
**A baker has 4 1/2 lb of dough and makes loaves using 3/4 lb each. How many loaves?**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

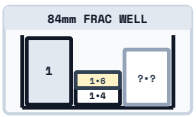
**A 3D printer uses 1.25 meters of filament per gear. How much filament for 6 gears?**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

**Which is taller in the 84 mm Fraction Comparator: three 1/4 blocks or four 1/6 blocks?**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

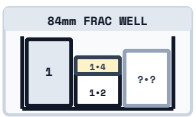
**Write the Two-Dot SC-6 keystroke sequence to enter the proper fraction 5/8:**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Spot the Mistake: Ava divided 1/2 ÷ 1/4 by multiplying 1/2 × 1/4 = 1/8. Find 1/2 ÷ 1/4!**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Practice D — Fractions of a Whole, Ratios & Percent Benchmarks

Name: _____ Date: _____ Score: _____ / 10

**TIER 2 • LEVEL 2: STORY & DETECTIVE LAB • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE D (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Find 3/5 of 40.**

Divide by denominator first: 40 ÷ 5 = 8, then multiply by 3 → 24.

Verified Answer: 24

**EXAMPLE 2** **Convert 3/8 to an exact decimal and percent.**

3 ÷ 8 = 0.375 = 37.5%.

Verified Answer: 0.375 (37.5%)

**1** Tier 2 • Detective

**Find the fractional part: 2/3 of 24 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 2 • Detective

**Find the fractional part: 3/4 of 36 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 2 • Detective

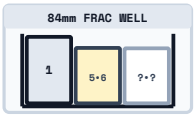
**Find the fractional part: 4/5 of 35 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 2 • Detective

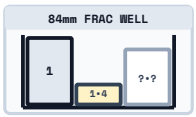
**Find the fractional part: 5/6 of 30 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 2 • Detective

**Convert fraction to percent: 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 2 • Detective

**Convert fraction to percent: 3/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 2 • Detective

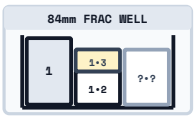
**Convert fraction to percent: 7/10 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 2 • Detective

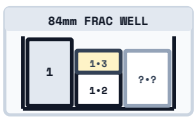
**Find 25% of 80:**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 2 • Detective

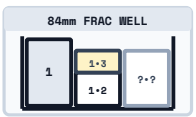
**Find 10% of 150:**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 2 • Detective

**Unit rate: 180 miles in 3 hours = how many miles per hour?**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Practice E — Multi-Step Fraction Expressions, Complex Fractions & Scaling

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE E (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **Evaluate (1/2 + 1/3) × 12.**

Distribute 12: 12(1/2) + 12(1/3) = 6 + 4 = 10!

Verified Answer: 10

**EXAMPLE 2** **Evaluate (3/4) ÷ (1/2) (Table Pod: 3/4 is 1/2 of full tower).**

Double three 1/4 blocks (63mm × 2 = 126mm): six 1/4 blocks = 1 1/2!

Verified Answer: 1 1/2

**1** Tier 3 • Challenge

**Clear denominators by distributing: (1/3 + 1/4) × 12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**Clear denominators by distributing: (1/2 + 1/6) × 12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

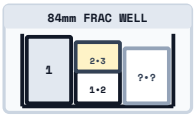
**Clear denominators by distributing: (3/4 - 1/3) × 12 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

**Telescoping fraction product: 1/2 × 2/3 × 3/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

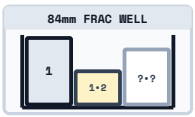
**Telescoping fraction product: 2/3 × 3/4 × 4/5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

**Mixed number multiplication: 2 1/2 × 4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

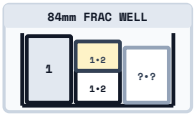
**Mixed number division: 4 1/2 ÷ 1 1/2 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

**Reciprocal sum: 1 ÷ (1/2 + 1/2) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

**Reciprocal sum: 1 ÷ (1/3 + 1/6) =**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Combine fraction and decimal: 3/4 + 0.5 =**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Practice F — 84 mm Fraction Tower Design Challenges & Proportional Reasoning

Name: _____ Date: _____ Score: _____ / 10

**TIER 3 • LEVEL 3: BOSS CHALLENGE • STUDY THESE 2 WORKED EXAMPLES BEFORE SOLVING PRACTICE F (#1–#10) ON PAPER FIRST**

**EXAMPLE 1** **If 1 Whole = 84 mm, find the combined height of 1/2 + 1/3 + 1/6 in mm.**

42 mm + 28 mm + 14 mm = 84 mm (1 full Whole!).

Verified Answer: 84 mm (1)

**EXAMPLE 2** **A recipe for 4 people uses 2/3 cup of sugar. How much for 12 people?**

12 is 3 × 4, so multiply 3 × (2/3) = 2 cups.

Verified Answer: 2 cups

**1** Tier 3 • Challenge

**84 mm Tower Height: How tall is a 3/4 tower in millimeters?**




Paper Step: _____ [ ] SC-6 Ans: _____

**2** Tier 3 • Challenge

**84 mm Tower Height: How tall is a 5/6 tower in millimeters?**




Paper Step: _____ [ ] SC-6 Ans: _____

**3** Tier 3 • Challenge

**Flush Tower Check: 1/2 + 1/3 + [ ? ] = 1 Whole**




Paper Step: _____ [ ] SC-6 Ans: _____

**4** Tier 3 • Challenge

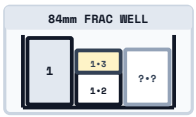
**Flush Tower Check: 1/2 + 1/4 + 1/8 + [ ? ] = 1 Whole**




Paper Step: _____ [ ] SC-6 Ans: _____

**5** Tier 3 • Challenge

**Evaluate three-fraction sum: 1/2 + 1/3 + 1/6 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**6** Tier 3 • Challenge

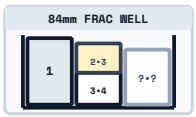
**Evaluate mixed number sum: 1 3/4 + 2 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**7** Tier 3 • Challenge

**Evaluate fraction of a fraction: 3/4 of 2/3 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**8** Tier 3 • Challenge

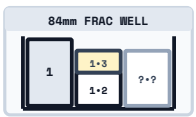
**Divide mixed numbers: 3 3/4 ÷ 1 1/4 =**




Paper Step: _____ [ ] SC-6 Ans: _____

**9** Tier 3 • Challenge

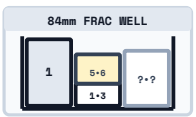
**Scale a ratio: If 3 gears weigh 4.5 oz, how much do 10 gears weigh?**




Paper Step: _____ [ ] SC-6 Ans: _____

**10** Tier 3 • Challenge

**Comparator Split & Rebuild Challenge: Subtract a 1/3 tower from a 5/6 tower, then find how many 1/4 towers fit inside:**




Paper Step: _____ [ ] SC-6 Ans: _____

# Unit 10 Mastery Check: Fraction Multiplication & Division

Name: _____ Date: _____ Score: _____ / 100

**100-POINT UNIT MASTERY ASSESSMENT — 10 UNSCAFFOLDED PROBLEMS (10 POINTS EACH)**

Solve each problem on paper first in the dot-grid workspace. Write your final answer on the line, then verify your result on the SC-6 stack. Record your total score in Appendix B (Score Tracker) and verify complete worked keystrokes online at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Find the fractional part of a bar model: 3/4 of 24 =**

[ ] +10 pts [ ] SC-6 Ans: _____

**2** Core Fluency • 10 pts

**Multiply fraction by fraction: 2/3 × 3/4 =**

[ ] +10 pts [ ] SC-6 Ans: _____

**3** Core Fluency • 10 pts

**Divide a whole number by a unit fraction: 5 ÷ 1/3 =**

[ ] +10 pts [ ] SC-6 Ans: _____

**4** Core Fluency • 10 pts

**Multiply fractions and simplify: 3/5 × 10/9 =**

[ ] +10 pts [ ] SC-6 Ans: _____

**5** Core Fluency • 10 pts

**Tower Division in the 84 mm Well: 3/4 ÷ 3/8 =**

[ ] +10 pts [ ] SC-6 Ans: _____

**6** Core Fluency • 10 pts

**Convert 5/8 to an exact decimal:**

[ ] +10 pts [ ] SC-6 Ans: _____

**7** Application • 10 pts

**A ribbon is 4 1/2 yards long. How many 3/4-yard pieces can be cut from it?**

[ ] +10 pts [ ] SC-6 Ans: _____

**8** Application • 10 pts

**Pythagoras's Perfect Fifth: Stop a 30 cm harp string at 2/3 of its length. Vibrating length?**

[ ] +10 pts [ ] SC-6 Ans: _____

**9** Application • 10 pts

**In the 84 mm Fraction Studio, each 1/12 block is 7 mm tall. How tall is 5/12?**

[ ] +10 pts [ ] SC-6 Ans: _____

**10** Error Analysis • 10 pts

**Error Analysis: Why is 1/2 ÷ 3/4 equal to 2/3 instead of 3/8?**

[ ] +10 pts [ ] SC-6 Ans: _____

# Unit 10 STEM: Suspension Bridge Tower Cable Geometry

Name: _____ Date: _____ Score: _____ / 4

**REAL-WORLD STEM APPLICATION & WORKSHOP: SUSPENSION BRIDGE TOWER CABLE GEOMETRY**

**Civil engineers calculate tower tension ratios using fraction multiplication and division.**

Companion Book (Vol. I: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fraction Scaling & Split-and-Rebuild Division) • Pythagoras's Musical Strings & Split-and-Rebuild Towers: Pythagoras found that 1/2, 2/3, and 3/4 string lengths create pure musical chords—and splitting and rebuilding 84 mm towers proves fraction division by hand!

Block Studio: Fraction Studio (84 mm Proportional Towers) • Stack Shortcut: [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷] = 2/3 (56 mm tower)

**CASE INVESTIGATION 1** STEM Modeling

**Secondary Suspender Scaling: A main bridge cable carries 3/4 of a kilonewton unit load, and a secondary suspender carries 1/2 of that load (1/2 × 3/4). Find the suspender load:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 2** STEM Modeling

**Three-Strand Load Sharing: A 1/2-meter anchor rod is split equally among 3 steel tie-down brackets (1/2 ÷ 3). What fraction of a meter does each bracket hold?**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 3** Applied Calculation

**Full Tower Reconstruction: A 1/2-meter scale test cable represents 3/4 of the design span. Use Split & Rebuild (1/2 ÷ 3/4) to find the full 1-tower design span:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

**CASE INVESTIGATION 4** Error Analysis Workshop

**Error Analysis: A structural intern computed 1/2 ÷ 3/4 = 3/8 by multiplying instead of dividing. Use the 84 mm Split & Rebuild model to explain why 1/2 ÷ 3/4 = 2/3:**




1. Model / Expression: _____

2. Paper Calculation: _____

**3. Final Answer:** _____

[ ] Verified on SC-6 Stack (Online Key: stackcalc.io/curriculum/answers)

### STEM Design Lab & Engineering Connection

Architects, designers, and engineers solve multi-step problems on paper first using fraction towers, prime factors, and area models before verifying on the SC-6 stack. Verify all worked solutions online at stackcalc.io/curriculum/answers.

# Certificate of Upper Elementary Stack Mastery

Awarded for completing all 10 units of the Upper Elementary Math Lab (840 Problems).

---

## CERTIFICATE OF UPPER ELEMENTARY STACK MASTERY

**This certifies that:** _____

has completed all 10 Hands-On Math Lab units, leveled practice worksheets (Practice A through F), 100-Point Unit Mastery Checks, and STEM Applied Cases—solving on paper first with physical manipulative models and verifying step-by-step on the 4-level StackCalc SC-6 calculator.

**[ ] Upper Elementary Stack Practitioner (Units 01–10 • Grades 3–5 • SC-6)**

**Portfolio Summary: 10 Lab Units • 840 Verified Problems • 100-Pt Mastery Avg: _____ %**

**[ ] CRA Manipulative Mastery [ ] Paper-First Predictions [ ] 4-Level Stack Verification**

**Teacher / Parent Signature:** _____ **Date:** _____

## ESSENTIAL SC-6 KEYCAP SEQUENCES TO REMEMBER

**Two-Dot Fraction Addition: 1/2 + 3/8 = 7/8**

[ • ] [ 1 ] [ • ] [ 2 ] [ ENTER ] [ • ] [ 3 ] [ • ] [ 8 ] [ + ]

Fraction Studio: Stacks 1/2 (42 mm) + three 1/8 blocks (31.5 mm) = 73.5 mm (7/8).

**Fraction Division (Split & Rebuild): 1/2 ÷ 3/4 = 2/3**

[ • ] [ 1 ] [ • ] [ 2 ] [ ENTER ] [ • ] [ 3 ] [ • ] [ 4 ] [ ÷ ]

Fraction Studio: Splits 1/2 into 3 equal 1/6 blocks (÷3) and stacks 4 of them (×4 = 2/3).

**Inside-Out Order of Operations: (14 + 6) × 3 = 60**

[ 1 ] [ 4 ] [ ENTER ] [ 6 ] [ + ] [ 3 ] [ × ]

Place Value Studio: Evaluates inner sum 14 + 6 = 20 in Row 0 before scaling by 3.

**Prime Tower Factorization Product: 2 × 2 × 3 = 12**

[ 2 ] [ ENTER ] [ 2 ] [ × ] [ 3 ] [ × ]

Prime Factor Studio: Stacks [2] + [2] + [3] logarithmic blocks to stand flush at 12.

**Long Division Verification: 21 × 4 + 3 = 87**

[ 2 ] [ 1 ] [ ENTER ] [ 4 ] [ × ] [ 3 ] [ + ]

Place Value Studio: Verifies 87 ÷ 4 = 21 R 3 via Quotient × Divisor + Remainder.

**CONGRATULATIONS! YOUR NEXT STEP IN THE STACKCALC LIBRARY (GRADES 6–8):**

- Next Curriculum Workbooks: Book 3 — Middle School Math Lab (Two-Method Ratios, Algebra & Geometry) & Book 5 — Stack Speed & Fluency.
- Companion Story Book: Read the complete mathematical discovery stories in Volume I: From Blocks to the Stack at https://stackcalc.io!

# Upper Elementary Math Lab Score Tracker & Standards Log

Record your scores for the Lab Investigation, Practice Worksheets A–F, Mastery Checks, and STEM Cases.

| Unit & Mathematical Topic | Standards | Lab (/4) | Prac A–F | Quiz (/100) | STEM (/4) |
|---|---|---|---|---|---|
| **Unit 01: Multiplication & Area Models**<br>Grade 3 • pp. 5–15 | CCSS.MATH.3.OA.A.1<br>CCSS.MATH.3.OA.B.5 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 02: Division Concepts & Remainders**<br>Grade 3 • pp. 16–26 | CCSS.MATH.3.OA.A.2<br>CCSS.MATH.3.OA.B.6 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 03: Unit Fractions & Proportional Heights**<br>Grade 3 – 4 • pp. 27–37 | CCSS.MATH.3.NF.A.1<br>CCSS.MATH.3.NF.A.2 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 04: Multi-Digit Multiplication & Partial Products**<br>Grade 4 • pp. 38–48 | CCSS.MATH.4.NBT.B.5 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 05: Long Division & Remainders**<br>Grade 4 • pp. 49–59 | CCSS.MATH.4.NBT.B.6 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 06: Fraction Equivalence & Denominators**<br>Grade 4 • pp. 60–70 | CCSS.MATH.4.NF.A.1<br>CCSS.MATH.4.NF.A.2 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 07: Prime Factors, GCF & LCM**<br>Grade 4 – 5 • pp. 71–81 | CCSS.MATH.4.OA.B.4<br>CCSS.MATH.6.NS.B.4 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 08: Order of Operations & The Stack**<br>Grade 4 – 5 • pp. 82–92 | CCSS.MATH.5.OA.A.1<br>CCSS.MATH.5.OA.A.2 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 09: Adding & Subtracting Fractions**<br>Grade 5 • pp. 93–103 | CCSS.MATH.5.NF.A.1<br>CCSS.MATH.5.NF.A.2 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |
| **Unit 10: Fraction Multiplication & Division**<br>Grade 5 • pp. 104–114 | CCSS.MATH.5.NF.B.4<br>CCSS.MATH.5.NF.B.7 | ____ / 4 | __/__/__ | ____ / 100 | ____ / 4 |

## ONLINE ANSWER KEY & COMPLETE STEP-BY-STEP KEYSTROKE PORTAL

In accordance with modern evidence-based mathematics instruction, complete worked answers are hosted online rather than printed in the student lab book to encourage authentic problem-solving and paper-first predictions.

• Online Portal: https://stackcalc.io/curriculum/answers#upper-elementary

**1. PAPER PREDICTION**

Solve by hand & write your prediction before touching keys.

**2. STACK VERIFICATION**

Enter your steps on the SC-6 to check for agreement.

**3. ERROR ANALYSIS**

If X differs, inspect where your paper step or key diverged.

APPENDIX C (EXAM 1 OF 4) • MID-BOOK BENCHMARK ASSESSMENT (UNITS 01–05 • SC-6)

# Cumulative Exam 1: Units 01–05 Benchmark

Name: _____ Date: _____ Score: _____ / 100

**100-POINT CUMULATIVE EXAM 1 — UNITS 01–05 BENCHMARK (10 PROBLEMS • 10 PTS EACH)**

Covers Area-Model Multiplication, Division, 84 mm Unit Fraction Towers, Multi-Digit Place Value Trays, and Long Division. Solve on paper first, verify on the SC-6 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Area Model Multiplication: Decompose 16 × 7 into Tens and Ones partial products on your grid and find 16 × 7:**

[ ] +10 pts [ ] SC-6 Ans: _____

**2** Core Fluency • 10 pts

**Equal-Sharing Division: Divide 96 blocks equally into 8 columns on the comparator tray. How many per column?**

[ ] +10 pts [ ] SC-6 Ans: _____

**3** Core Fluency • 10 pts

**Fraction Tower Height: One whole tower is 84 mm tall. How tall is a 3/4 tower in millimeters?**

[ ] +10 pts [ ] SC-6 Ans: _____

**4** Core Fluency • 10 pts

**Two-Dot Unit Fraction Sum: Stack three 1/8 blocks and two 1/8 blocks in the Fraction Studio comparator well:**

[ ] +10 pts [ ] SC-6 Ans: _____

**5** Core Fluency • 10 pts

**Multi-Digit Place Value Addition: Combine 468 + 375 using Gold [+1] Carry Tokens across Tens and Hundreds:**

[ ] +10 pts [ ] SC-6 Ans: _____

**6** Core Fluency • 10 pts

**Zero-Borrowing Vault: Unbundle across the zero Tens column on the Place Value Tray to solve 704 − 258 = ?**

[ ] +10 pts [ ] SC-6 Ans: _____

**7** Application • 10 pts

**Two-Digit Area Product: Multiply 24 × 15 using a 4-quadrant area model and verify on the stack:**

[ ] +10 pts [ ] SC-6 Ans: _____

**8** Application • 10 pts

**Long Division Quotient: Divide 378 ÷ 14 on paper using partial quotients and check on the SC-6:**

[ ] +10 pts [ ] SC-6 Ans: _____

**9** Application • 10 pts

**Multi-Step Crate Packing: 6 crates hold 24 books each, plus 18 extra books on the cart. How many books in all?**

[ ] +10 pts [ ] SC-6 Ans: _____

**10** Error Analysis • 10 pts

**Error Spotter: Maya solved 500 − 163 and wrote 463 without unbundling Hundreds. What is 500 − 163?**

[ ] +10 pts [ ] SC-6 Ans: _____

# Cumulative Exam 2: Units 06–10 Benchmark

Name: _____ Date: _____ Score: _____ / 100

**100-POINT CUMULATIVE EXAM 2 — UNITS 06–10 BENCHMARK (10 PROBLEMS • 10 PTS EACH)**

Covers Fraction Equivalence Towers, Logarithmic Prime Factor Blocks, GCF/LCM, 4-Register PEMDAS, and Fraction Multiplication & Division. Solve on paper first, verify on the SC-6 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Common Denominator Tower Duel: Add 1/3 + 1/4 using 1/12 towers in the Fraction Studio comparator:**

[ ] +10 pts [ ] SC-6 Ans: _____

**2** Core Fluency • 10 pts

**Unlike Fraction Difference: Compare a 3/4 tower and a 1/6 tower in the comparator well: 3/4 − 1/6 = ?**

[ ] +10 pts [ ] SC-6 Ans: _____

**3** Core Fluency • 10 pts

**Prime Factor Tower Product: Stack two [2] blocks, one [3] block, and one [5] block in the Prime Factor Studio. What composite number is formed?**

[ ] +10 pts [ ] SC-6 Ans: _____

**4** Core Fluency • 10 pts

**Prime Factor GCF Overlap: Find the Greatest Common Factor of 24 and 36 from their shared prime blocks:**

[ ] +10 pts [ ] SC-6 Ans: _____

**5** Core Fluency • 10 pts

**Prime Factor LCM Union: Find the Least Common Multiple of 12 and 18 using their prime factor towers:**

[ ] +10 pts [ ] SC-6 Ans: _____

**6** Core Fluency • 10 pts

**4-Register Two-Branch PEMDAS: Evaluate (18 + 6) ÷ (9 − 5) from the inside out without parentheses:**

[ ] +10 pts [ ] SC-6 Ans: _____

**7** Application • 10 pts

**Mixed Number Addition: Combine 1 1/2 and 2 1/4 using Fraction Studio towers and two-dot entry:**

[ ] +10 pts [ ] SC-6 Ans: _____

**8** Application • 10 pts

**Fraction Tower Scaling (Multiplication): Find 2/3 of a 3/4 tower: (2/3) × (3/4) = ?**

[ ] +10 pts [ ] SC-6 Ans: _____

**9** Application • 10 pts

**Split & Rebuild Fraction Division: Divide a 1/2 tower by 3/4 using Split & Rebuild: (1/2) ÷ (3/4) = ?**

[ ] +10 pts [ ] SC-6 Ans: _____

**10** Error Analysis • 10 pts

**Error Spotter: Ethan evaluated 8 + 4 × 5 left-to-right as 60 instead of multiplying 4 × 5 first. Find 8 + (4 × 5):**

[ ] +10 pts [ ] SC-6 Ans: _____

# Cumulative Exam 3: Multi-Step Synthesis

Name: _____ Date: _____ Score: _____ / 100

**100-POINT CUMULATIVE EXAM 3 — FULL-CURRICULUM SYNTHESIS (10 PROBLEMS • 10 PTS EACH)**

Synthesizes Fraction Tower Operations, Prime Factor Trees, Place Value Decimal Shifts, Volume, and Nested Order of Operations. Solve on paper first, verify on the SC-6 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Two-Branch Product: Evaluate $(14 + 11) \times (19 - 11)$ on the 4-level stack:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**2** Core Fluency • 10 pts

**Four-Fraction Flush Tower: Add $1/2 + 1/4 + 1/6 + 1/12$ on the Fraction Studio lid:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**3** Core Fluency • 10 pts

**Fraction of a Whole Quantity: Find $3/8$ of 64 miles:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**4** Core Fluency • 10 pts

**Fraction Division by a Unit Fraction: How many $1/8$ blocks fit inside a $3/4$ tower?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**5** Core Fluency • 10 pts

**Decimal Place Value Shift: Multiply $3.45 \times 100$ by sliding digits two columns left on the Place Value Tray:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**6** Core Fluency • 10 pts

**Rectangular Prism Volume: A 3D block storage bin measures 8 cm × 5 cm × 6 cm. Find its volume $V = l \times w \times h$:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**7** Application • 10 pts

**Nested Order of Operations: Evaluate $120 \div (6 + 3 \times 2)$ from the inside out:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**8** Application • 10 pts

**Prime Tower Quotient: Remove the prime factor blocks of 12 from the prime factor tower of 84. What factor block remains?**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**9** Application • 10 pts

**Combined Recipe Scaling: A baker triples a 2/3-cup flour scoop and adds 1/2 cup. Find the total flour in cups:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

**10** Error Analysis • 10 pts

**Error Spotter: To solve $(2/3) \div (4/5)$, a student multiplied $(2/3) \times (4/5) = 8/15$ instead of splitting ÷4 and rebuilding ×5. Find $(2/3) \div (4/5)$:**

[ ] +10 pts [ ] SC-6 **Ans:** _____

# Cumulative Exam 4: Final Capstone Mastery

Name: _____ Date: _____ Score: _____ / 100

**100-POINT CUMULATIVE EXAM 4 — FINAL CAPSTONE & 3D BLOCK STUDIO EXAM (10 PTS EACH)**

Final Grades 3–5 mastery assessment uniting the Place Value Studio, Fraction Studio, and Prime Factor Studio. Solve on paper first, verify on the SC-6 stack, and check worked keystrokes at stackcalc.io/curriculum/answers.

**1** Core Fluency • 10 pts

**Place Value & Decimal Sum: Combine 14.75 + 8.50 − 3.25 on the docked Whole & Decimal Place Value Trays:**

[ ] +10 pts [ ] SC-6 Ans: _____

**2** Core Fluency • 10 pts

**Three-Tower Flush Sum: Stack 1/2 + 1/3 + 1/6 in the Fraction Studio comparator well:**

[ ] +10 pts [ ] SC-6 Ans: _____

**3** Core Fluency • 10 pts

**Split & Rebuild Mastery: Divide a 3/4 tower by 3/8 using Split & Rebuild: (3/4) ÷ (3/8) = ?**

[ ] +10 pts [ ] SC-6 Ans: _____

**4** Core Fluency • 10 pts

**Fundamental Theorem Check: Compute $2^3 \times 3 \times 5$ by chaining prime factors on the stack:**

[ ] +10 pts [ ] SC-6 Ans: _____

**5** Core Fluency • 10 pts

**GCF × LCM Symmetry Law: Verify that GCF(12, 18) × LCM(12, 18) equals 12 × 18:**

[ ] +10 pts [ ] SC-6 Ans: _____

**6** Core Fluency • 10 pts

**Composite Surface Area: Find the total area of two 9 × 4 walls and two 6 × 4 walls:**

[ ] +10 pts [ ] SC-6 Ans: _____

**7** Application • 10 pts

**Average of Four Lab Trials: Find the mean of {18, 22, 26, 30} by summing and dividing by 4:**

[ ] +10 pts [ ] SC-6 Ans: _____

**8** Application • 10 pts

**Multi-Step Fraction Scaling: Evaluate (3/4 + 1/2) × (4/5) on the SC-6 stack:**

[ ] +10 pts [ ] SC-6 Ans: _____

**9** Application • 10 pts

**Engineering Beam Cut: From a 10-foot beam, saw off three pieces of length 2 1/2 feet. How much beam remains?**

[ ] +10 pts [ ] SC-6 Ans: _____

**10** Error Analysis • 10 pts

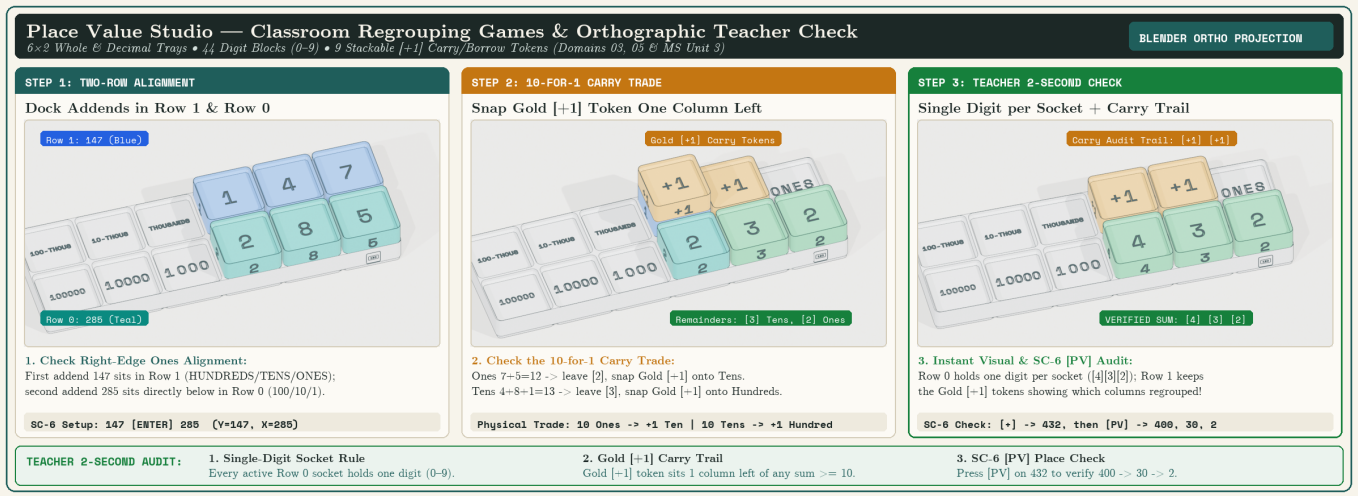
**Capstone Error Analysis: A student computed (36 − 12) ÷ (4 + 2) as 36 − (12 ÷ 4) + 2 = 35. Find the true value of (36 − 12) ÷ (4 + 2):**

[ ] +10 pts [ ] SC-6 Ans: _____

APPENDIX D (PART 1 OF 4) • OFFICIAL 3D BLOCK KIT 1 — PLACE VALUE STUDIO

# Place Value Studio: 3D Blocks & Desk Games

Hands-on activities and 2-second visual checks for the 6×2 Whole/Decimal Trays, Digit Blocks (0–9), and Gold [+1] Carry Tokens.




## HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

### Activity 1: 'Race to 1,000 Carry Relay' (2-Row Tray & Gold [+1] Carry Tokens)

Start at 000 on the 6×2 Whole Number Tray. Place a 3-digit addend in Row 0 (Line X) and add it to the waiting quantity in Row 1 (Line Y). Whenever a column sum reaches 10 or more, leave the ones digit in Row 0 and snap a Gold [+1] Carry Token one column to the left, then verify on the stack!

**Stack Check (147 + 285 = 432):** `1` `4` `7` `ENTER` `2` `8` `5` `+`

### Activity 2: 'The Decimal Point Shifter (×10 & ÷10 Slide)' (Whole & Decimal Trays)

Dock the Whole Number Tray (100,000 to 1) and Decimal Tray (0.1 to 0.000001) side-by-side at the raised decimal point (•). Build 4.75 and multiply by 10 by physically sliding every digit block one column left across the stationary decimal point—seeing why digits shift columns!

**Stack Check (4.75 × 10 = 47.5):** `4` `•` `7` `5` `ENTER` `1` `0` `×`

### Activity 3: 'The Borrowing Bank Vault (Zero-Placeholder Unbundling)'

Set 502 in Row 1 (Line Y) and 168 in Row 0 (Line X). Because Ones (2 − 8) and Tens (0 − 6) cannot subtract directly, unbundle 1 Hundred ([5] → [4]), trade across the zero Tens ([0] → [9]), and snap the Gold [+1] token onto Ones (12 − 8 = 4) to reveal 334.

**Stack Check (502 − 168 = 334):** `5` `0` `2` `ENTER` `1` `6` `8` `-`

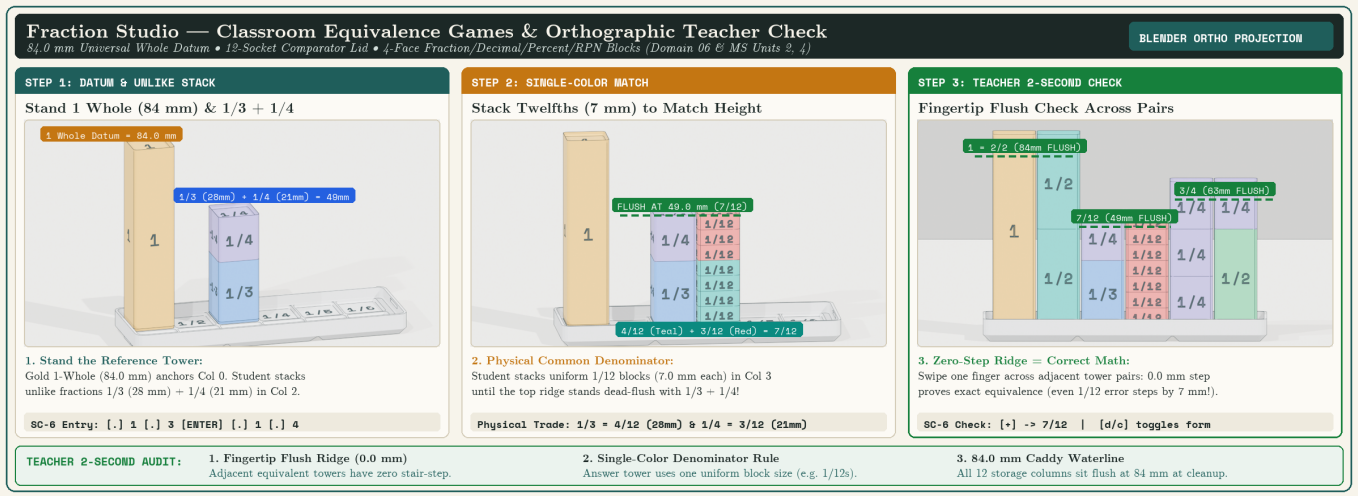
### WHY PHYSICAL 3D PLACE VALUE BLOCKS BEAT FLAT PAPER (2-SECOND VISUAL CHECK)

- Check 1 — Single-Digit Socket Rule: Each machined socket on the Place Value Tray holds one digit block (0–9). Whenever a column reaches 10, the physical tray forces a 10-for-1 trade into the next column left.
- Check 2 — Tactile Gold [+1] Carry Trail: Every Gold [+1] token sits visibly one column to the left of the column that crossed 10—eliminating forgotten carries.
- Check 3 — Direct 1-to-1 Stack Mapping: Row 1 of the physical tray maps directly to Register Y (Waiting Shelf) and Row 0 maps directly to Register X (Workbench).

APPENDIX D (PART 2 OF 4) • OFFICIAL 3D BLOCK KIT 2 — FRACTION STUDIO

# Fraction Studio: 84 mm Towers & Flush Checks

Proportional 84.0 mm fraction towers (1 Whole down to 1/12) and the 12-Socket Comparator Lid.




## HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

### Activity 1: 'The 84.0 mm Flush Tower Challenge (Make 1 Whole)'

Stand the Gold 1-Whole block (84.0 mm) in Column 0 of the 12-Socket Comparator Lid. Build an equal 84.0 mm tower in Column 1 using four blocks from different denominator families: 1/2 (42 mm) + 1/4 (21 mm) + 1/6 (14 mm) + 1/12 (7 mm) = 84.0 mm. Swipe a finger across the top ridge to feel 0.0 mm step!

**Stack Check (1/2 + 1/4 + 1/6 + 1/12 = 1):** [.] [1] [.] [2] [ENTER] [.] [1] [.] [4] [+] [.] [1] [.] [6] [+] [.] [1] [.] [1] [2] [+]

### Activity 2: 'Two-Dot Common Denominator Duel (1/3 + 1/4 = 7/12)'

Stack 1/3 (28.0 mm) + 1/4 (21.0 mm) = 49.0 mm in Column 2. Next, find a single-color block family (twelfths, 7.0 mm each) and stack seven 1/12 blocks (49.0 mm) in Column 3 so both towers stand 100% flush!

**Stack Check (1/3 + 1/4 = 7/12):** [.] [1] [.] [3] [ENTER] [.] [1] [.] [4] [+]

### Activity 3: 'Split & Rebuild Fraction Division (1/2 ÷ 3/4 = 2/3)'

Stand the 1/2 block (42.0 mm) in Column 1. Since it represents 3 fourths of the mystery tower, split 42.0 mm into 3 equal 1/6 blocks (÷ 3 → 14.0 mm is 1 fourth), then stack 4 of those 1/6 blocks in Column 3 (× 4 → 56.0 mm = two 1/3 blocks = 2/3)!

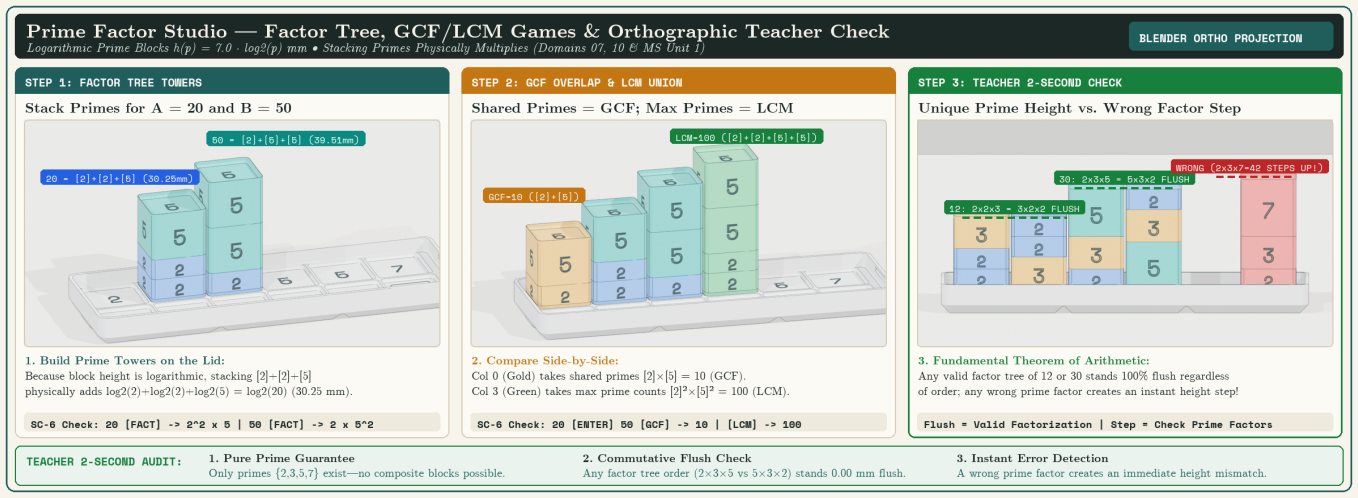
**Stack Check (1/2 ÷ 3/4 = 2/3):** [.] [1] [.] [2] [ENTER] [.] [3] [.] [4] [÷]

### WHY 84.0 MM 3D FRACTION TOWERS BEAT FLAT DRAWINGS (2-SECOND FLUSH AUDIT)

- Check 1 — The 0.0 mm Fingertip Flush Test: Because $84 = 2^2 \times 3 \times 7$, halves, thirds, fourths, sixths, sevenths, and twelfths all have exact integer millimeter heights! Even a 1/12 error creates a bold 7.0 mm physical step.
- Check 2 — Single-Color Answer Tower Rule: Adding unlike fractions requires trading into one uniform block color (common denominator) that stands dead-flush with the mixed tower.
- Check 3 — Four-Face Rotational Engraving: Rotate any Fraction Studio block 90° in your hand to read its Fraction (1/4), Decimal (0.25), Percent (25%), and Two-Dot Key (.1.4).

# Prime Factor Studio: Logarithmic Prime Blocks

Logarithmic prime factor blocks ($h(p) = 7.0 \cdot \log_2(p)$ mm for primes 2, 3, 5, 7) where multiplication becomes physical height addition.




## HANDS-ON 3D BLOCK STUDIO ACTIVITIES & STACK VERIFICATION

### Activity 1: 'The Commutative Skyscraper Race ($2 \times 3 \times 5 = 30$)'

Factor 30 two different ways: Tower A uses $6 \times 5 \rightarrow [2] + [3] + [5]$ in Column 2, while Tower B uses $10 \times 3 \rightarrow [5] + [2] + [3]$ in Column 3. Because each block's height is proportional to $\log_2(p)$, both towers stand 100% flush at 34.35 mm!

**Stack Check ($2 \times 3 \times 5 = 30$):** 2 ENTER 3 × 5 ×

### Activity 2: 'Fundamental Theorem of Arithmetic Challenge ($12 = 2 \times 2 \times 3$)'

Stack the prime tower for 12 ([2] + [2] + [3] = 25.10 mm) in Column 1, then try to match its exact height in Column 2 using different primes (e.g., [5] + [2] = 10 = 23.25 mm or [2] + [7] = 14 = 26.65 mm). No other prime combination ever stands flush!

**Stack Check ($2 \times 2 \times 3 = 12$):** 2 ENTER 2 × 3 ×

### Activity 3: 'Side-by-Side GCF Overlap & LCM Union Duel (20 & 50)'

Build Tower $A = 20$ ([2]+[2]+[5]) and Tower $B = 50$ ([2]+[5]+[5]). Copy the shared prime blocks ([2]+[5] = 10) to build GCF(20, 50) = 10 on the left, and stack the tallest count of each prime ([2]+[2]+[5]+[5] = 100) to build LCM(20, 50) = 100 on the right!

**Stack Check (GCF = 10, LCM = 100 → $10 \times 100 = 1000$):** 2 ENTER 5 × 2 ENTER 2 × 5 × 5 × ×

### HOW LOGARITHMIC 3D PRIME BLOCKS MAKE FACTORING & LOGARITHMS VISIBLE

- Check 1 — Pure Prime Guarantee: Because the Prime Factor Studio contains only prime blocks ($\{2, 3, 5, 7\}$), you can never stop a factor tree early at a composite number like 4, 6, 8, or 9.
- Check 2 — Physical Logarithm Law: Stacking blocks adds their heights $h(a) + h(b) = 7.0 \cdot \log_2(a) + 7.0 \cdot \log_2(b) = 7.0 \cdot \log_2(a \cdot b)$, turning abstract logarithms into a 3D tower!
- Check 3 — GCF + LCM Height Symmetry: On the comparator lid, $h(\text{GCF}) + h(\text{LCM}) = h(A) + h(B)$, proving visually why GCF(A, B) × LCM(A, B) = $A \times B$.

# Official StackCalc Block Studios, Calculators & Library Guide

Order official 3D Manipulative Block Studios, distraction-free calculators, and companion books at https://stackcalc.io.

## 1. THE OFFICIAL STACKCALC HANDS-ON HARDWARE & 3D BLOCK STUDIO ECOSYSTEM (STACKCALC.IO)

| | | |
|---|---|---|
| STUDIO 1 | **Place Value Studio** | Core Kit • K–6 |

6×2 Whole & Decimal Trays, 40 Digit Blocks (0–9), Gold [+1] Carry Tokens & Base-10 Set

| | | |
|---|---|---|
| STUDIO 2 | **Fraction Studio** | Core Kit • 2–8 |

78 Proportional Fraction Towers (84.0 mm Whole down to 1/12) & 12-Socket Comparator Lid

| | | |
|---|---|---|
| STUDIO 3 | **Prime Factor Studio** | Core Kit • 3–12 |

Logarithmic Prime Blocks ($h(p) = 7.0 \cdot \log2(p)$ mm for 2, 3, 5, 7) & Comparator Lid

| | | |
|---|---|---|
| SC-6 | **Elementary RPN Calculator** | Core Calc • K–8 |

4-Register Visible Stack (T, Z, Y, X), Two-Dot Fractions (.a.b), [PV], [FACT], [GCF] + Web Sim

| | | |
|---|---|---|
| SC-32 | **Scientific STEM Calculator** | Grades 8–12 & AP |

32-Key Scientific RPN: [POLY], [$\rightarrow\theta,r$], [SIN/COS/TAN], [LN/LOG], [BASE] + iOS/watchOS App

## 2. COMPANION TRADE BOOKS & WORKBOOKS IN THE STACKCALC LIBRARY (STACKCALC.IO)

| | |
|---|---|
| Volume I & II • Illustrated Trade Books: | Vol. I: From Blocks to the Stack (Grades K–6) • Vol. II: Thinking in the Stack (Grades 7–12 & AP) |
| Book 1 • Teaching with the Stack: | Master K–12 Teacher & Homeschool Guide, Word-for-Word Scripts & Misconception Matrix (46 pp.) |
| Book 2A & 2B • Elementary Math Labs: | Book 2A: Early Elementary (K–2) • Book 2B: Upper Elementary (3–5) (10 Units + 4 Exams • 840 Probs) |
| Book 3 • Middle School Math Lab: | Grades 6–8 Two-Method Block & SC-6 Workbook (10 Units, 45 Sections & 10 Quizzes • 1,000 Problems) |
| Book 4 & 5 • High School & Speed Labs: | Book 4: High School & STEM Lab (SC-32) • Book 5: Stack Speed & Fluency (180 Timed Tests) |

## 3. ONLINE ANSWER KEY, FREE SIMULATOR & HARDWARE ORDERING PORTAL

- Online Worked Solutions & Keystrokes (840 Verified Problems): https://stackcalc.io/curriculum/answers#upper-elementary
- Order Physical 3D Block Studios & Distraction-Free SC-6 Calculators: https://stackcalc.io
- Free Web Simulator & iOS/watchOS App: Practice on the interactive SC-6 / SC-32 simulator or download StackCalc 32 on the App Store!